ZOO202

Animal Diversity: Chordates

(Lesson 1- 43)

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Lesson 1

Protochordates: General characters

Why are protochordates not classified as true Chordata?

Like the remaining subphylum of the chordates, the Vertebrata, the protochordates have;

- A hollow dorsal nerve cord
- Gill slits,
- A stiff supporting rod,
- The notochord,
- The forerunner of the backbone.
- The protochordates differ chiefly from the vertebrates in **not having a backbone**.

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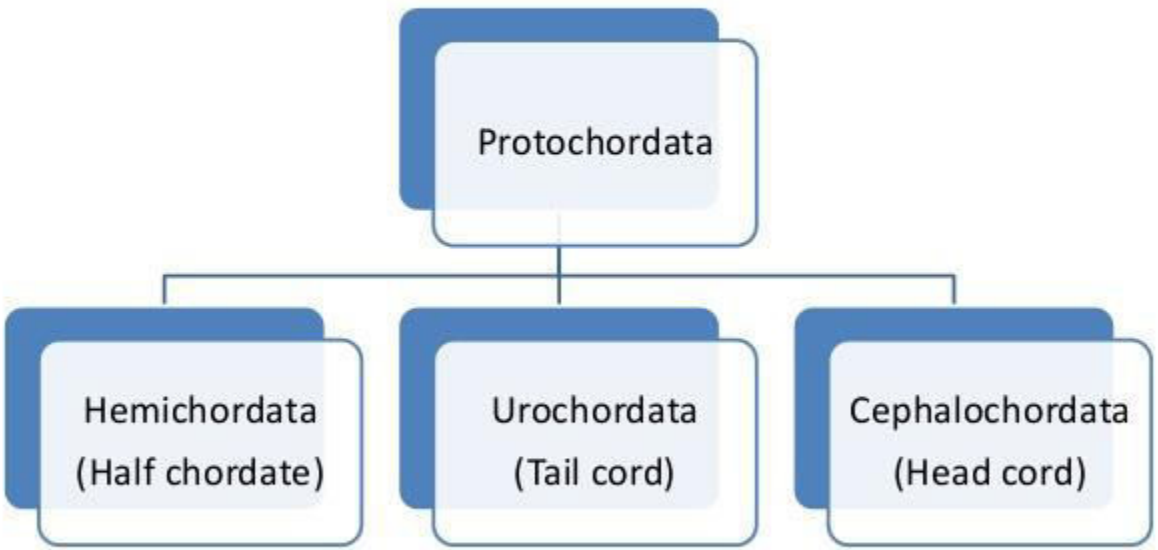
Characteristics of Protochordata (Tunicates & Amphioxus)

- The organisms belonging to the Protochordata are generally known as the **lower chordates**.
- They don't form a "proper" taxonomic group and are only classified as such for convenience purposes.
- However, they do form a major division of Chordata.
- They are also known as **Acraniata** because they lack a true skull.

Characteristics of Protochordata

- They are generally found in **marine water**.
- Their body is bilaterally **symmetrical, triploblastic**, and **coelomated**.
- At a certain stage of their lives, their body develops a long, rod-like structure for support called **notochord.**
- They exhibit organ system level of organization.
- Eg., **Herdmania, Amphioxus**.

Classification of Protochordata



Lesson 2

Phylum Hemichordata

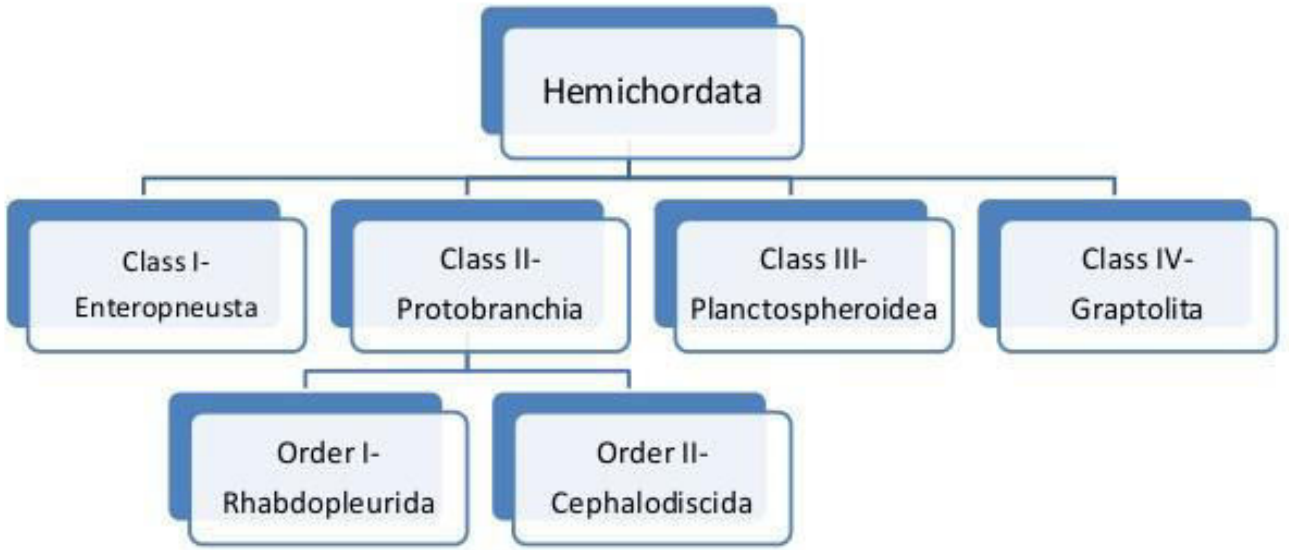
Hemichordata (Tunicates & Amphioxus)



- Marine, Solitarily
- **Deuterostomate** animals with a body divided into three regions: proboscis, collar, and trunk; coelom divided into three cavities
- Body is divided into **proboscis, collar, and trunk**.
- Bilaterally symmetrical
- Primitive notochord restricted to proboscis only, thus called **stomochord**.
- Triploblastic

- True coelom present
- Straight or U shaped gut with anus
- Nervous system normally diffuse and variable
- Open circulatory system
- Blood has no color and corpuscles.
- Excretory organ- Glomerulus
- Filter feeding
- Four classes- Enteropneusta, Pterobranchia, Planctosphaeroidea, Graptolita
- Eg., **Cephalodiscus, Rhabdopeura**

Classification of Hemichordata



Lesson 3

Class Enteropneusta

Class I – Enteropneusta (Acorn worms)



General Characters

- Marine worms
- Usually range in size between 10 and 40 cm
- More than 70 species
- most occupy U-shaped burrows in sandy

and muddy substrates between the limits of high and low tides.

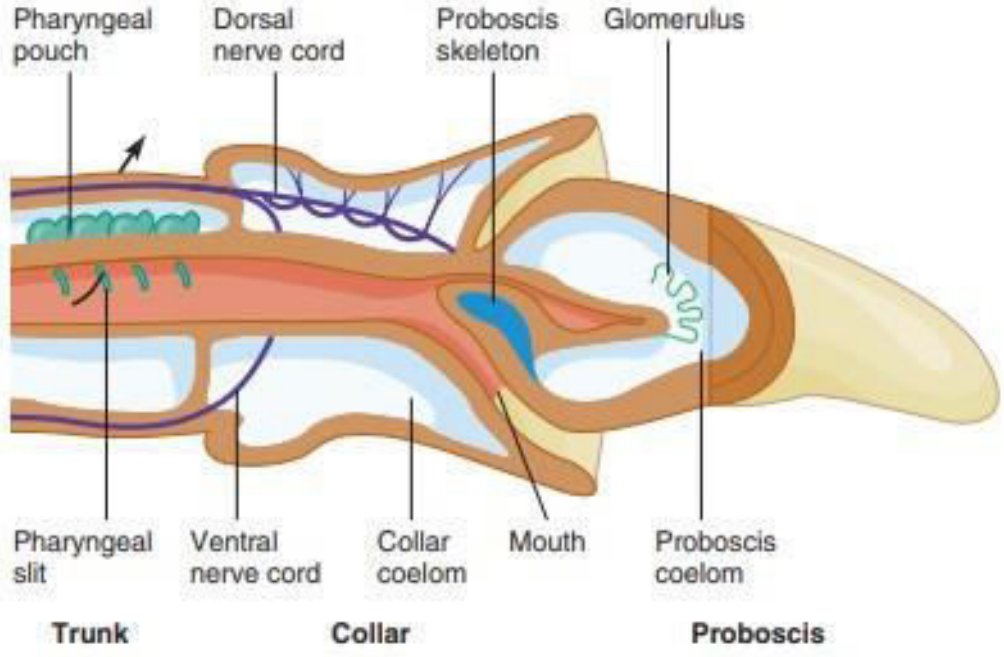
Proboscis is a short, conical projection at the worm's anterior end.

A **ring like collar** is posterior to the proboscis

An **elongate trunk** is the third division of the body

A **ciliated epidermis** and **gland cells** cover acorn worms.

The **mouth** is located ventrally between the proboscis and the collar.



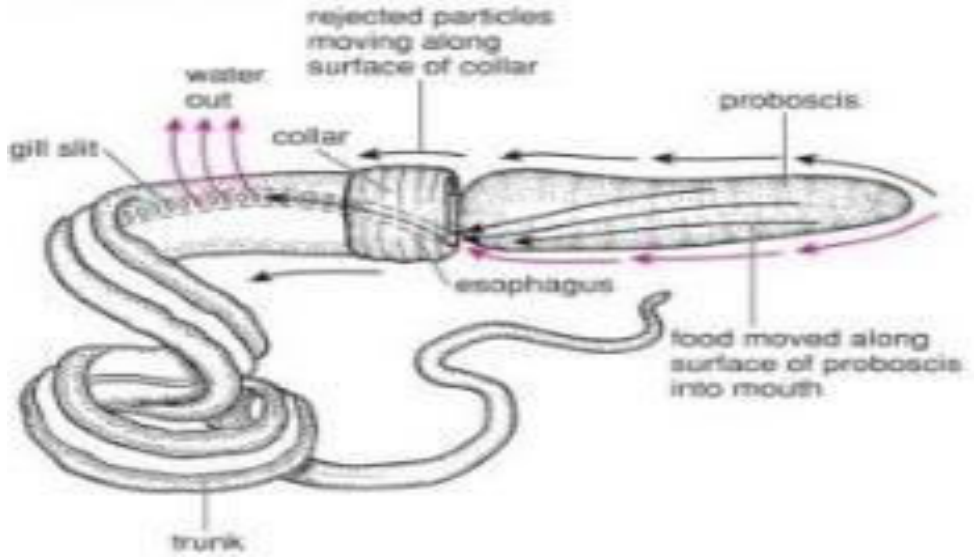
- A variable number of **pharyngeal slits**, from a few to several hundred, are positioned laterally on the trunk.
- Pharyngeal slits are openings between the anterior region of the digestive tract, called the pharynx, and the outside of the body.

Lesson 4

Enteropneusta : Feeding, Digestion and Nervous system

Enteropneusta : Feeding

- **Cilia** and **mucus** assist acorn worms in feeding.
- Detritus and other particles adhere to the mucus-covered proboscis. Tracts of cilia transport food and mucus posteriorly and ventrally.



- **Ciliary tracts** converge near the mouth and form a mucoid string that enters the mouth.
- Acorn worms may reject some substances trapped in the **mucoid string** by pulling the proboscis against the collar.
- **Ciliary tracts** of the collar and trunk transport rejected material and discard it posteriorly.

Enteropneusta : Digestion

The digestive tract of enteropneusts is a simple tube.

Food is digested as diverticula of the gut, called hepatic sacs, release enzymes.

The worm extends its posterior end out of the burrow during defecation. At low tide, coils of fecal material, called castings, lie on the substrate at burrow openings.

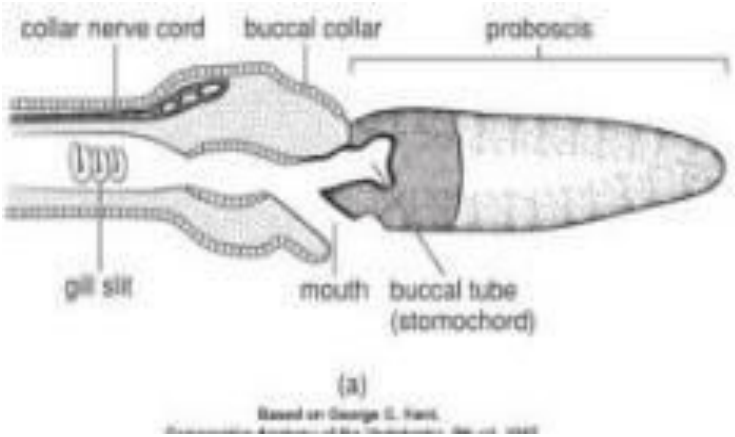
Enteropneusta : Nervous system

The nervous system of enteropneusts is **ectodermal in origin**

Lies at the base of the ciliated epidermis.

Consists of dorsal and ventral nerve tracts

and a network of epidermal nerve cells, called a **nerve plexus**.



There are **no major ganglia**.

Sensory receptors are unspecialized and widely distributed over the body.

Lesson 5

Enteropneusta : Respiration, Circulation and Excretion

Enteropneusta : Respiration

- Because acorn worms are small, respiratory gases and metabolic waste products (principally ammonia) probably are exchanged by diffusion across the body wall.
- In addition, respiratory gases are exchanged at the pharyngeal slits.
- Cilia associated with pharyngeal slits circulate water into the mouth and out of the body through the pharyngeal slits.
- As water passes through the pharyngeal slits, gases are exchanged by diffusion between water and blood sinuses surrounding the pharynx.

Enteropneusta : Circulation

The circulatory system of acorn worms consists of **one dorsal and one ventral contractile vessel**.

Blood moves **anteriorly in the dorsal vessel** and **posteriorly in the ventral vessel**.

Branches from these vessels lead to **open sinuses**.

All blood flowing anteriorly passes into a series of blood sinuses, called the **glomerulus**, at the base of the proboscis.

Enteropneusta : Excretion

- Excretory wastes may be filtered through the glomerulus, into the coelom of the proboscis,
- Later on release to the outside through one or two pores in the wall of the proboscis.
- The blood of acorn worms is colorless, lacks cellular elements, and distributes nutrients and wastes.

Lesson 6

Enteropneusta : Reproduction and Development

Enteropneusts: Reproduction and Development

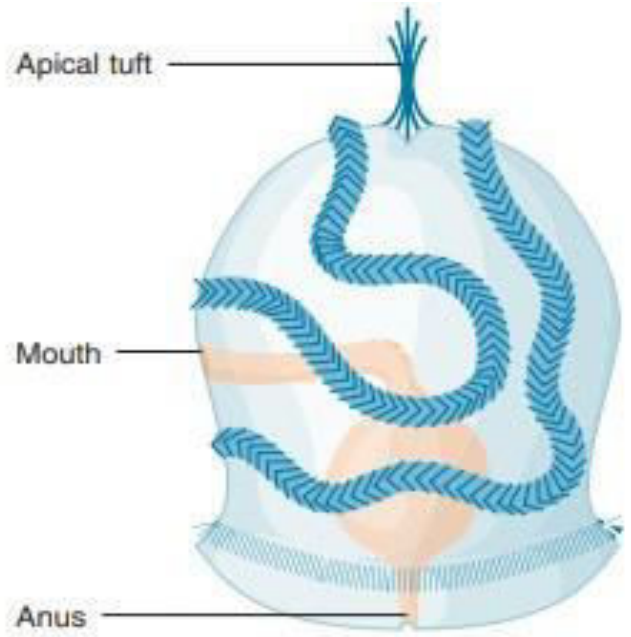
Enteropneusts are **dioecious**.

Two rows of gonads lie in the body wall in the anterior region of the trunk, and each gonad opens separately to the outside.

Fertilization is **external**.

Spawning by one worm induces others in the area to spawn—behavior that suggests the presence of spawning pheromones.

- Enteropneusts normally reproduce sexually via external fertilization, and develop either directly or via **tornaria larvae**.
- The indirect developers, including ***Balanoglossus*** and ***Ptychodera*** **species**, are in the majority



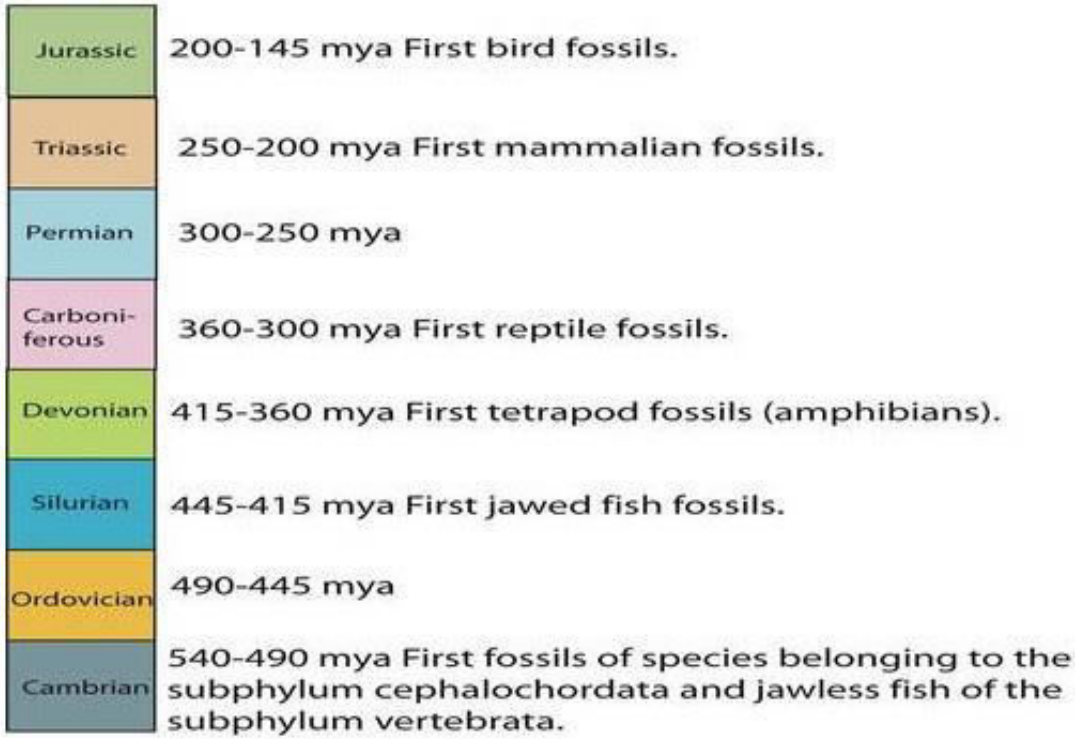
- These species develop from egg to planktonic tornaria larva to adult form.
- The tornaria larvae eventually become **sessile**, with the burrow-dwellers Developing tails behind the anus that they use to anchor themselves in their mucus-lined tunnels.
- Direct developers, on the other hand, hatch into adult animals, by passing the planktonic phase.
- An example is *Saccoglossus kowalevskii*.

Lesson 7

Phylum Chordata

Chordates

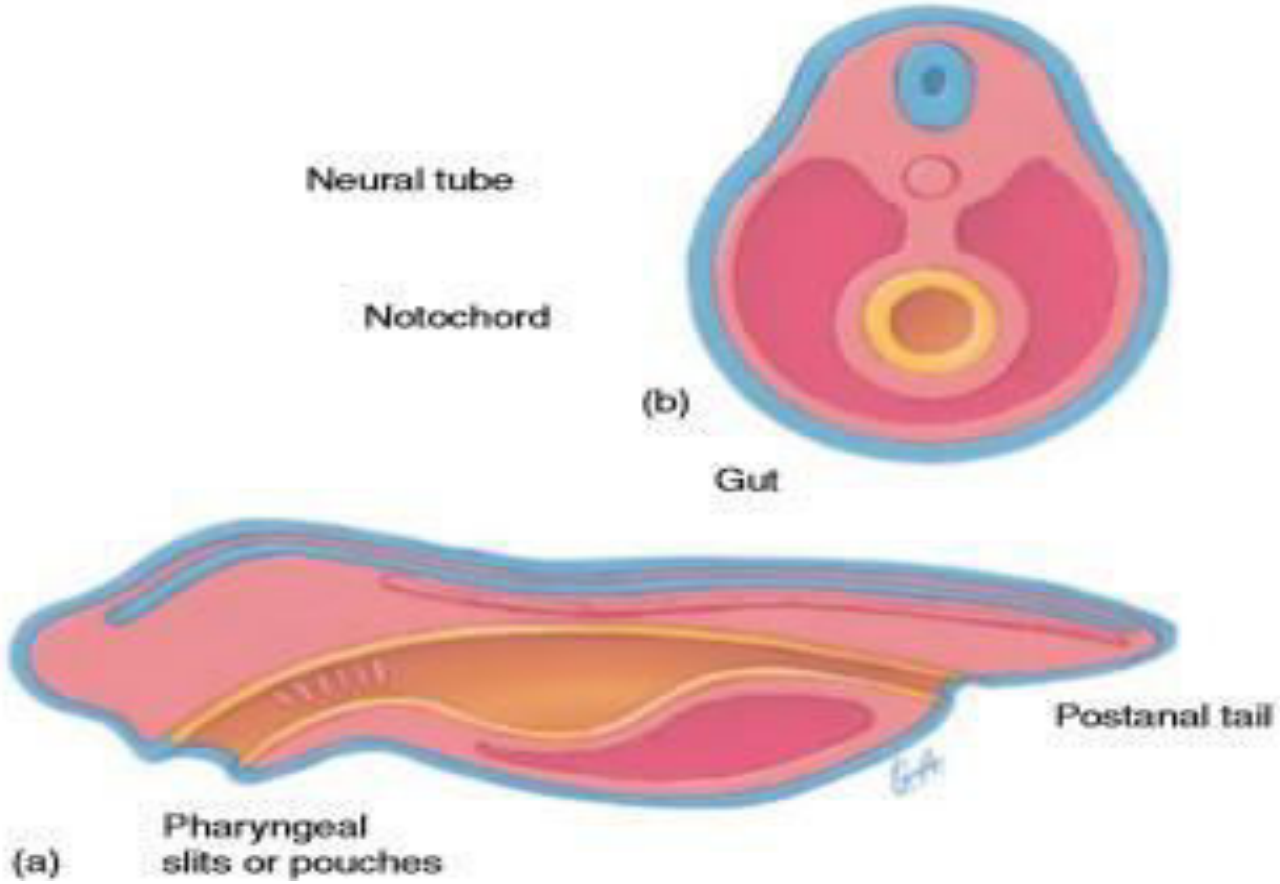
- The earliest tetrapods are the amphibians, with fossils dating back to the Devonian period
- Shortly thereafter, reptile-like tetrapods arose.
- It was some time after the appearance of reptiles, approximately 80 million years, that mammals entered the scene. The first mammalian fossils date back to the Triassic period, about 208 million years ago.
- Finally, bird fossils appear 146 million years ago in the Jurassic period.



Chordate Characteristics

- Bilaterally symmetrical
- deuterostomate animals
- Four unique characteristics present at some stage in development: notochord, pharyngeal slits or pouches, dorsal tubular nerve cord, and postanal tail Presence of an endostyle or thyroid gland
- Complete digestive tract.
- Ventral, contractile blood vessel (heart)

- The four characteristics listed in number 2 are unique to chordates and are discussed further in the paragraphs that follow (figure 17.6).
- The phylum is named after the notochord
- It consists of a connective-tissue sheath that encloses cells
- This arrangement gives the notochord some turgidity, which prevents compression along the anteroposterior axis.
- At the same time, the notochord is flexible enough to allow lateral bending
- In most adult vertebrates, cartilage or bone partly or entirely replaces the notochord.



Chordate Body Plan

Lesson 8

Subphylum: Urochordates

Subphylum Urochordata

Urochordates have a sparse fossil record. A Precambrian fossil known as *Yarnemia* has been referred to the Urochordata, but this assignment is doubtful. Complete body fossils of tunicates are rare, but tunicates in some families generate microscopic *spicules* that may be preserved as microfossils. Such spicules have occasionally been described from Jurassic and later rocks. Few paleontologists are familiar with them; tunicate spicules may be mistaken for sponge spicules.

Urochordates: General Characteristics

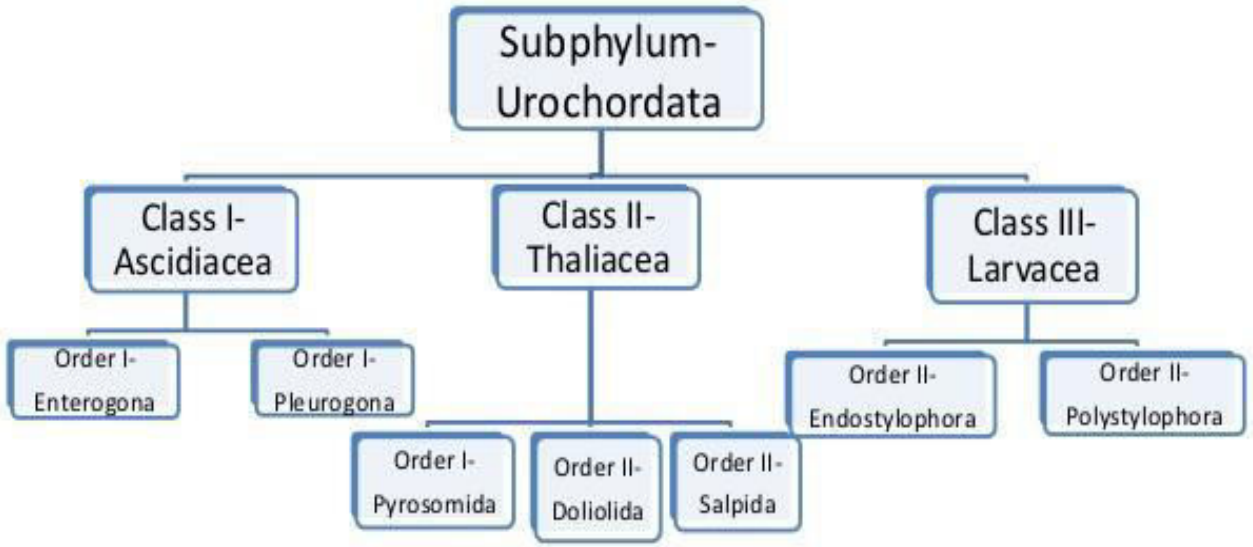
- Tail chordate
- Most primitive of the chordates
- They are found in the **marine environment**.
- They are sessile and **filter-feeders**.
- They are also known as tunicates because their body is surrounded by a leathery sheath composed of **tunicin** (cellulose).
- **Retrogressive metamorphosis** : The notochord appears in the larval stage in the tail of the larva and disappears in the adult.

- Triploblastic
- U shaped gut
- Coelom absent
- Hermophroditic
- Respiration occurs through gills.
- They have an Incomplete closed circulatory syste
- Haemocyanin as blood pigment
- The excretory organs are absent.
- They reproduce asexually by budding.

Eg., **tunicates or sea squirts**



Clasification of Urochordata



Lesson 9

Introduction to Tunicates and their body organization

Ascidians: tunicates

- The ascidians comprise the largest class of tunicates.
- They are sessile as adults and are either solitary or colonial.
- The appendicularians and thaliaceans are planktonic as adults. In some localities, tunicates occur in large enough numbers to be considered a dominant life-form.
- Sessile urochordates attach their saclike bodies to rocks, pilings, ship hulls, and other solid substrates.
- The unattached end of urochordates contains two **siphons** that permit seawater to circulate through the body.

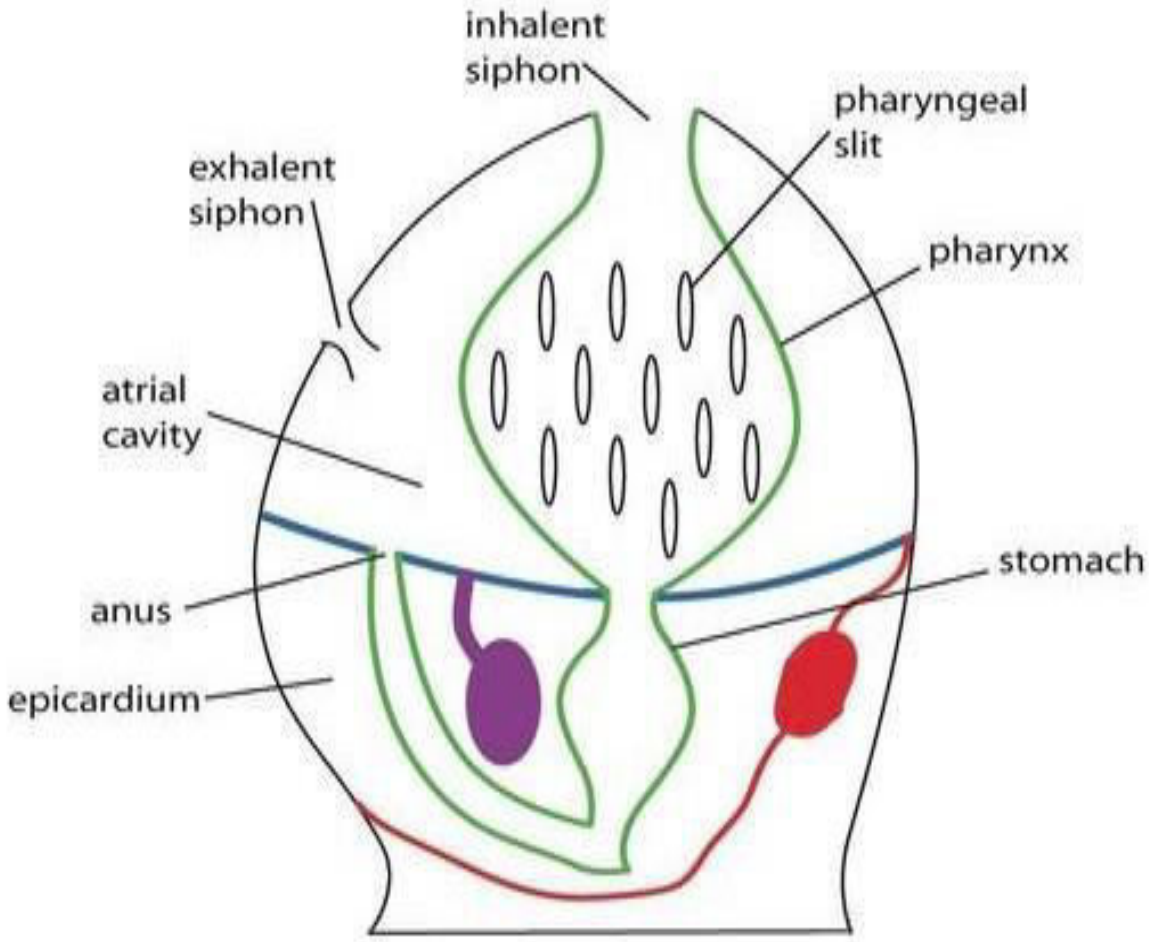
- One siphon is the oral siphon, which is the inlet for water circulating through the body and is usually directly opposite the attached end of the ascidian.
- It also serves as the **mouth opening**. The second siphon, the atrial siphon, is the opening for **excurrent water**.





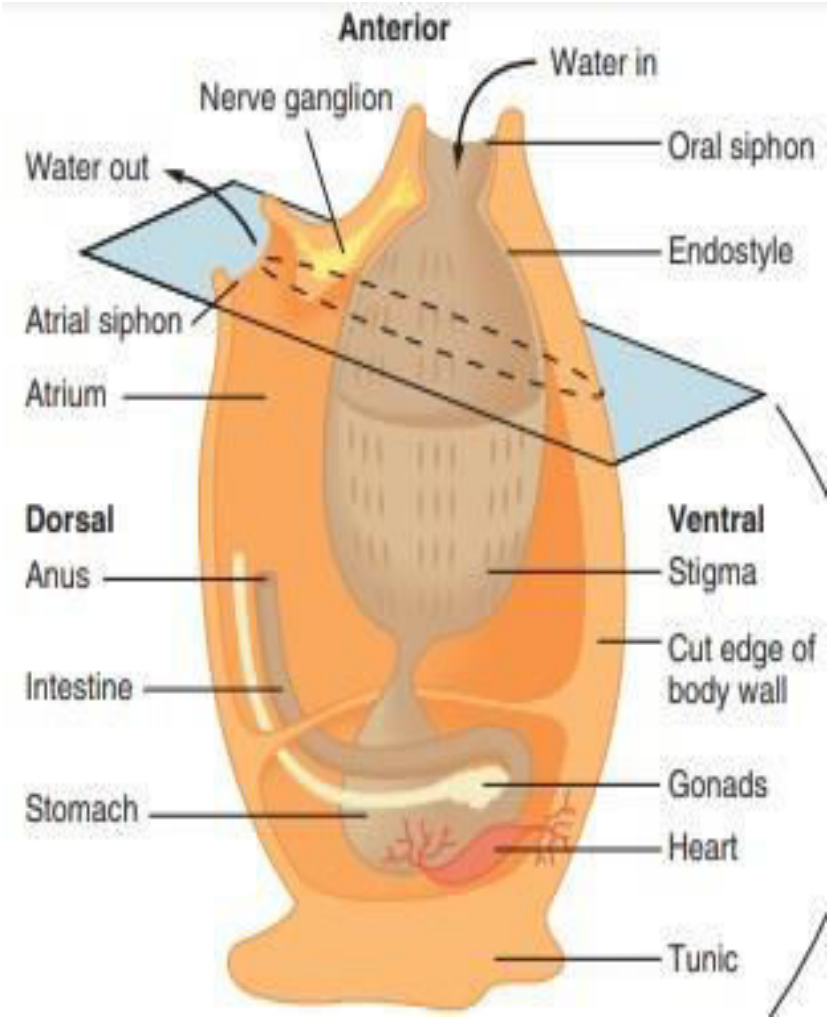
- The body wall of most tunicates (L. tunicatus, to wear a tunic or gown) is a connective-tissue-like covering, called the **tunic**, that appears gel-like but is often quite tough.
- Secreted by the **epidermis**, it is composed of proteins, various salts, and cellulose.
- Some mesodermally derived tissues, including blood vessels and blood cells, are incorporated into the tunic.
- Rootlike extensions of the tunic, called **stolons**, help anchor a tunicate to the substrate and may connect individuals of a colony.

Tunicates derive their name from a thick extracellular body covering, called a **tunic,** that is made up of cellulose-like, complex sugars. Another interesting feature of tunicates is their circulatory system. Instead of blood vessels, the heart pumps blood through a number of small pockets called **sinuses.** In addition, the heart periodically changes the direction that it pumps blood.



Tunicates: Body organization

- Tunicates derive their name from a thick extracellular body covering, called a tunic that is made up of cellulose-like, complex sugars
- Heart pumps blood through a number of small pockets called sinuses



- The adult tunicate body is shaped like a small barrel with two large openings called siphons.
- Water enter through inhalaent siphone and exit through exhalent siphone
- The majority of the organs lie within a body cavity below the atrial cavity called the **epicardium**

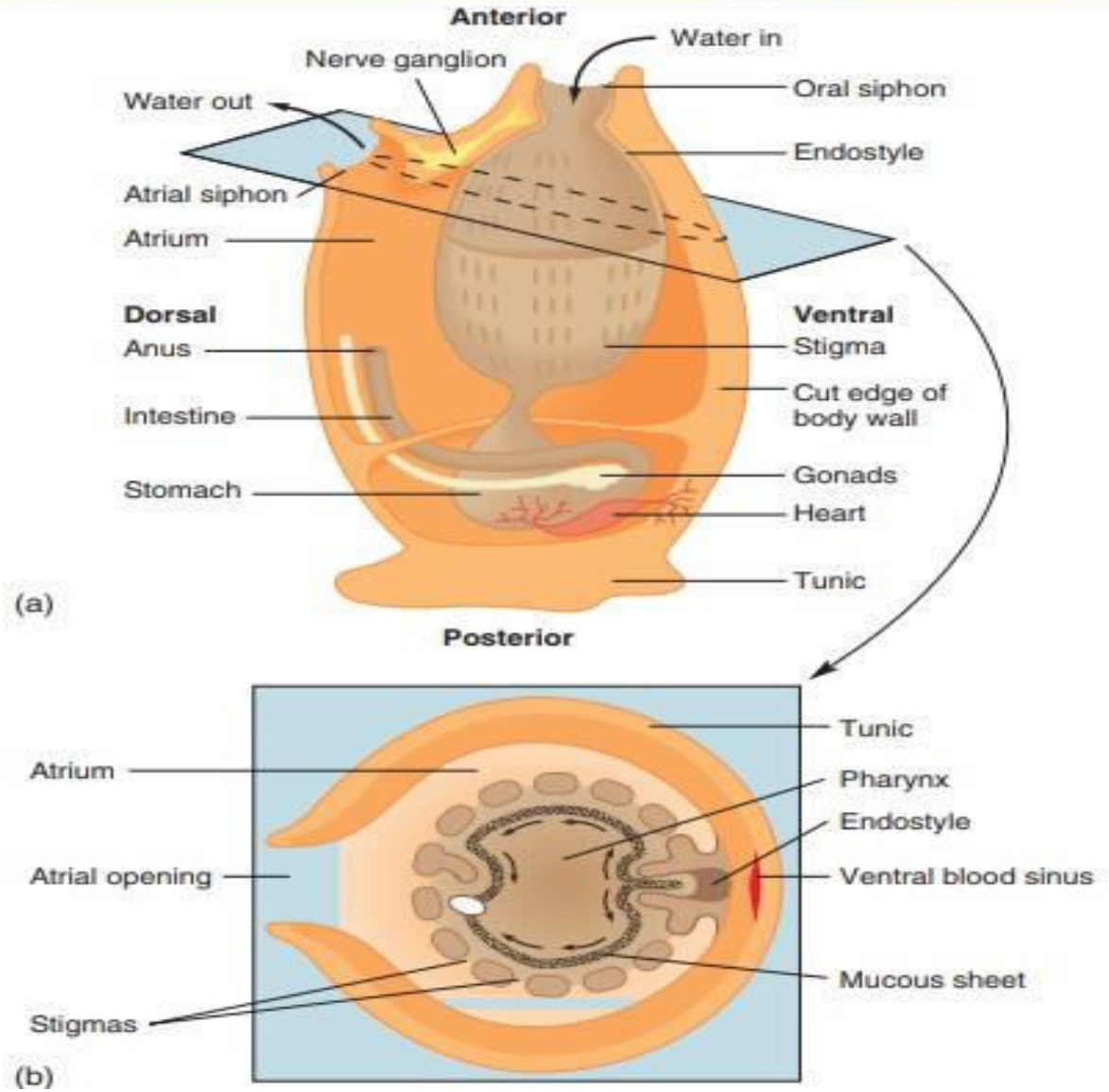
Lesson 10

Tunicates: Digestion and Respiration

Tunicates mainly eat phytoplankton, zooplankton, detritus and other small organisms in the water, these organisms are filtered through the tunicate. They have cilia in the pharynx and they also produce mucus. Their food gets stuck in the mucus and with the help of beating cilia the mucus moves toward the digestive system. There are also predatory tunicates, which eat small creatures that get pulled through their siphon and get stuck on their mucous.

Tunicates: Digestion

- The most obvious internal structures of the urochordates are a very large pharynx and a cavity, called the atrium, that surrounds the pharynx laterally and dorsally.
- The pharynx of tunicates originates at the oral siphon and is continuous with the remainder of the digestive tract.



- The oral margin of the pharynx has tentacles that prevent large objects from entering the pharynx.
- Numerous pharyngeal slits called stigmas perforate the pharynx. Cilia associated with the stigmas cause water to circulate into the pharynx, through the stigmas, and into the surrounding atrium. Water leaves the tunicate through the atrial siphon.
- The digestive tract of adult tunicates continues from the pharynx and ends at the anus near the atrial siphon.

- During feeding, cells of a ventral, ciliated groove, called the **endostyle**, form a mucous sheath.
- Cilia move the mucous sheet dorsally across the pharynx.
- Food particles, brought into the oral siphon with incurrent water, are trapped in the mucous sheet and passed dorsally.
- Food is incorporated into a **string of mucus** that ciliary action moves into the next region of the gut tract.

- Digestive enzymes are secreted in the **stomach**, and most absorption occurs across the walls of the intestine.
- **Excurrent water** carries digestive wastes from the anus out of the atrial siphon.
- In addition to its role in feeding, the pharynx also functions in **gas exchange**.
- Gases are exchanged as water circulates through the tunicate.

Lesson 11

Tunicates: Circulation and Excretion

Tunicates: Circulation

- The tunicate heart lies at the **base of the pharynx**.
- One vessel from the heart runs anteriorly under the endostyle, and another runs posteriorly to the digestive organs and gonads.
- Blood flow through the heart is **not unidirectional**.
- Peristaltic contractions of the heart may propel blood in one direction for a few beats; then the direction is reversed.
- The significance of this reversal is not understood.
- Tunicate **blood plasma is colorless** and contains various kinds of amoeboid cells.

Tunicates: Excretion

- Ammonia diffuses into water that passes through the pharynx and is excreted.
- In addition, **amoeboid cells** of the circulatory system accumulate uric acid and sequester it in the intestinal loop.
- **Pyloric glands** on the outside of the intestine are also thought to have excretory functions.

Tunicates: Reproduction

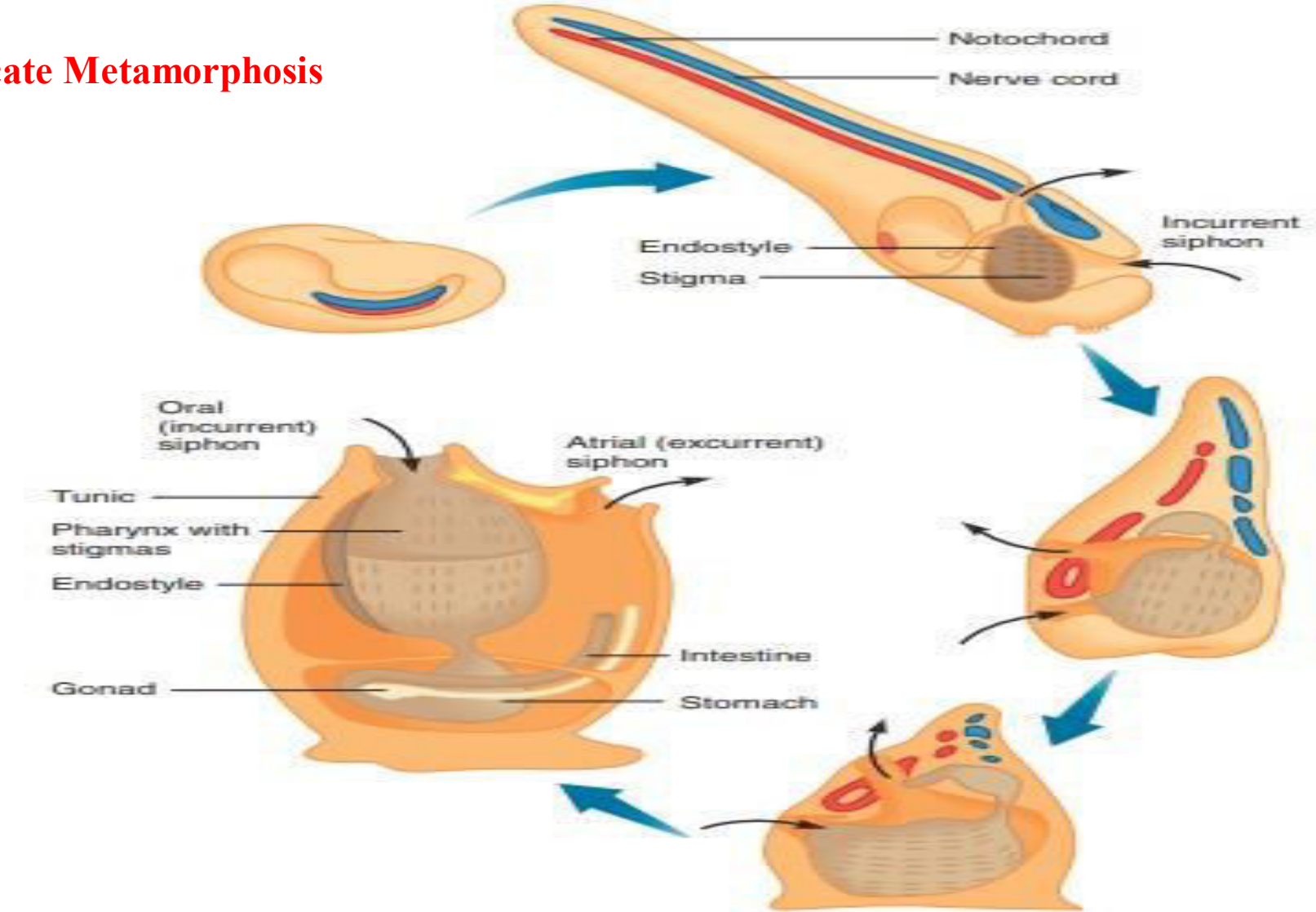
- Urochordates are **monoecious**.
- Gonads are located near the loop of the intestine
- Genital ducts open near the **atrial siphon**.
- Gametes may be shed through the atrial siphon for **external fertilization** or eggs may be retained in the atrium for fertilization and early development.
- Although self-fertilization occurs in some species, **cross-fertilization** is the rule.

Tunicates: Development

- Development results in the formation of a **tadpole like larva** with all four chordate characteristics.
- Metamorphosis begins after a brief free-swimming larval existence, during which the larva does not feed.

- The larva settles to a firm substrate and attaches by **adhesive papillae** located below the mouth.
- During metamorphosis, the outer epidermis shrinks and pulls the notochord and other tail structures internally for reorganization into adult tissues.
- The internal structures rotate 180°, positioning the oral siphon opposite the adhesive papillae and bending the digestive tract into a **U shape.**

Tunicate Metamorphosis



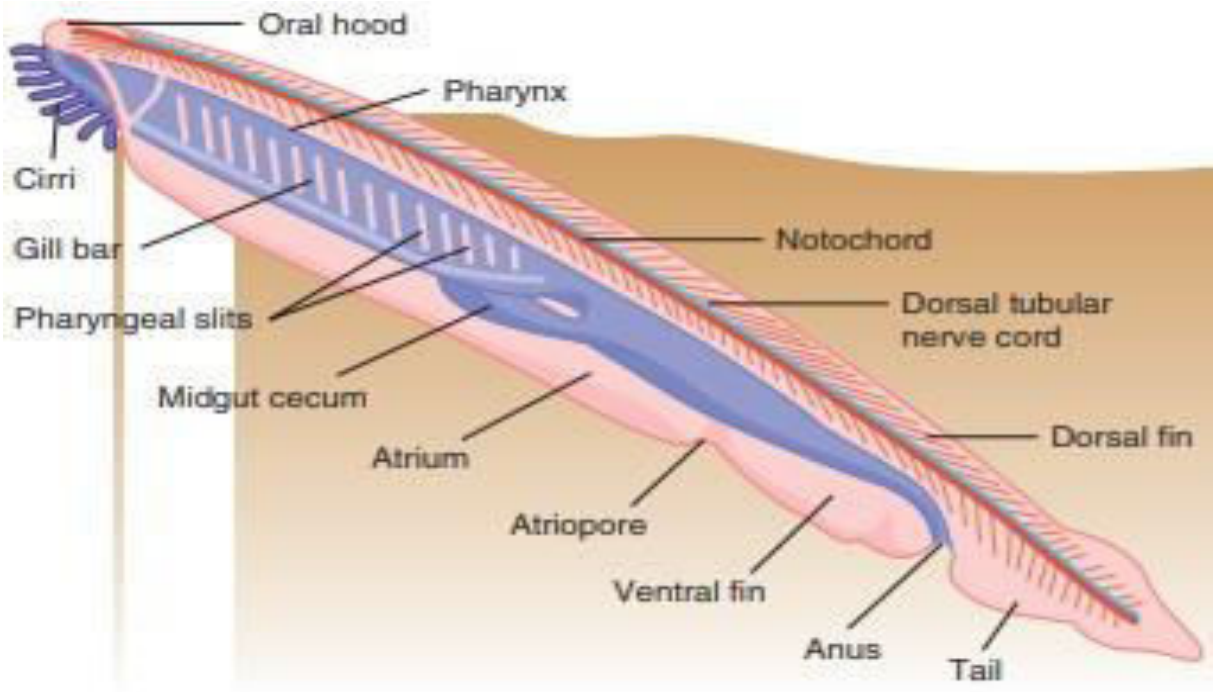
Lesson 12

Subphylum Cephalochordata

Cephalochordata: General Characteristics

- Members of the subphylum Cephalochordata are called **lancelets**.
- The cephalochordates consist of two genera, **Branchiostoma (amphioxus)** and **Asymmetron**, and about 45 species.
- They are distributed throughout the world's oceans in shallow waters that have clean sand substrates.
- Cephalochordates are small **(up to 5 cm long),** tadpole like animals.
- They are elongate, laterally flattened, and nearly transparent.
- In spite of their streamlined shape, cephalochordates are relatively **weak swimmers**

- They spend most of their time in a filter feeding position partly to mostly buried with their anterior end sticking out of the sand
- The notochord of cephalochordates extends from the tail to the head, giving them their name.
- Unlike the notochord of other chordates, most of the cells are muscle cells, making the notochord somewhat contractile.
- Both of these characteristics are probably adaptations to burrowing.



Internal structure of Branchiostoma (amphioxus)

- Contraction of the muscle cells increases the **rigidity of the notochord** by compressing the fluids within, giving additional support when pushing into sandy substrates.
- Relaxation of these muscle cells increases **flexibility for swimming**.
- Muscle cells on either side of the notochord cause **undulations** that propel the cephalochordate through the water.
- **Longitudinal, ventrolateral folds** of the body wall help stabilize cephalochordates during swimming
- **Median dorsal fin and a caudal fin** also aid in swimming.

- An **oral hood** projects from the anterior end of cephalochordates.
- Ciliated, fingerlike projections, called **cirri**, hang from the ventral aspect of the oral hood and are used in feeding.
- The posterior wall of the oral hood bears the **mouth opening** that leads to a large pharynx.

- Numerous pairs of **pharyngeal slits** perforate the pharynx and are supported by cartilaginous gill bars.
- Large folds of the body wall extend ventrally around the pharynx and fuse at the ventral midline of the body, **creating the atrium**.
- A chamber that surrounds the pharyngeal region of the body.
- It may protect the delicate, filtering surfaces of the pharynx from bottom sediments.
- The opening from the atrium to the outside is called the **atriopore**

Lesson 13

Cephalochordates: Feeding and Digestion

Cephalochordates: Feeding

- Cephalochordates are **filter feeders**.
- During feeding, they are partially or mostly buried in sandy substrates with their mouths pointed upward.
- Cilia on the lateral surfaces of gill bars sweep water into the mouth.
- Water passes from the pharynx, through pharyngeal slits to the atrium, and out of the body through the atriopore.
- Food is initially sorted at the **cirri**.

- Larger materials catch on cilia of the **cirri**.
- As these larger particles accumulate, contractions of the cirri throw them off.
- Smaller, edible particles are pulled into the mouth with water and are collected by cilia on the **gill bars** and in mucus secreted by the **endostyle**.

Cephalochordates: Digestion

- As in tunicates, the endostyle is a ciliated groove that extends longitudinally along the midventral aspect of the pharynx.
- Cilia move food and mucus dorsally, forming a food cord to the gut.
- A ring of cilia rotates the food cord, dislodging food.
- Digestion is both **extracellular and intracellular**.
- A diverticulum off the gut, called the **midgut cecum**, extends anteriorly.
- It ends blindly along the right side of the pharynx and secretes digestive enzymes.
- An anus is on the left side of the ventral fin.

Lesson 14

Cephalochordates: Circulation, Excretion, Reproduction and Development

Cephalochordates: Circulation

- Cephalochordates do not possess a true heart.
- Contractile waves in the walls of major vessels propel blood.
- Blood contains amoeboid cells and bathes tissues in open spaces.

Cephalochordates: Excretion

- Excretory tubules are **modified coelomic cells** closely associated with blood vessels.
- This arrangement suggests active transport of materials between the **blood and excretory tubules**.
- The coelom of cephalochordates is reduced, compared to that of most other chordates.
- It is restricted to canals near the gill bars, the endostyle, and the gonads.

Cephalochordates: Reproduction and Development

- Cephalochordates are **dioecious**.
- Gonads bulge into the atrium from the lateral body wall.
- Gametes are shed into the atrium and leave the body through the atriopore.
- External fertilization leads to a **bilaterally symmetrical larva**.
- Larvae are **free-swimming**
- But they eventually **settle to the substrate** before metamorphosing into adults.

Lesson 15

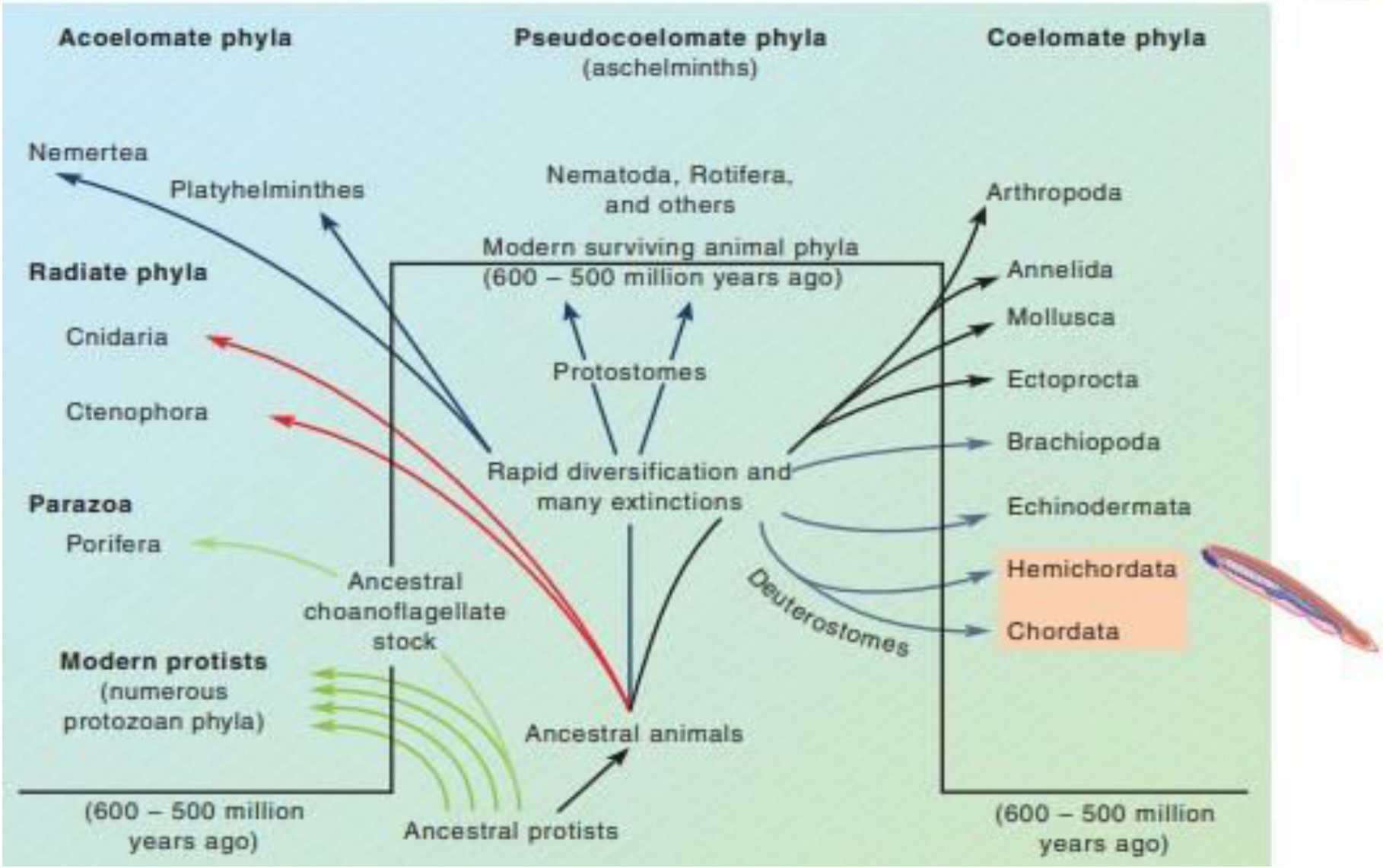
Phylogenetic Relationships between hemichordates and chordates

Phylogenetic Relationships

- Animals in the phyla Hemichordata and Chordata share **deuterostome characteristics** with echinoderms.
- Most zoologists, therefore, believe that ancestral representatives of these phyla were derived from a common, as yet undiscovered, **triploblastic ancestor**.
- The chordates are characterized by a **dorsal tubular nerve cord**, **a notochord**, **pharyngeal slits or pouches**, and **a postanal tail**.
- The only characteristics they share with the hemichordates are pharyngeal slits and, in some species, a dorsal tubular nerve cord.

- Therefore, most zoologists agree that the evolutionary ties between the **chordates and hemichordates** are closer than those between echinoderms and either phylum.
- Chordates and hemichordates, however, probably diverged from widely separated points along the **deuterostome lineage**.
- The diverse body forms and lifestyles present in these phyla support this generalization.

Phylogenetic Relationships among the Hemichordata and Chordata



Lesson 16
Deuterostomate Phylogeny

Phylogenetic Relationships

- The evolutionary relationships between the hemichordates and chordates are difficult to document with certainty.
- The **dorsal tubular nerve cord** and **pharyngeal slits** of hemichordates are evidence of evolutionary ties between these phyla.
- **Synapomorphies** that distinguish chordates from hemichordates include **tadpole larvae, a notochord, a postanal tail, and an endostyle**.

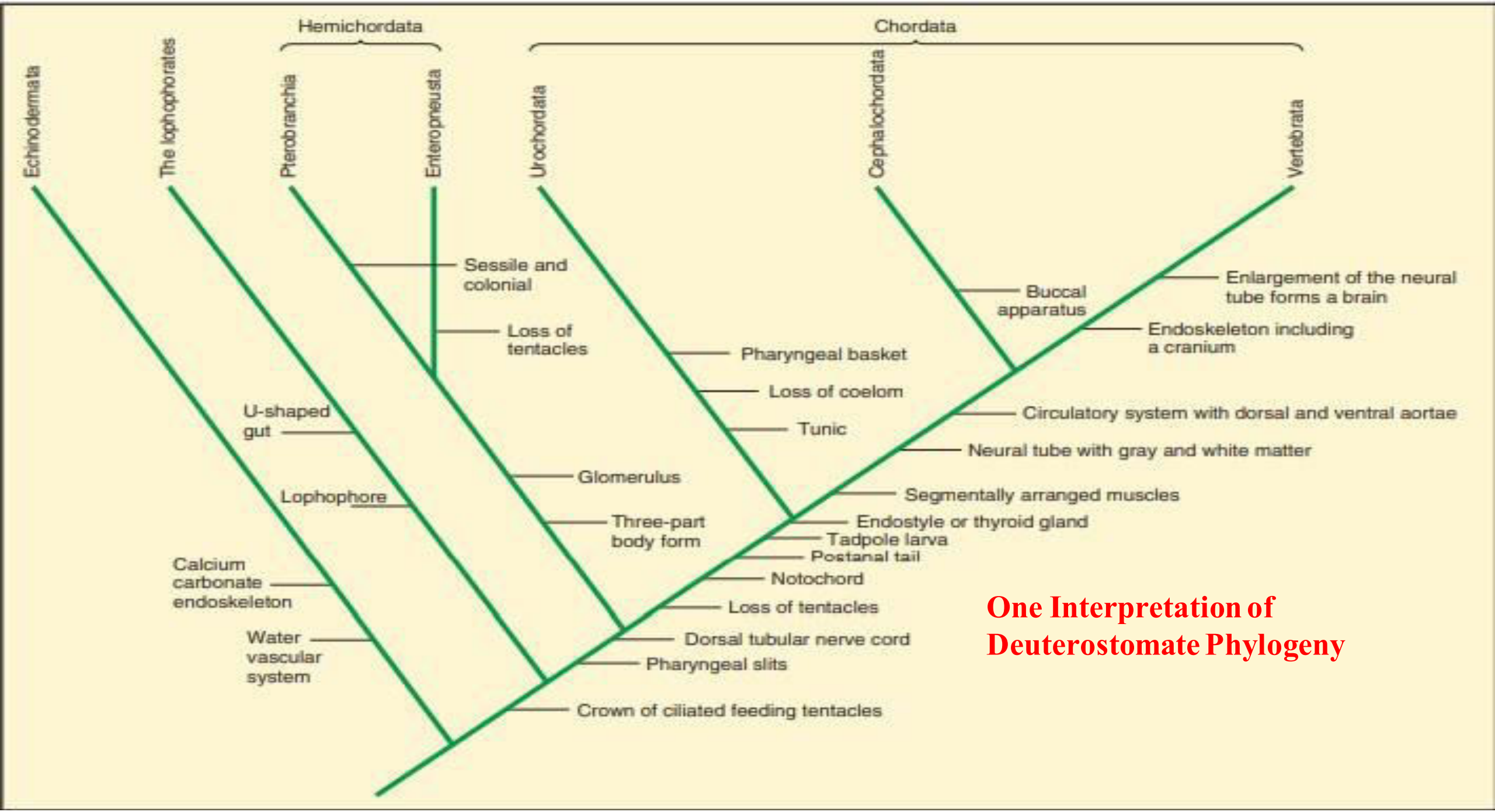
- The earliest echinoderms were probably **sessile filter feeders**.
- The lifestyle of **adult urochordates** suggests a similar ancestry (perhaps from a common ancestor with echinoderms) for chordates.
- The evolution of motile chordates from attached ancestors may have involved the development of a **tadpole like larva**.
- Increased larval mobility is often adaptive for species with sedentary adults because it promotes dispersal.

- The evolution of motile adults could have resulted from **paedomorphosis**
- Paedomorphosis is the development of sexual maturity in the larval body form.

  (Paedomorphosis is well documented in the animal kingdom, especially among amphibians.)
- Paedomorphosis could have led to a small, sexually reproducing, **fishlike chordate** that could have been the ancestor of higher chordates.
- The largest and most successful chordates belong to the subphylum Vertebrata.

- **Bony or cartilaginous vertebrae** that completely or partially replace the notochord characterize the vertebrates.
- The development of the anterior end of the nerve cord into a brain and the development of specialized sense organs on the head are evidence of a **high degree of cephalization**.
- The skeleton is modified anteriorly into a **skull or cranium**.
- There are **eight classes of vertebrates**.
- Because of their cartilaginous and bony endoskeletons, vertebrates have left an abundant fossil record.

- Ancient jawless fishes were common in the **Ordovician period**, approximately 500 million years ago.
- Over a period of approximately 100 million years, fishes became the **dominant vertebrates**.
- Near the end of the **Devonian period**, approximately 400 million years ago, **terrestrial vertebrates** made their appearance.
- Since that time, vertebrates have radiated into most of the earth's habitats.



One Interpretation of Deuterostomate Phylogeny

Lesson 17
The Fishes

Phylogenetic Relationship

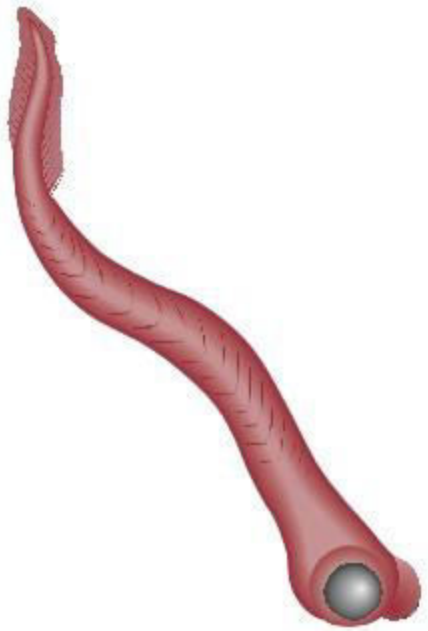
- Fishes are members of the chordate subphylum Vertebrata; thus,
- They have **vertebrae** that surround their spinal cord and provide the primary axial support.
- They also have a **skull** that protects the brain.
- Zoologists do not know what animals were the first vertebrates?
- Molecular evidence gathered by comparing gene similarities of cephalochordates and vertebrates suggests that the vertebrate lineage may go back about **750 million years**.



lionfish (Pterois)

- This date has not been confirmed by fossil evidence.
- Recent cladistic analysis of vertebrate evolution indicates that a group of fishes called **hagfishes are the most primitive vertebrates** known (living or extinct).
- Making a connection between this lineage and other vertebrates depends on the analysis of two key vertebrate characteristics, **the brain and bone**.
- Chinese researchers have recently unearthed the oldest alleged vertebrate fossils, a small, **lancelet-shaped animal** that has characteristics that suggest an active, predatory lifestyle.
- A brain is present that would have processed sensory information from the pair of eyes that are seen in fossils.

- **Muscle blocks** along the body wall suggest an active swimming existence. This evidence means that these **530 million-year-old animals located prey by sight** and then pursued it through prehistoric seas.
- The **origin of bone** in vertebrates is equally intriguing.
- A group of ancient eel-like animals, **the conodonts**, are known from fossils that date back about **510 million years**.
- They have been assigned to a variety of phyla, but recent evidence is bringing a growing number of zoologists to accept them as full fledged vertebrates.



Conodont (Clydagnathus)

- They have **two large eyes** and a **mouth filled with tooth like structures** made of dentine a component found in the vertebrate skeleton.
- These structures may represent the earliest occurrence of bone in vertebrates.
- Other hypotheses on the origin of bone suggest that it may have arisen as **denticles in the skin** (similar to those of sharks), in association with certain sensory receptors,
- or as structures for **mineral** (especially calcium phosphate) storage.
- Regardless of its origin, bone was well developed by **500 million years ago**.
- It was present in the bony armor of a group of fishes called **ostracoderms**.
- Ostracoderms were relatively inactive filter feeders that lived on the bottom of prehistoric lakes and seas

- They possessed **neither jaws nor paired appendages**
- However, the evolution of fishes resulted in **both jaws** and **paired appendages** as well as many other structures.
- The results of this adaptive radiation are described in this chapter.
- Did ancestral fishes live in freshwater or in the sea?
- The answer to this question is not simple.
- The **first vertebrates were probably marine**, because ancient stocks of other deuterostome phyla were all marine.
- Vertebrates, however, adapted to freshwater very early, and much of the evolution of fishes occurred there.

- Apparently, early vertebrate evolution involved the movement of fishes back and forth between marine and freshwater environments.
- The majority of the evolutionary history of some fishes took place in ancient seas, and most of the evolutionary history of others occurred in freshwater.
- The importance of freshwater in the evolution of fishes is evidenced by the fact that over **41% of all fish species are found in freshwater**.
- Even though freshwater habitats represent only a small percentage **(0.0093% by volume)** of the earth's water resources.

Other phylogenetic considerations

Lesson 18

Other phylogenetic consideration

Most cladistic and anatomical evidence indicates that the lungfish lineage gave rise to no other vertebrate taxa.

Two important series of evolutionary events occurred during the evolution of the **Osteichthyes**.

One of these was an evolutionary explosion that began about 150 million years ago and resulted in the vast diversity of **teleosts** living today.

The second series of events involves the evolution of **terrestrialism**.

The presence of functional lungs in modern lungfishes has led to the suggestion that the lungfish lineage may have been ancestral to modern terrestrial vertebrates.

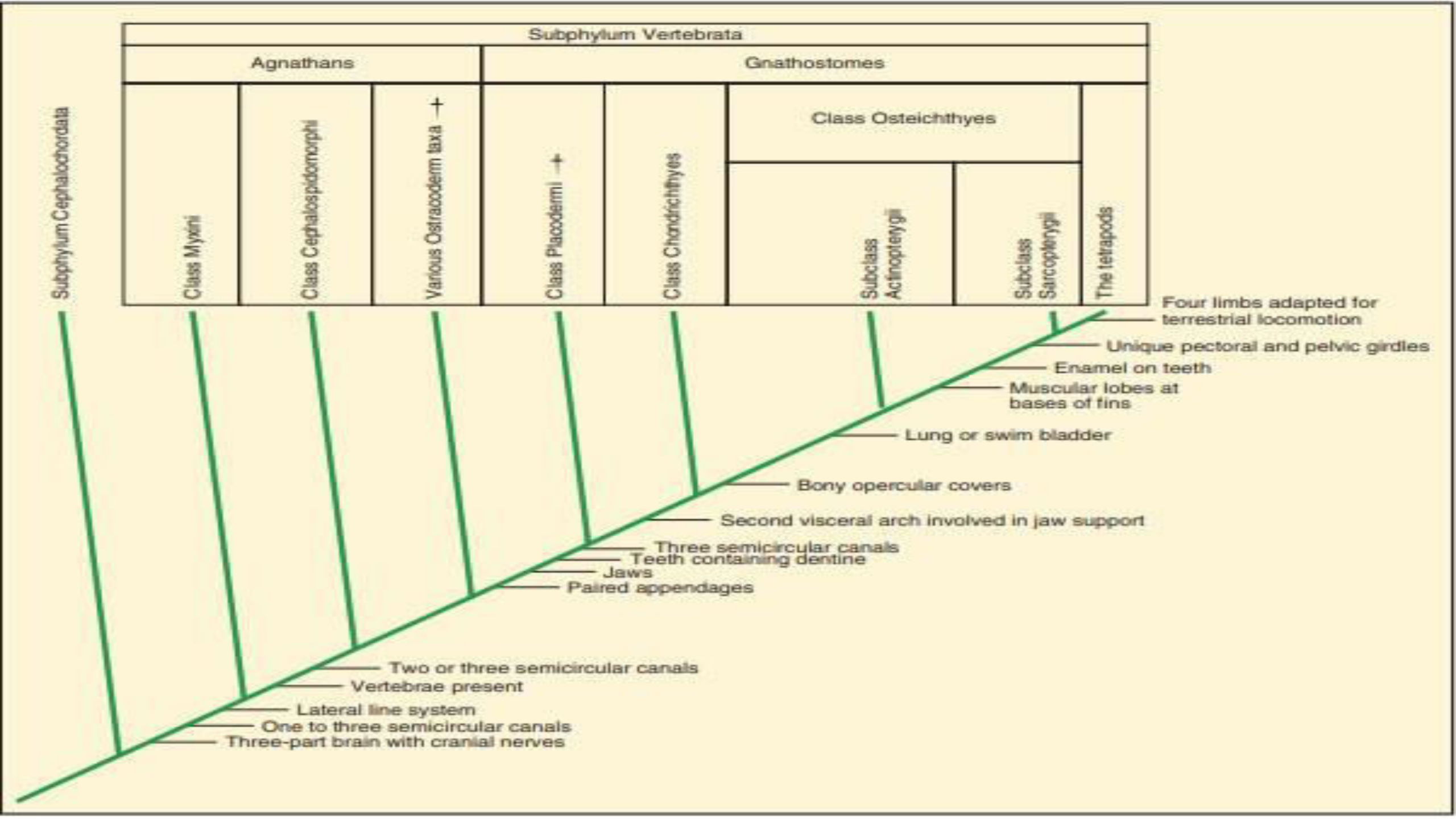
Another group of Scarcopterygii, the osteolepiforms, are now believed to be ancestors of terrestrial vertebrates.

These fish probably occupied relatively shallow-water habitats that would have allowed them to avoid falling prey to larger predatory fishes.

Young osteolepiforms may have inhabited very shallow water.

In the warm, shallow-water environments, depleted oxygen would have favored the evolution of sarcopterygian lungs.

Abundant food, in the form of arthropods and other invertebrates, at the water's edge would have favored shallow water and terrestrial feeding, appendages that allowed the fish to wallow at the land/water interface, and eye lenses that could function in air. While millions of years would elapse before any vertebrate could be considered terrestrial, brief excursions onto land allowed some vertebrates to exploit resources that for the previous 50 million years had only been available to terrestrial arthropods


Subphylum Vertebrata
Agnathans
Gnathostomes
Subphylum Cephalochordata
Class Myxini
Class Cephalaspidomorphi
Various Ostracoderm taxa †
Class Placodermi †
Class Chondrichthyes
Class Osteichthyes
Subclass Actinopterygii
Subclass Sarcopterygii
The tetrapods
Four limbs adapted for terrestrial locomotion
Unique pectoral and pelvic girdles
Enamel on teeth
Muscular lobes at bases of fins
Lung or swim bladder
Bony opercular covers
Second visceral arch involved in jaw support
Three semicircular canals
Teeth containing dentine
Jaws
Paired appendages
Two or three semicircular canals
Vertebrae present
Lateral line system
One to three semicircular canals
Three-part brain with cranial nerves

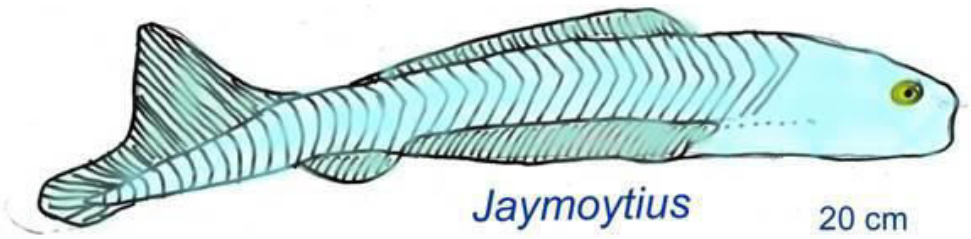
One Interpretation of the Phylogeny of Fishes. The evolutionary relationships among fishes are unsettled. This cladogram shows few selected ancestral and derived characters. Each lower taxon has numerous synapomorphies that are not shown. The position of the lampreys in fish phylogeny is debated. Recent evidence indicates that both lampreys and ostracoderms are more closely related to jawed vertebrates than to hagfishes. Most zoologists consider the ostracoderms a paraphyletic group (having multiple lineages). Their representation as a monophyletic group is an attempt to simplify this presentation. Daggers (†) indicate groups whose members are extinct.

Lesson 19

Classification of Fishes

Oldest known vertebrate fossils

- Tadpole – like (6 inches in length)
- Jawless fish were only vertebrates for more than 50 m.y.a



Vertebrates (Subphylum Vertebrata)

- Backbone enclosing a nerve cord, or spinal cord

Phylum Chordata

- Notochord
- Pharyngeal pouches
- Postanal tail
- Dorsal Nerve Chord

Phylum Chordata

Subphylum Vertebrata

Super Class Agnatha Jawless fishes

Class Myxini (Hagfishes)

Class Cephalaspidomorphi (Lampreys)

Super class Gnathostomes

Class Chondrichthyes, Cartilaginous fishes, (Sharks, Sharks and Rays, Ratfishes)

Class Osteichthyes, Bony fishes

Sub class Actinopterygii (Ray – finned)

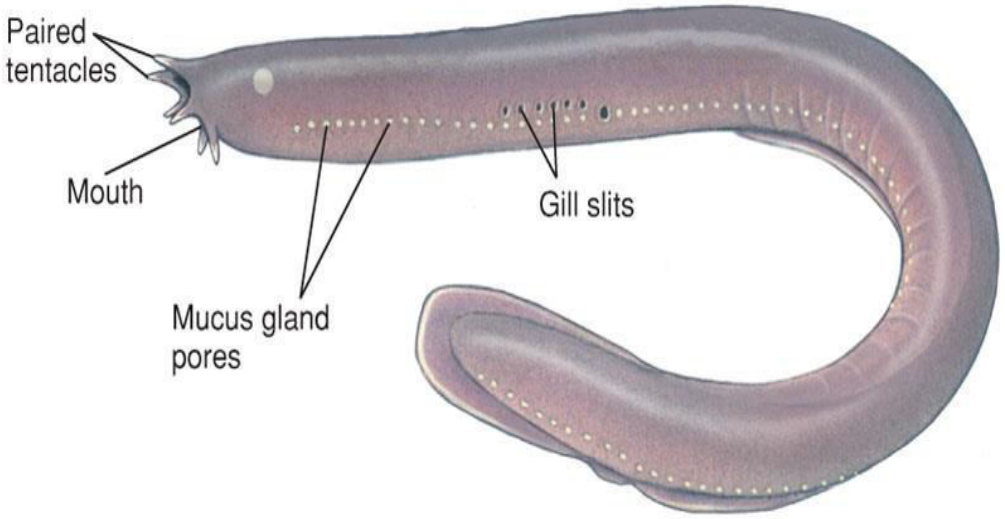
Sub class Sarcophterygii (Lobe – finned)

Class Agnatha: Jawless Fishes

Hagfishes and lampreys

- lack jaws, most primitive of living fishes
- Feed by suction
- Lack paired fins and scales

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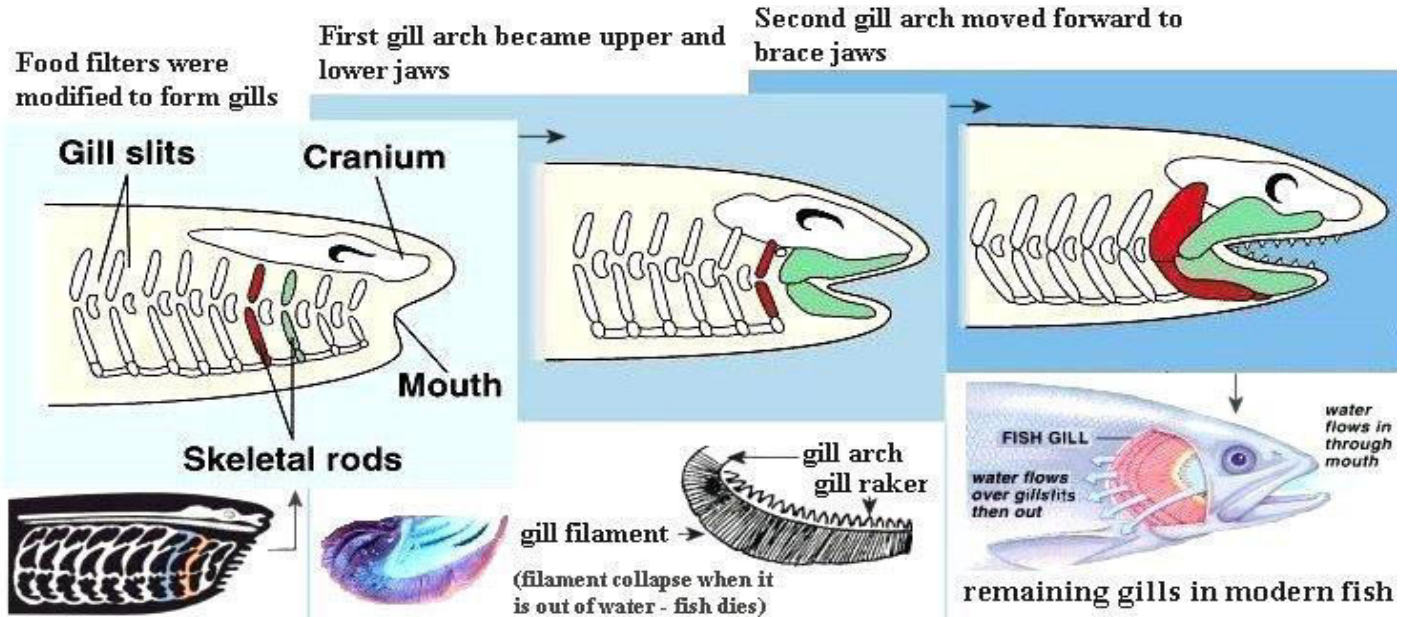
Hagfish (slime eels)

- No vertebrae
- Retains notochord

Lampreys, freshwater

- Primitive Vertebral column composed of cartilage

- Origin of **Jaws and Paired Fins** 450 m.y.a
- Paired fins increased fishes' stability and maneuverability
- Jaw allow to seize prey
- Jaws are thought to have evolved from **1st pair of gill arches** (support pharynx)



Class Chondrichthyes: Cartilaginous

Fishes

(Sharks, rays, skates, and ratfishes)

- Endoskeleton of cartilage
- Paired fins, movable jaws, gill slits
- Rough sandpaper – like skin

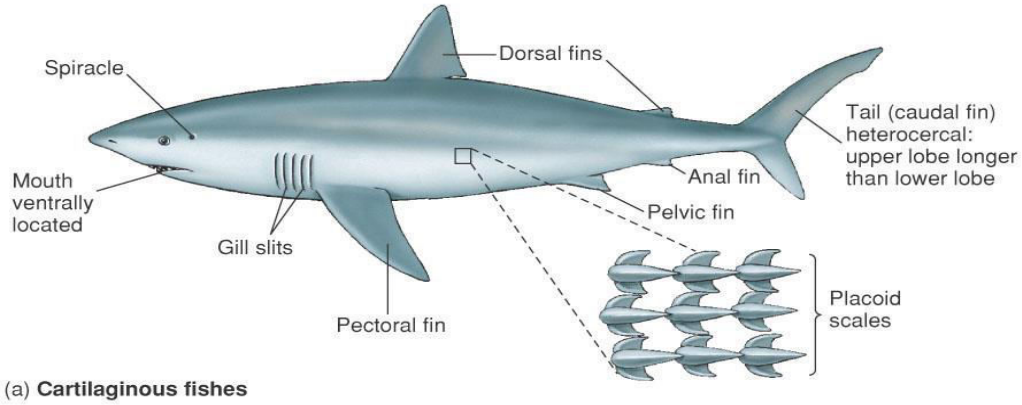
Placoid scales

  - Pointed tip that is directed backward
  - Same composition as teeth

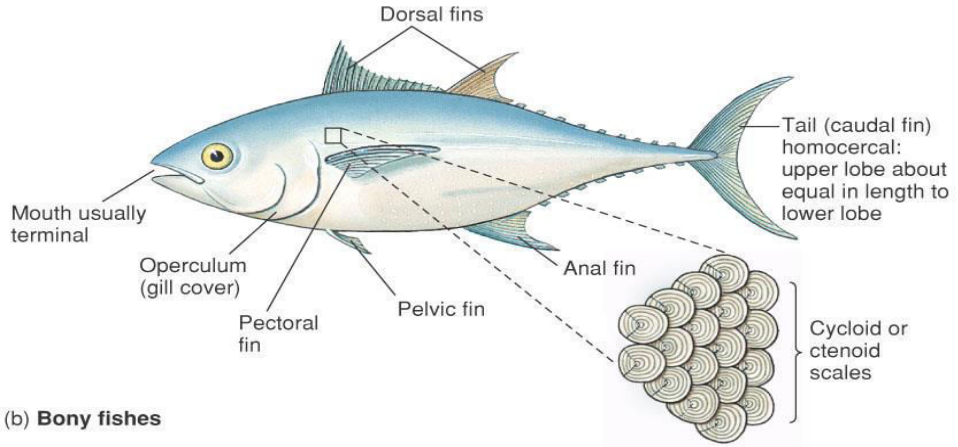
Super Class Osteichthyes: Bony Fish

- Largest group of living vertebrates
- Gills covered by operculum
- Usually swim bladder
- Highly maneuverable fins
- Cycloid scales

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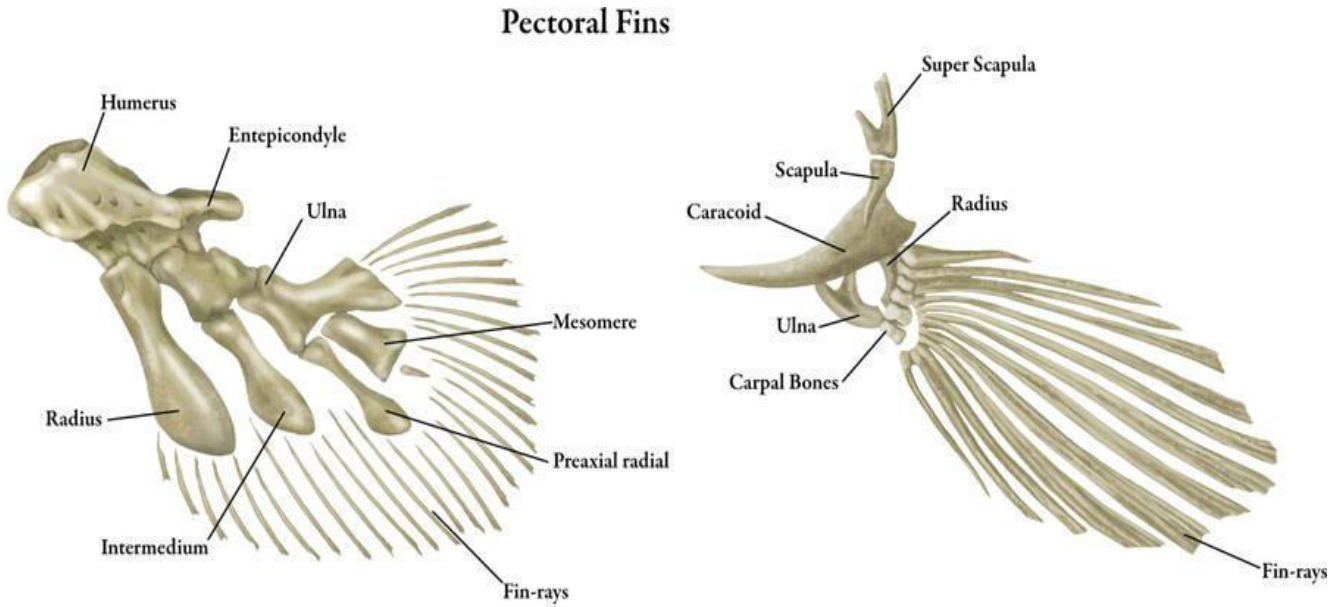
(a) **Cartilaginous fishes**



(b) **Bony fishes**

Superclass Osteichthyes: Bony Fish

- Class Sarcopterygii (lobed – fin finned)
- Class Actinopterygii (ray – finned)



Class Sarcopterygii (lobed – fin finned)

Fins supported by main axis of bone

Lungfishes

- Modified swim bladder
- Breath air

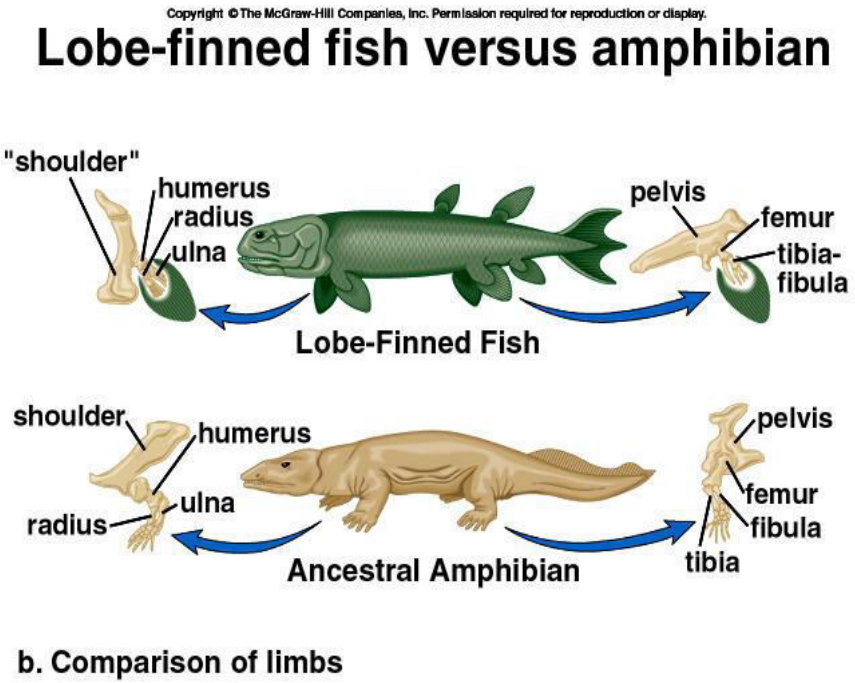


Class Sarcopterygii (lobed – fin finned)

Coelacanth

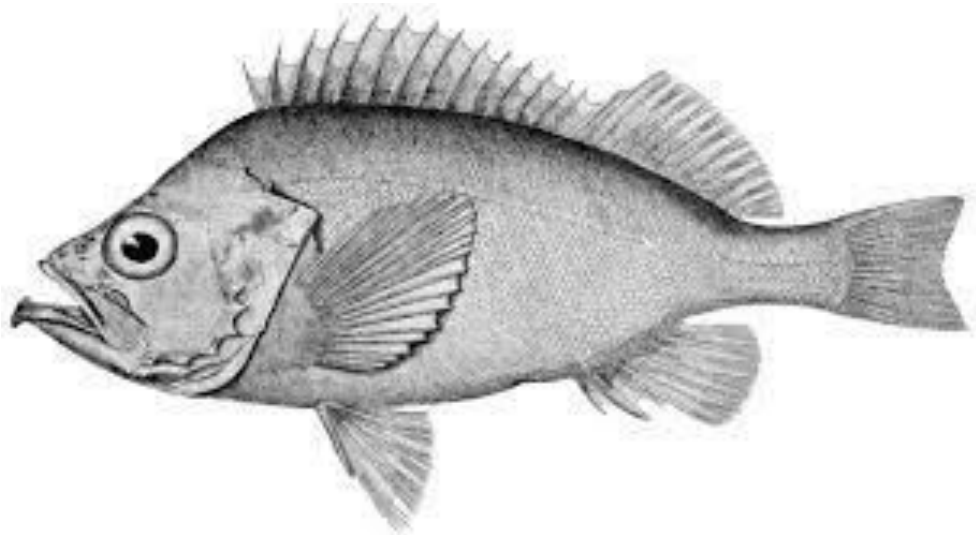
- Were thought to be extinct but were rediscovered in West African waters
- Coelacanth species are thought to be ancestors of amphibians





Class Actinopterygii (ray – finned)

- Most common of boney fish

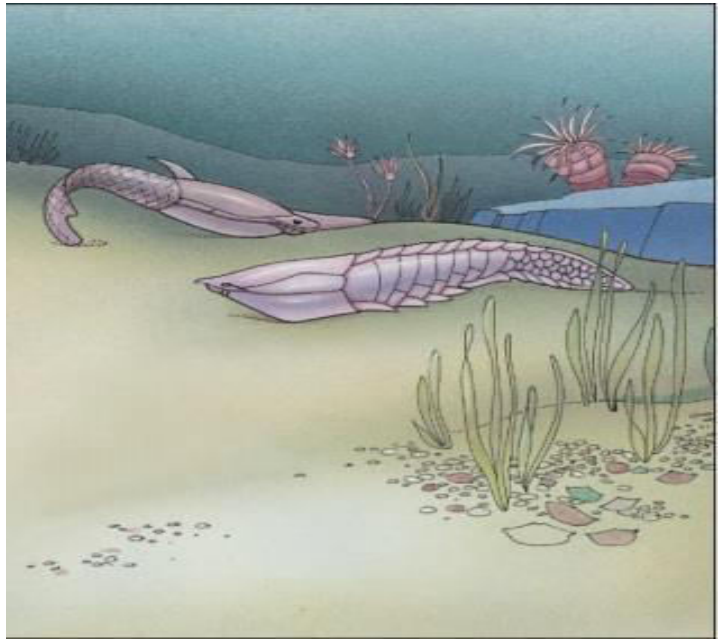


Lesson 20
Agnathans

Agnathans

- **Ostracoderms are extinct agnathans** that belonged to several classes.
- The fossils of predatory water scorpions (phylum Arthropoda) are often found with fossil ostracoderms.
- As sluggish as ostracoderms apparently were, **bony armor** was probably their only defense. Ostracoderms were **bottom dwellers**, often about **15 cm long**.
- Most were probably **filter feeders**, either filtering suspended organic matter from the water or extracting annelids and other animals from muddy sediments.

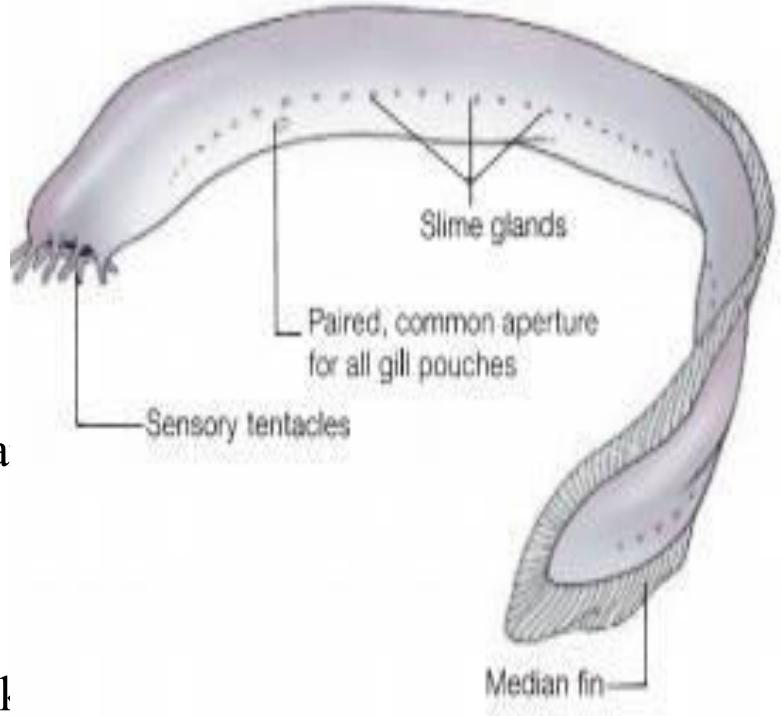
- Some ostracoderms may have used **bony plates around the mouth** in a jaw like fashion to crack gastropod shells or the exoskeletons of arthropods.



Artist's Rendering of an Ancient Silurian Seafloor

Class Myxini

- Hagfishes are members of the class Myxini (miksˇı-ne) (Gr. myxa, slime).
- Hagfishes live buried in the sand and mud of marine environments.
- They feed on soft-bodied invertebrates and scavenge dead and dying fish.
- When hagfishes find a suitable fish, they enter the fish through the mouth a eat the contents of the body, leaving only a sack of skin and bones.
- Anglers must contend with hagfishes because they will bite at a baited hool
- Most zoologists now consider the hagfishes to be the most primitive group of vertebrates.



Hagfish external structure

Lesson 21

Agnathans: Cephalaspidomorphi

Agnathans: Cephalaspidomorphi

Class Cephalaspidomorphi

- **Lampreys** are agnathans in the class Cephalaspidomorphi.
- They are common inhabitants of marine and freshwater environments in temperate regions.
- Most adult lampreys prey on other fishes, and the **larvae are filter feeders**.
- The mouth of an adult is sucker like and surrounded by lips that have sensory and attachment functions.
- Numerous epidermal teeth line the mouth and cover a movable tongue like structure.

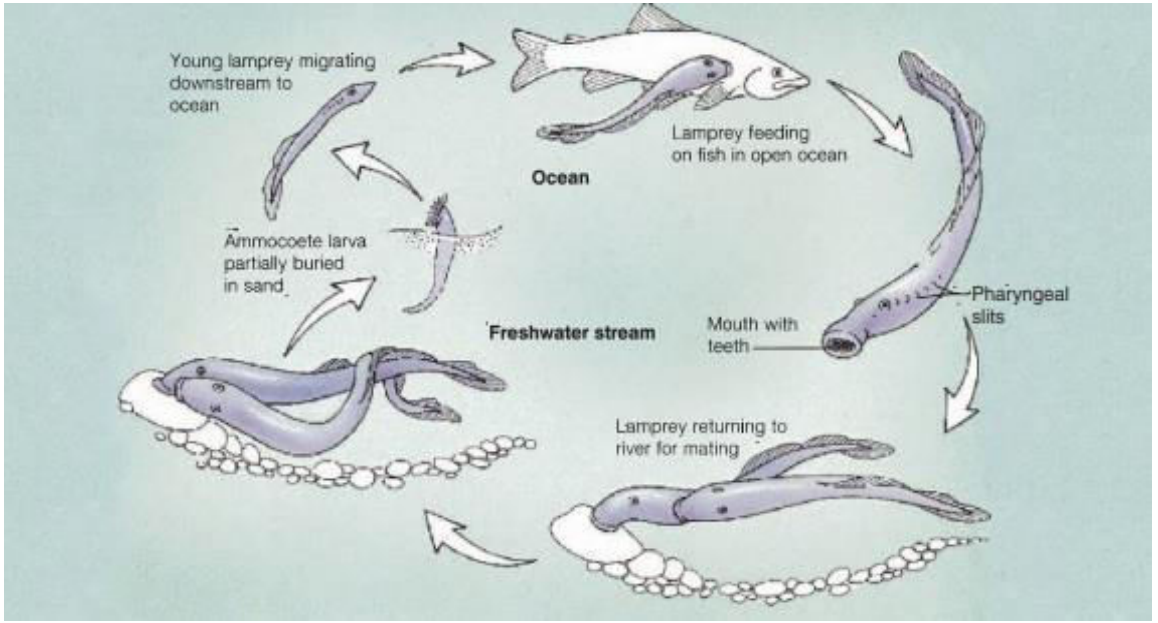
- Adults attach to prey with their lips and teeth and use their tongues to rasp away scales.
- Lampreys have **salivary glands** with anticoagulant secretions and feed mainly on the blood of their prey.
- Some lampreys, however, are not predatory.
- Members of the genus Lampetra are called **brook lampreys**.
- The larval stages of brook lampreys last for about three years, and the adults neither feed nor leave their stream.
- They reproduce soon after metamorphosis and then die.

- Adult sea lampreys live in the ocean or the Great Lakes.
- Near the end of their lives, they migrate, sometimes hundreds of miles, to a spawning bed in a freshwater stream.
- Once lampreys reach their spawning site, usually in relatively shallow water with swift currents, they begin building a nest by making small depressions in the substrate.
- When the nest is prepared, a female usually attaches to a stone with her mouth.
- A male uses his mouth to attach to the female's head and wraps his body around the female.



lamprey (Petromyzon marinus)

- Eggs are shed in small batches over a period of several hours, and fertilization is external.
- The relatively sticky eggs are then covered with sand.



External Structure and Life History of a Seal Lamprey

- Eggs hatch in approximately three weeks into **ammocoete larvae**.
- The larvae drift downstream to softer substrates, where they bury themselves in sand and mud and filter feed in a fashion similar to amphioxus.
- Ammocoete larvae grow from **7 mm to about 17 cm** over three years.
- During later developmental stages, the larvae metamorphose to the adult over a period of several months.

- The **mouth becomes sucker like**, and the teeth, tongue, and feeding musculature develop.
- Lampreys eventually leave the mud permanently and begin a journey to the sea to begin life as predators.
- Adults return only once to the headwaters of their stream to **spawn and die**.

Lesson 22

GNATHOSTOMES

Gnathostomes

- Two major developments in vertebrate evolution were the **appearance of jaws and paired appendages.**
- Jaws are used in feeding and are partly responsible for a transition to more active, predatory lifestyles.
- Pectoral fins of fishes are appendages usually just behind the head, and pelvic fins are usually located ventrally and posteriorly.
- Both sets of paired fins give fishes a more precise steering mechanism and increase fish agility.

- Two classes of gnathostomes still have living members: the **cartilaginous fishes** (class Chondrichthyes) and the **bony fishes** (class Osteichthyes).
- Another class, the armored fishes, or **placoderms**, contained the earliest jawed fishes.
- They are now extinct and apparently left no descendants.
- A fourth group of ancient, extinct fishes, the **acanthodians**, may be more closely related to the bony fishes.

Class Chondrichthyes

Members of the class Chondrichthyes include the **sharks, skates, rays, and ratfishes**.

Most chondrichthians are carnivores or scavengers, and most are marine.

In addition to their biting mouthparts and paired appendages, chondrichthians possess epidermal placoid scales and a cartilaginous endoskeleton.

The subclass **Elasmobranchii** which includes the sharks, skates, and rays, has about seven hundred species



gray reef shark (Carcharhinus)



manta ray (Manta hamiltoni)



bullseye stingray (Urolophus concentricus)

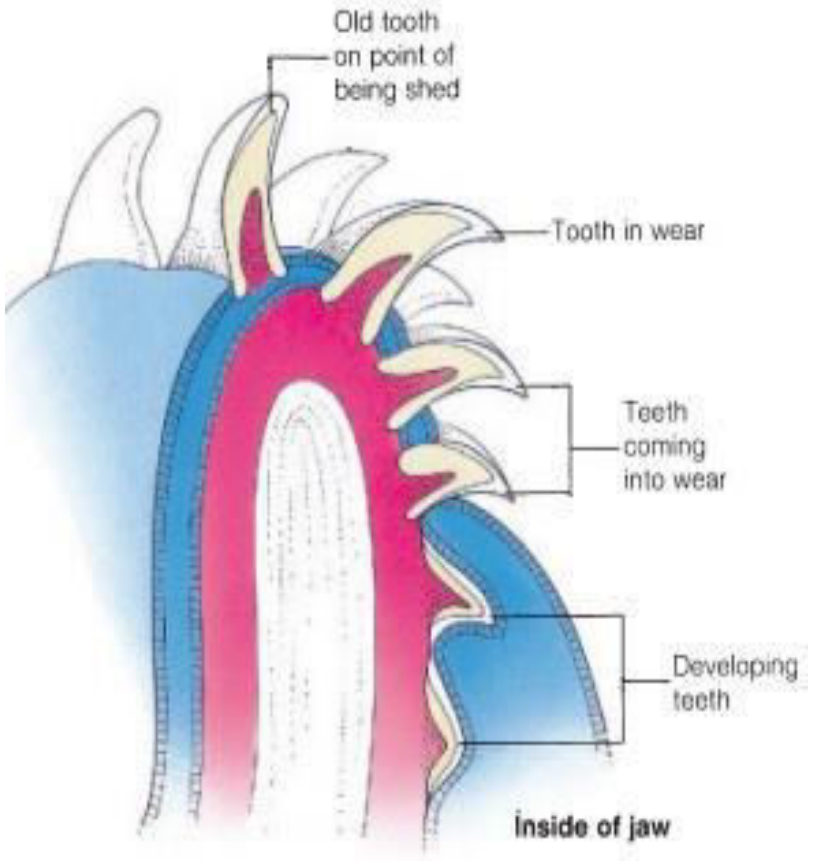
- Sharks arose from early jawed fishes midway through the Devonian period, about 375 million years ago.
- The absence of certain features characteristic of bony fishes (e.g., a swim bladder to regulate buoyancy, a gill cover, and a bony skeleton) is sometimes interpreted as evidence of the primitiveness of elasmobranchs.

- This interpretation is mistaken, as these characteristics simply resulted from different adaptations in the two groups to similar selection pressures.
- Tough skin with dermal, placoid scales covers sharks.
- These scales project posteriorly and give the skin a tough, sandpaper texture.
- In fact, dried shark skin has been used for sandpaper.
- Posteriorly pointed scales also reduce friction with the water as a shark swims



Section of shark skin

- Shark teeth are actually modified placoid scales.
- The row of teeth on the outer edge of the jaw is backed up by rows of teeth attached to a ligamentous band that covers the jaw cartilage inside the mouth.
- As the outer teeth wear and become useless, newer teeth move into position from inside the jaw and replace them.
- In young sharks, this replacement is rapid, with a new row of teeth developing every seven or eight days.



The teeth of sharks

- Crowns of teeth in different species may be adapted for shearing prey or for crushing the shells of molluscs.
- Sharks range in size from less than 1 m (e.g., Squalus, the laboratory dissection specimen) to greater than 10 m (e.g., basking sharks and whale sharks).
- The largest sharks are not predatory but are filter feeders.
- They have pharyngeal-arch modifications that strain plankton.
- The fiercest and most feared sharks are the **great white shark (Carcharodon)** and the **mako (Isurus).**
- Extinct specimens may have reached lengths of 15 m or more.

- Skates and rays are specialized for life on the ocean floor.
- They usually inhabit shallow water, where they use their blunt teeth to feed on invertebrates.
- Their most obvious modification for life on the ocean floor is a lateral expansion of the pectoral fins into wing like appendages.
- Locomotion results from dorsoventral muscular waves that pass posteriorly along the fins.

- Frequently, elaborate color patterns on the dorsal surface of these animals provide effective camouflage.
- The **sting ray (Dasyatis)** has a tail modified into a defensive lash; the dorsal fin persists as a venomous spine.
- Also included in this group are the **electric rays (Narcine and Torpedo)** and **manta rays (Manta).**
- A second major group of chondrichthians, in the subclass **Holocephali** contains about 30 species.

- A frequently studied example, **Chimaera**, has a large head with a small mouth surrounded by large lips.
- A narrow, tapering tail has resulted in the common name "**ratfish**."
- Holocephalans diverged from other chondrichthians nearly 300 million years ago.
- Since that time, specializations not found in other elasmobranchs have evolved, including a gill cover, called an operculum.
- Teeth modified into large plates for crushing the shells of molluscs.
- Holocephalans lack scales.

Lesson 23

Class Osteichthyes

- Members of the class Osteichthyes are characterized by having at least some **bone in their skeleton and/or scales, bony operculum covering the gill openings, and lungs or a swim bladder**.
- Any group that has at least 20,000 species and is a major life-form in most of the earth's vast aquatic habitats must be judged very successful from an evolutionary perspective.

- The first fossils of bony fishes are from **late Silurian** deposits (approximately 405 million years old).
- By the **Devonian period** (350 million years ago), the two subclasses were in the midst of their adaptive radiations.
- Members of the subclass **Sarcopterygii** have muscular lobes associated with their fins and usually use lungs in gas exchange.
- One group of sarcopterygians are the **lungfishes**.

- They can survive when rivers or lakes are dry by burrowing into the mud.
- They keep a narrow air pathway open by bubbling air to the surface.
- After the substrate dries, the only evidence of a lungfish burrow is a small opening in the earth.
- **Lungfishes** may remain in aestivation for six months or more.
- (Aestivation is a dormant state that helps an animal withstand hot, dry periods.) When rain again fills the lake or riverbed, lungfishes emerge from their burrows to feed and reproduce.

- A second group of sarcopterygians are the **coelacanths**.
- The most recent coelacanth fossils are over 70 million years old.
- In 1938, however, people fishing in deep water off the coast of South Africa brought up one fish that was identified as a coelacanth.
- Since then, numerous other specimens have been caught in deep water around the Comoro Islands off Madagascar.
- The discovery of this fish, ***Latimeria chalumnae***, was a milestone event because Latimeria is probably the closest living fish relative of terrestrial vertebrates.

- It is large—up to 80 kg—and has heavy scales.
- Ancient coelacanths lived in freshwater lakes and rivers
- Thus, the ancestors of **Latimeria** must have moved from freshwater habitats to the deep sea.
- A third group of sarcopterygians, called **osteolepiforms**, became extinct before the close of the Paleozoic period.
- They are believed to have been the ancestors of ancient amphibians.
- The subclass Actinopterygii contains fishes that are sometimes called the **ray finned fishes** because their fins lack muscular lobes.

- They usually possess swim bladders, gas-filled sacs along the dorsal wall of the body cavity that regulate buoyancy.
- Zoologists now realize that there have been many points of divergence in the evolution of the Actinopterygii.
- One modern classification system divides the Actinopterygii into two infraclasses.
- One group of actinopterygians, the chondrosteans, contains many species that lived during the Permian, Triassic, and Jurassic periods (215 to 120 million years ago), but only 25 species remain today.

Lesson 24
Chondrosteans

- **Ancestral chondrosteans** had a bony skeleton, but living members, the sturgeons and paddlefishes, have cartilaginous skeletons.
- Chondrosteans also have a tail with a large upper lobe.
- Most **sturgeons** live in the sea and migrate into rivers to breed.
- (Some sturgeons live in freshwater but maintain the migratory habits of their marine relatives.)
- They are large (up to 1,000 kg), and bony plates cover the anterior portion of the body.
- **Heavy scales** cover the tail.
- The sturgeon mouth is small, and **jaws are weak**.

- Sturgeons feed on invertebrates that they stir up from the sea or riverbed using their snouts.
- Because sturgeons are valued for their caviar (eggs), they have been severely overfished.



Shovelnose sturgeons (Scaphirhynchus platorynchus)

- **Paddlefishes** are large, freshwater chondrosteans.
- They have a large, **paddle like rostrum** that is innervated with sensory organs believed to detect weak electrical fields.
- They swim through the water with their large mouths open, filtering crustaceans and small fishes.
- They are found mainly in lakes and large rivers of the **Mississippi River** basin and are also known from western North America and China



Paddlefish (Polydon spathula)

- The second group of actinopterygians (**Neopterygii**) flourished in the Jurassic period and succeeded most chondrosteans.
- Two very primitive genera occur in temperate to warm freshwaters of North America. **Lepisosteus**, the garpike, has thick scales and long jaws that it uses to catch fishes.
- **Amiais** commonly referred to as the dogfish or bowfin.
- Most living fishes are members of this group and are referred to as **teleosts** or modern bony fishes.
- They have a **symmetrical caudal fin** and a **swim bladder** that has lost its connection to the digestive tract.

- After their divergence from ancient marine actinopterygians in the late Triassic period, teleosts experienced a remarkable evolutionary diversification and adapted to nearly every available aquatic habitat.
- The number of teleost species exceeds 20,000.



Bottom fish
(*Pseudopleuronectes americanus*)



Yellowtail snapper
(Ocyurus chrysurus)



Sarcastic fringehead
(Neoclinus blanchardi)

Lesson 25
Locomotion, Nutrition and the Digestive system

Locomotion

- Fins and body wall to push against the incompressible surrounding water.

  Most fishes are arranged in a **zigzag pattern**.

- Muscles extend posteriorly and anteriorly in a zigzag fashion

- **Tuna** and **mackerel**, supplement body movements with a **vertical caudal (tail) fin**

- The **forked shape of the caudal fin** reduces surface area that could cause turbulence and interfere with forward movement.

Nutrition and Digestive System

- The earliest fishes were probably **filter feeders** and **scavengers**
- Fish nutrition dramatically changed with the **evolution of jaws**
- Most modern fishes are **predators**
- Some fishes feed on **invertebrate animals**
- Kinds of food that one fish eats at different times in its life varies.
- Fishes usually **swallow prey whole**.

- **Teeth** capture and hold prey
- Teeth are for crushing the shells of molluscs or the exoskeletons of arthropods.
- Fishes often use the **suction** which develops a negative pressure
- **Herring, paddlefishes, and whale sharks** are filter feeders
- **Gill rakers**, trap plankton while the fish is swimming through the water
- **Carp** feed on a variety of plants and small animals.

- Lamprey, are external parasites for at least a portion of their lives.
- A few are primarily **herbivores**
- The fish **digestive tract** is similar to that of other vertebrates.
- **Stomach** stores large, often infrequent, meals.
- Primary site for enzyme secretion
- **Sharks** and other **elasmobranchs** have a **spiral valve**
- **Pyloric ceca** increase absorptive and secretory surfaces.

Lesson 26
Circulation

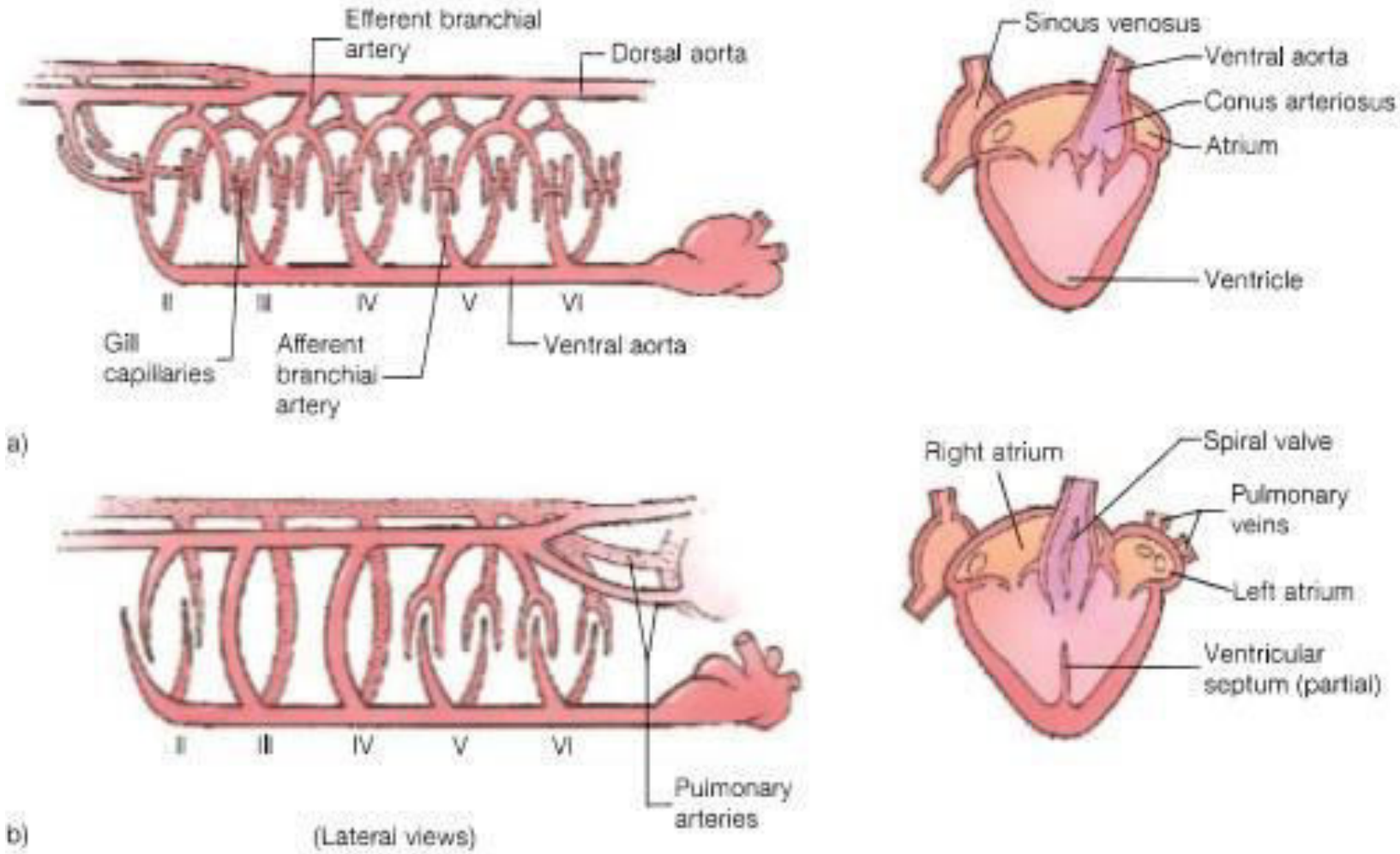
Circulation

- Closed circulatory system
- Heart pumps blood
- Red blood cells containing **hemoglobin**
- The evolution of lungs are associated with;

i. the loss of gills

ii. delivery of blood to the lungs

iii. separation of oxygenated and unoxygenated blood



Diagrammatic representation of the circulatory systems of (a) bony fishes and (b) lungfishes.

- Heart develops from four embryological enlargements of a ventral aorta.
- Blood flows from the venous system through the sinus venosus, the atrium, the ventricle, the conus arteriosus, and into the ventral aorta
- **Five afferent vessels** carry blood to the gills
- Blood is collected by efferent vessels, delivered to the dorsal aorta, and distributed to the body.
- Lungfishes provide a good example of how lungs alter the circulatory pattern.
- A vessel to the lungs has developed as a branch off **aortic arch VI**

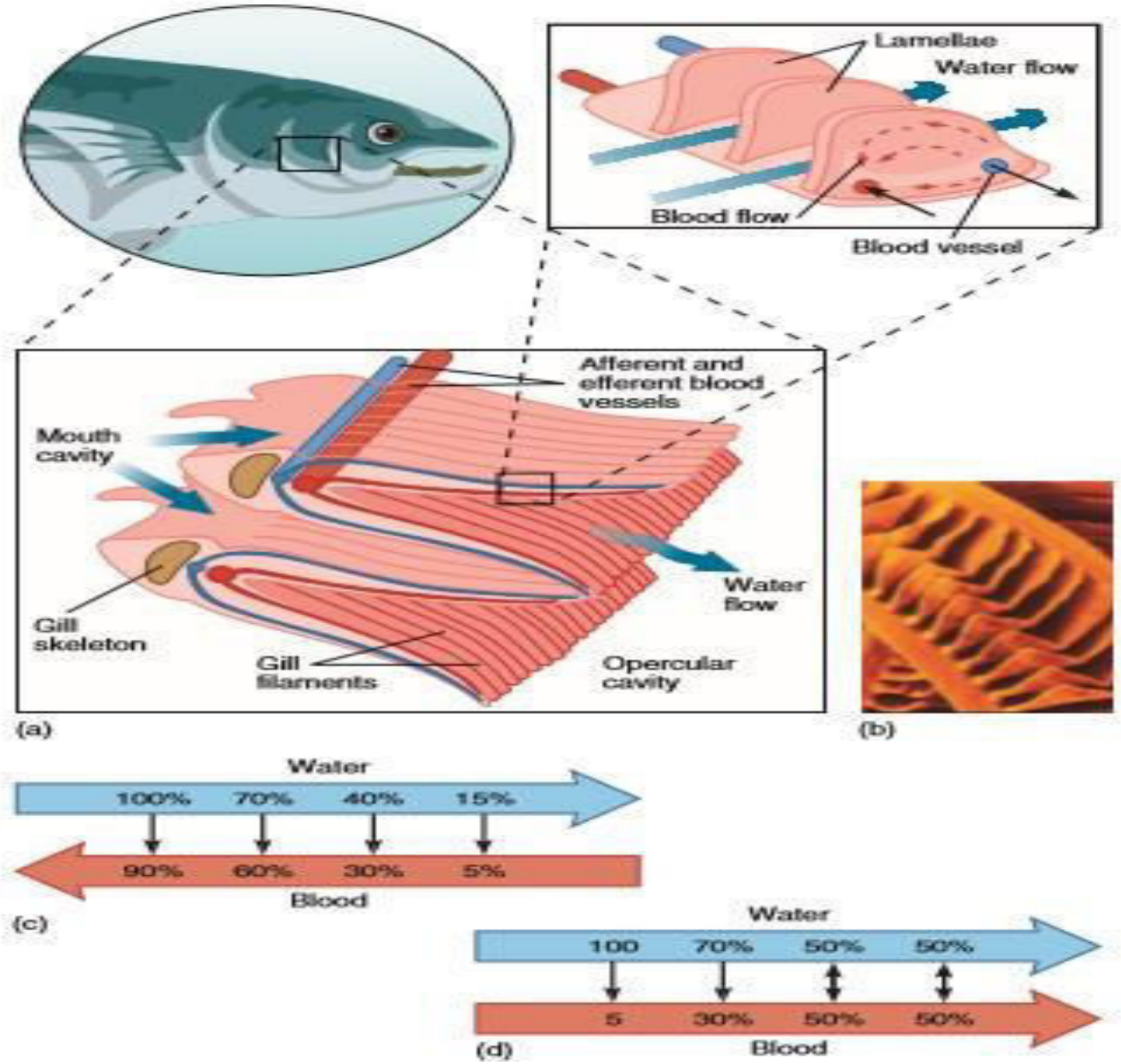
- This vessel is called the **pulmonary artery**.
- Blood from the lungs returns to the heart through pulmonary veins and enters the left side of the heart.
- The atrium and ventricle are **partially divided**.
- Partial divisions help keep unoxygenated blood from oxygenated blood
- A **spiral valve** in the conus arteriosus help pump blood
- Lungfishes show a distinction between a **pulmonary circuit and a systemic circuit**

Lesson 27
Gas Exchange

- Less than 2.5% of the oxygen present in air.
- Fishes pass water across gill surfaces and extract the small amount of oxygen present in the water.
- Muscular pumping mechanism
- Muscles surrounding the pharynx and the opercular cavity, power this pump.
- Maintain water flow by holding their mouths open
- This method is called ram ventilation
- Elasmobranchs do not have opercula to help pump water

- Others move water over their gills with a pumping mechanism
- Gill bars with external flaps that close and form a cavity Spiracles are modified pharyngeal slits that open just behind the eyes of elasmobranchs
- Gas exchange across gill surfaces is very efficient.
- Gill (visceral) arches support gills.
- Gill filaments extend from each gill arch called pharyngeal lamellae

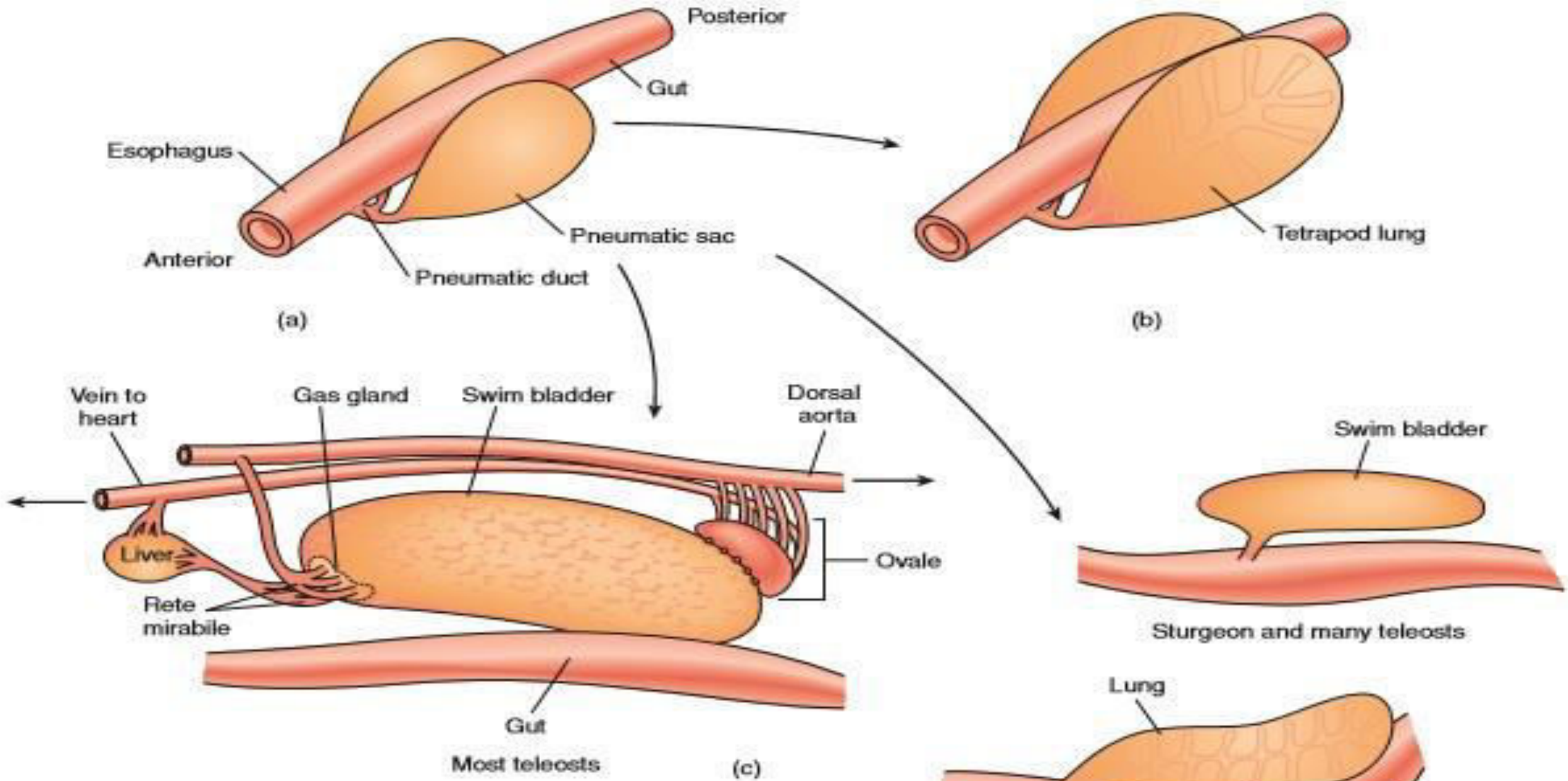
- Branchial arteries carry blood to the gills and into gill filaments.
- The arteries break into capillary beds in pharyngeal lamellae.
- Gas exchange occurs as blood and water move in opposite directions
- This countercurrent exchange mechanism provides very efficient gas exchange



Gas Exchange at the Pharyngeal Lamellae

Lesson 28
Swim Bladders and Lungs
Buoyancy Regulation

- Lungs are more primitive
- Much of the early evolution of bony fishes occurred **Devonian period.**
- These bodies of water became dried.
- Lungs in these habitats meant the difference between life and death.
- Later evolution of modern bony fishes resulted in the use of **pneumatic sacs** in buoyancy regulation



(Possible Sequence in the Evolution of Pneumatic Sacs)

Buoyancy Regulation

- Maintain their vertical position in a column of water
- Low-density compounds into their tissues.
- Saturated with buoyant oils.
- Maintain vertical position to provide lift.
- Pectoral fins help to create lift
- Large upper lobe of a shark’s caudal fin provides upward thrust

- A third adaptation is the **reduction of heavy tissues** in fishes.
- Bones of fishes are generally less dense
- Elasmobranch cartilaginous skeleton probably results from cartilage

The fourth adaptation is the **swim bladder**.

- Precisely controlling the volume of gas in its swim bladder.
- The pneumatic duct connects the swim bladders of garpike,

sturgeons

- Fishes gulp air at the surface to force air into their swim bladders.

- Teleosts have **swim bladders**
- The blood secretes gases into the swim bladder in a vascular network called the **rete mirabile** (“miraculous net”).
- Gases reabsorb into the blood at the posterior end of the bladder, the **ovale**

Lesson 29
NERVOUS AND SENSORY FUNCTIONS

- The central nervous system consists of a **brain and a spinal cord**.
- **Sensory receptors** are widely distributed over the body.
- Also possess specialized receptors for olfaction, vision, hearing, equilibrium, balance, and for detecting water movements.
- **External nares** lead to olfactory receptors.
- Mostly receptors are in **blind-ending olfactory sacs**.
- In a few fishes, the **external nares open to nasal passages**
- Some fishes rely heavily on their sense of smell e.g. **salmon** and **lampreys**

- The **eyes** of fishes are similar
- They **are lidless**, **lenses are round**.
- Move the lens forward or backward in the eye.
- Receptors for equilibrium, balance, and hearing are in the inner ears
- **Semicircular canals** detect rotational movements
- Other sensory patches help with equilibrium and balance.
- Vibrations may pass from the water through the bones of the skull to the middle ear

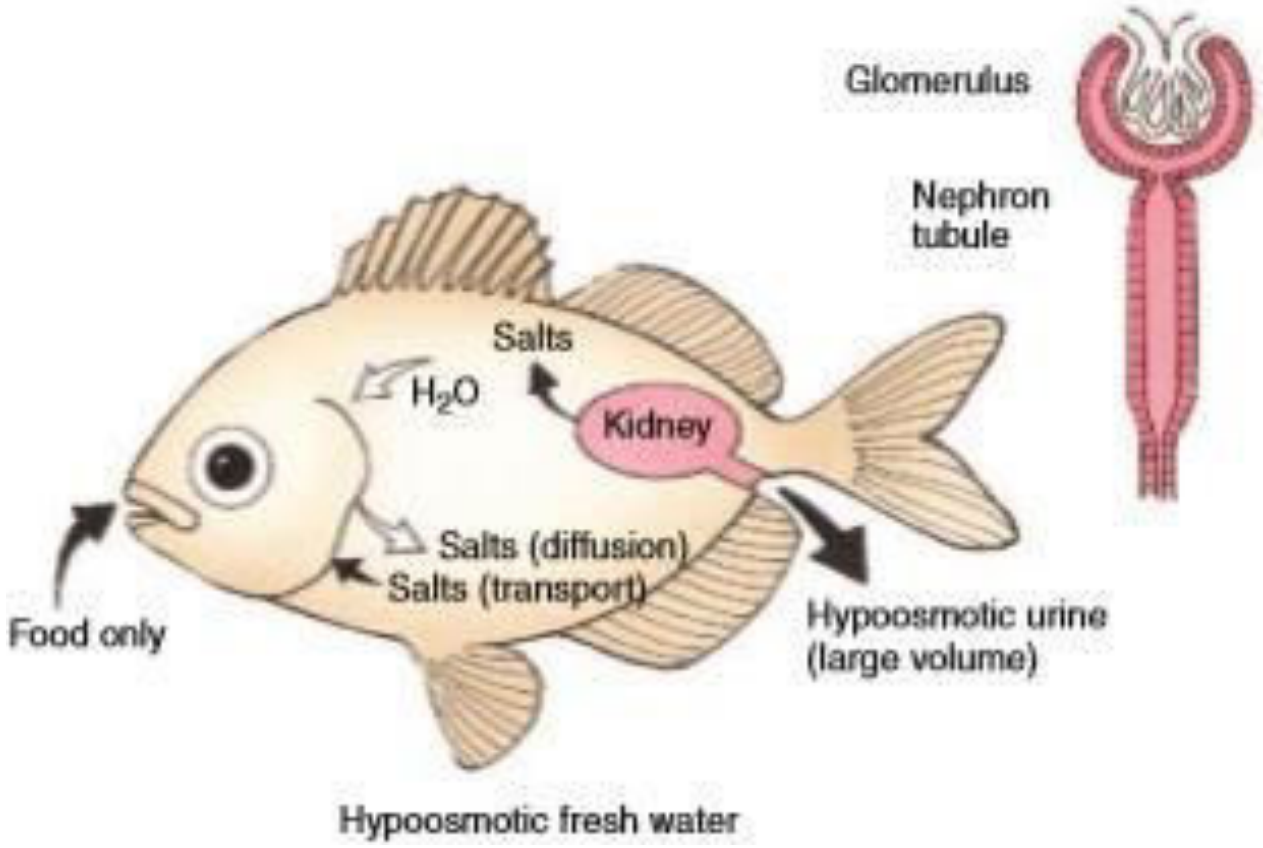
- Fishes lack the outer and/or middle ear.
- Few fishes have chains of bony ossicles
- Vibrations are amplified by the swim bladder
- fishes has a lateral-line system.
- The lateral-line system consists of sensory pits in the epidermis In these pits are receptors that are stimulated by water
- Lateral lines are used to detect either water currents, or apredator
- Fishes may also detect low-frequency sounds with these receptors.

Lesson 30
EXCRETION

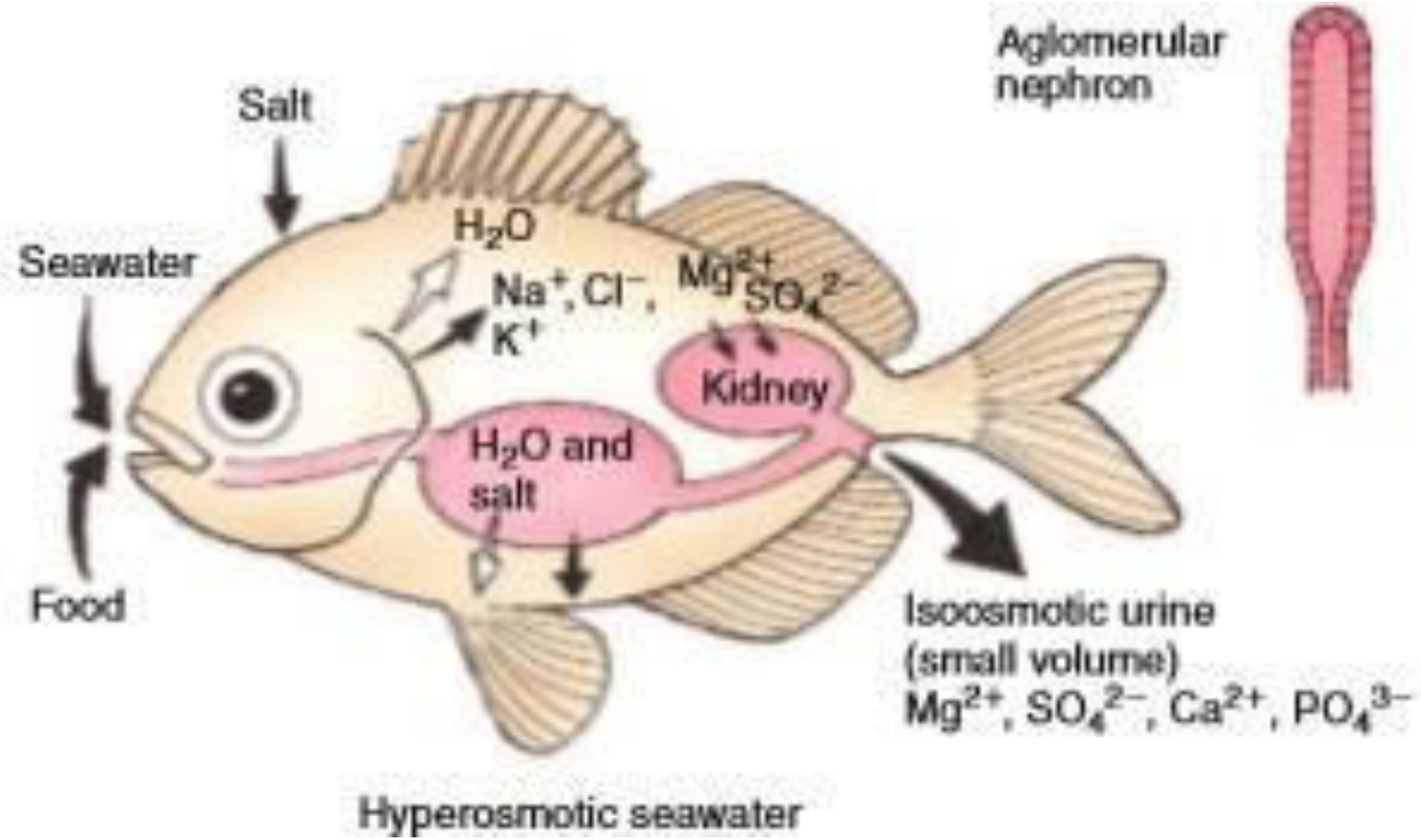
- Fishes must maintain a proper balance of electrolytes (ions) and water.
- **Osmoregulation** is a major function of the kidneys and gills Kidneys are located near the midline of the body
- Excretory structures in the kidneys are called **nephrons**.
- Nephrons filter blood borne nitrogenous wastes across a glomerulus.
- The filtrate then reabsorbed into the blood while passing through tubules.
- The filtrate remaining in the tubule system is then excreted.

- Freshwater fishes live in an environment containing few dissolved substances.
- Osmotic uptake of water across gill, oral, and intestinal surfaces
- To control excess water buildup and ion loss
- Numerous nephrons of freshwater fishes frequently possess large glomeruli and relatively short tubule systems
- Reabsorption of some ions and organic compounds follows filtration.
- Little water is reabsorbed.
- Freshwater fishes produce large quantities of very dilute urine. Ions are still lost through the urine and by diffusion

- Active transport of ions into the blood at the gills compensates for this ion loss.
- Freshwater fishes also get some salts in their food



- Marine fishes face the opposite problems.
- Their environment contains 3.5% ions and tissues contain 0.65%
- Marine fishes, therefore, must combat water loss and accumulation of excess ions.
- They drink water and eliminate excess ions by excretion
- The nephrons of marine fishes frequently possess small glomeruli and long tubule systems.
- Much less blood is filtered than in freshwater fishes


Aglomerular nephron
Salt
Seawater
H_2O
Na^+, Cl^-, K^+
Mg^{2+} SO_4^{2-}
Kidney
H_2O and salt
Food
Isoosmotic urine (small volume)
Mg^{2+}, SO_4^{2-}, Ca^{2+}, PO_4^{3-}
Hyperosmotic seawater

Lesson 31
Osmoregulation

- Elasmobranchs have a unique **osmoregulatory mechanism**.
- Convert nitrogenous wastes into urea
- Urea is sequestered in tissues all over the body.
- Enough urea is stored to make body tissues isosmotic with seawater
- Most marine fishes face problem of losing water to their environment
- Energy that does not have to be devoted to water conservation This adaptation required the development of tolerance to high levels of urea

- Elasmobranchs must still regulate the ion concentrations in their tissues.
- Possess a rectal gland that removes excess sodium chloride from the blood and excretes it into the cloaca.
- Diadromous fishes migrate between freshwater and marine environments.
- Salmon (e.g., Oncorhynchus) and marine lampreys (Petromyzon)migrate from the sea to freshwater
- Freshwater eel (Anguilla)migrates from freshwater to marine environments.

- Diadromous migrations require gills capable of coping with both uptake and secretion of ions
- Osmoregulatory powers may not be developed in all life-history stages.
- Fishes have few problems getting rid of the nitrogenous byproducts of protein metabolism.
- Up to 90% of nitrogenous wastes are eliminated as **ammonia**
- The remaining 10% of nitrogenous wastes are excreted as **urea, creatine, or creatinine**.
- These wastes are produced in the liver and are excreted via the kidneys.

Lesson 32
Reproduction

- Produe overwhelming numbers of eggs
- Some fishes show mating behavior or nesting behavior
- Mating may occur in large schools
- Huge masses of eggs and sperm released into the open ocean
- The vast majority of fishes are oviparous
- Some elasmobranchs are ovoviviparous, and their embryos develop in a modified oviduct of the female.

- Nutrients are supplied from yolk
- Other elasmobranchs, including gray reef sharks and hammerheads, are viviparous.
- A placenta-like outgrowth diverts nutrients from the female to the yolk sacs
- Internal development of viviparous bony fishes occurs in ovarian follicles
- In guppies (Lebistes), eggs are retained in the ovary.
- Embryos are then released into a cavity

- Some fishes have specialized structures that aid in sperm transfer
- Male elasmobranchs have modified pelvic fins called claspers.
- Male inserts a clasper into the cloaca of a female.
- Sperm travel along grooves of the clasper.
- Fertilization occurs in the female’s reproductive tract
- Fishes with internal fertilization usually produce fewer eggs.

Lesson 33
Development

- Care of the embryos is limited or nonexistent.
- Some fishes construct and tend nests
- Some carry embryos during development.
- Clusters of embryos may be brooded in **special pouches**
- Some of the best-known brooders include the **seahorses (Hippocampus)** and **pipefishes (e.g., Syngnathus).**
- Males of these fishes carry embryos in ventral pouches.
- The male **Brazilian catfish (Loricaria typhys)** broods embryos in an enlarged lower lip.



Male Garibaldi (Hypsypops rubicundus)

Most fishes care little

Sunfishes and sticklebacks provide short-term care of posthatching young.

Male sticklebacks assemble fresh plant material

Male snaps if one offspring wanders too far from the nest

Sunfish males do the same for young that wander from schools

- The **Cichlidae** engage in longer-term care
- In some species, young are mouth brooded, and other species tend young in a nest.
- After hatching, the young venture from the parent's mouth or nest



Dogtooth cichlid (Cynotilapia afra)



Fontosa (Cyphontilapia fontosa)

Lesson 34
Amphibians

Amphibians?

“These foul and loathsome animals are abhorrent because of their cold body, pale color, cartilaginous skeleton, filthy skin, fierce aspect, calculating eye, offensive smell, harsh voice, squalid habitation, and terrible venom; and so their Creator has not exerted his powers to make many of them.”

- Linnaeus, 1758

Amphibians General Characteristics

- Amphibians are cold blooded vertebrates which can hue on land and in water.
- Amphibians show four limbs with which they can swim in water and jump or walk on the land
- In Amphibians animals exoskeleton is absent but in apoda animals small cycloid scales are present.
- In Amphibians the adult animals lungs are present.
- Amphibians skin is a respiratory organ
- Skull is **dicondylic**.

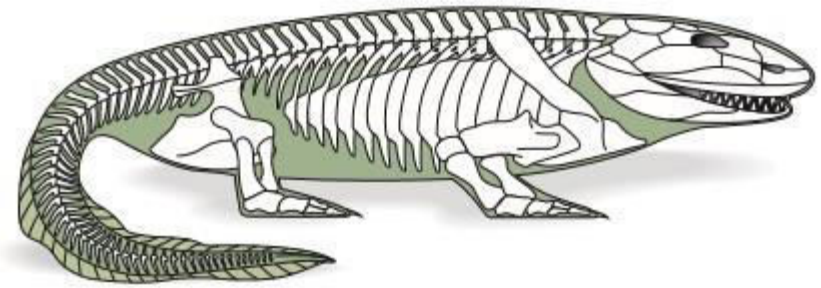
- Ribs are absent.
- Body divisible into head and trunk
- Tail is present in urodela animals.
- Digestive system is well developed.
- External ear is absent. Middle and inner ears are present, the middle ear **columella auris** is present.
- Amphibians Heart is 3 chambered with 2 auricles and 1 ventricle
- The blood contains R.B.C. Blood vascular system contain hepatic and renal portal systems.

- 10 pairs of cranial nerves arise.
- Sexes are separate.
- Male and female can be indentified - **Sexual dimorphism**.
- In Amphibians the life history a larva stage may be present.
- Amphibian Eggs are telolecithal
- Cleavage is holoblastic unequal.

Lesson 35
Phylogenetic relationships

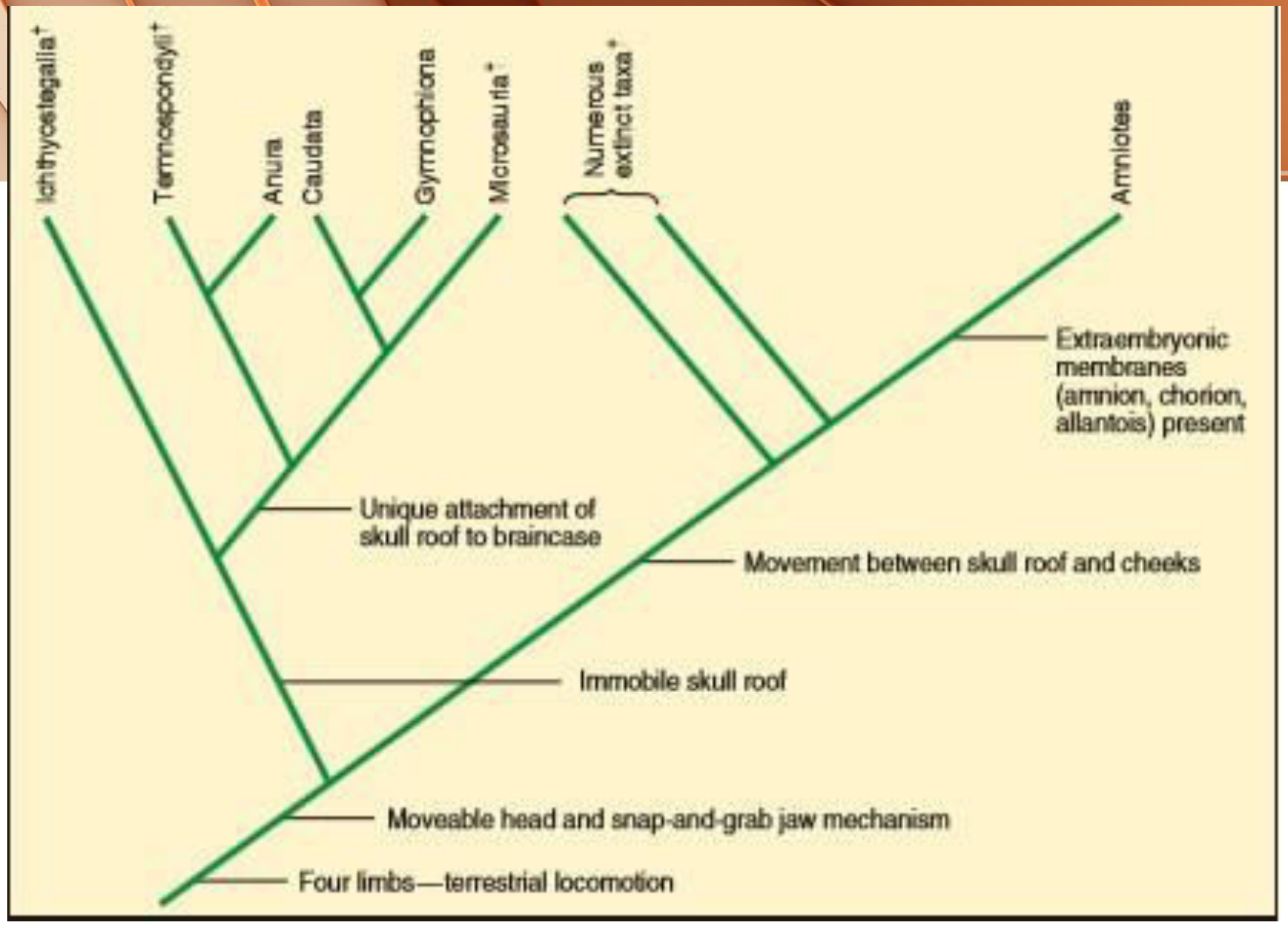
- Adaptive radiation resulted in vertebrates
- Prehistoric waters contained many active, powerful predators. Land was free of vertebrates and free of predators.
- Animals that moved around the water's edge were not likely to be prey for other animals.
- These animals probably found ample food in the arthropods that lived there.
- The major component of the diet is arthropods.

- Adaptive radiation resulted in a much greater variety
- Later convergent and parallel evolution and widespread extinction clouded evolutionary pathways.
- No one knows, therefore, what animal was the first amphibian But the structure of limbs, skulls, and teeth suggests that **Ichthyostega** is probably similar to the earliest amphibians



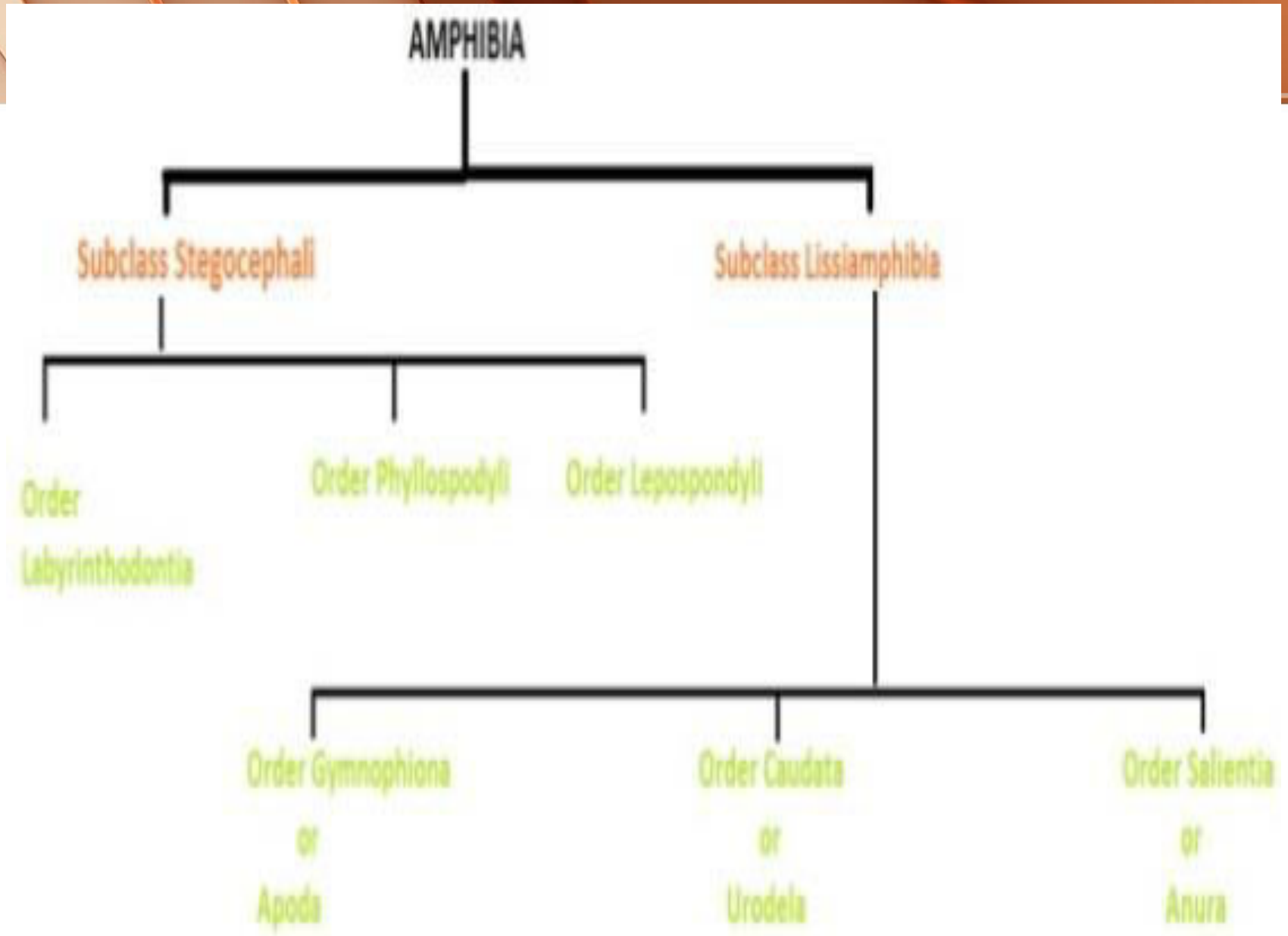
Ichthyostega: An Early Amphibian

- Two lineages of early amphibians can be distinguished by how the roof and the posterior portion of the skull
- One lineage of amphibians became extinct late in the Carboniferous period.
- An amniotic egg that resisted drying evolved in this group.
- This lineage left as its descendants
- A second lineage flourished into the Jurassic period.
- Most of this lineage became extinct
- This lineage is called the nonamniote lineage.



Evolutionary Relationships among the Amphibians

Lesson 36
Classification of Amphibians


AMPHIBIA
Subclass Stegocephali
Subclass Lissiamphibia
Order Labyrinthodontia
Order Phyllospodyli
Order Lepospondyli
Order Gymnophiona or Apoda
Order Caudata or Urodela
Order Salientia or Anura

Sub class Stegocephali

- Extinct species
- Pendatactyl limbs
- Skin with scales and bony plates

Order Labyrinthodontia

- Primitive amphibians
- Also known as “Stem Amphibia”
- Teeth large with folded dentine same as crossoptergian (fishes)
- E.g. Eryops (Extinct)

Order Phyllospondyli

- Small Salamander like
- Notochord and spinal cord housed in common cavity
- E.g, Branchiosaurs

Order Lepospondyli

- Small eel like
- Vertebrea of single piece
- Neural arch and centrum continuous
- Ex. Diplocaulus (Extint)

Subclass Lissiamphibia

- Modern Aphibia
- No scales present
- Teeth small and simple

Order Gymnophiona (Apoda)

- No limbs or limb girdles
- Blind
- Wormlike, legless amphibians with indistinct or even hidden eyes.

- A series of **annular grooves (annuli)** along the length of the body
- Most caecilians lead a burrowing existence, although members of one genus, *Typhlonectes*, are aquatic.



Caecilian (Ichthyophis glutinosus)

- Eyes hidden beneath the bones of the skull
- At larval stages they have teeth, are carnivores and have limbs during most of their development
- Further subdivided into five suborders
- Cryptobranchoidea, Salamandriodea, Ambystomatoidea, Proteidea, Meantes
- E.g. **Ambystoma, Sirencaecilian (Ichthyophis glutinosus)**

Order Anura (Salientia)

- Tail absent in adult
- Eyelids and tympanum present
- Skin scaleless and loose
- Further divided into five suborders
- Amphicoela, Opisthocoela, Procoela, Diplasiocela
- E.g. Frog and Toad



American toad (Bufo americanus)

- Shortened **vertebral column** (nine or fewer presacral vertebrae)
- Presence of a **urostyle** formed from developing tail vertebrae.
- Absence of **tail** in adults.
- Hind limb longer than forelimb.
- Fusion of **radius and ulna** into a single element.
- Fusion of **tibia and fibula** into a single element.

Order caudata

- Members of the order Caudata are the **salamanders**.
- Tail throughout life
- Pairs of legs relatively unspecialized
- Approximately 115 of the 350 described species of salamanders live in North America.
- Most live in **moist forest-floor litter**



Order Caudata

- A number of families live in **caves**
- Salamanders in the family Plethodontidae are the most fully terrestrial
- Members of the family Salamandridae are commonly called **newts**.
- Spend most of their lives in water and frequently retain **caudal fins**.
- From few centimeters to 1.5 m
- The largest North American salamander is the **hellbender (Cryptobranchus alleganiensis)**, about 65 cm.

- Internal fertilization
- A pyramidal, gelatinous spermatophore
- Females pick up the sperm cap with the cloaca and store in **spermatheca**
- Fertilized eggs are deposited singly, in clumps, or in strings



Order Caudata (Eggs)

- Larvae are similar to adults.
- Possess external gills, a tail fin, larval dentition, and a rudimentary tongue
- Metamorphoses into a terrestrial adult
- Many other salamanders undergo incomplete metamorphosis and are paedomorphic (e.g., Necturus)



Order Caudata (Larva)



Order Caudata (Adult)

Lesson 37
EXTERNAL STRUCTURE AND LOCOMOTION

External Structure

- Skin protects against infective microorganisms
- Also functions in gas exchange, temperature regulation, and absorption and storage of water.
- Lacks a covering of scales, feathers, or hair.
- Highly glandular
- Glands keep the skin moist
- Produce sticky secretions that help a male cling to a female

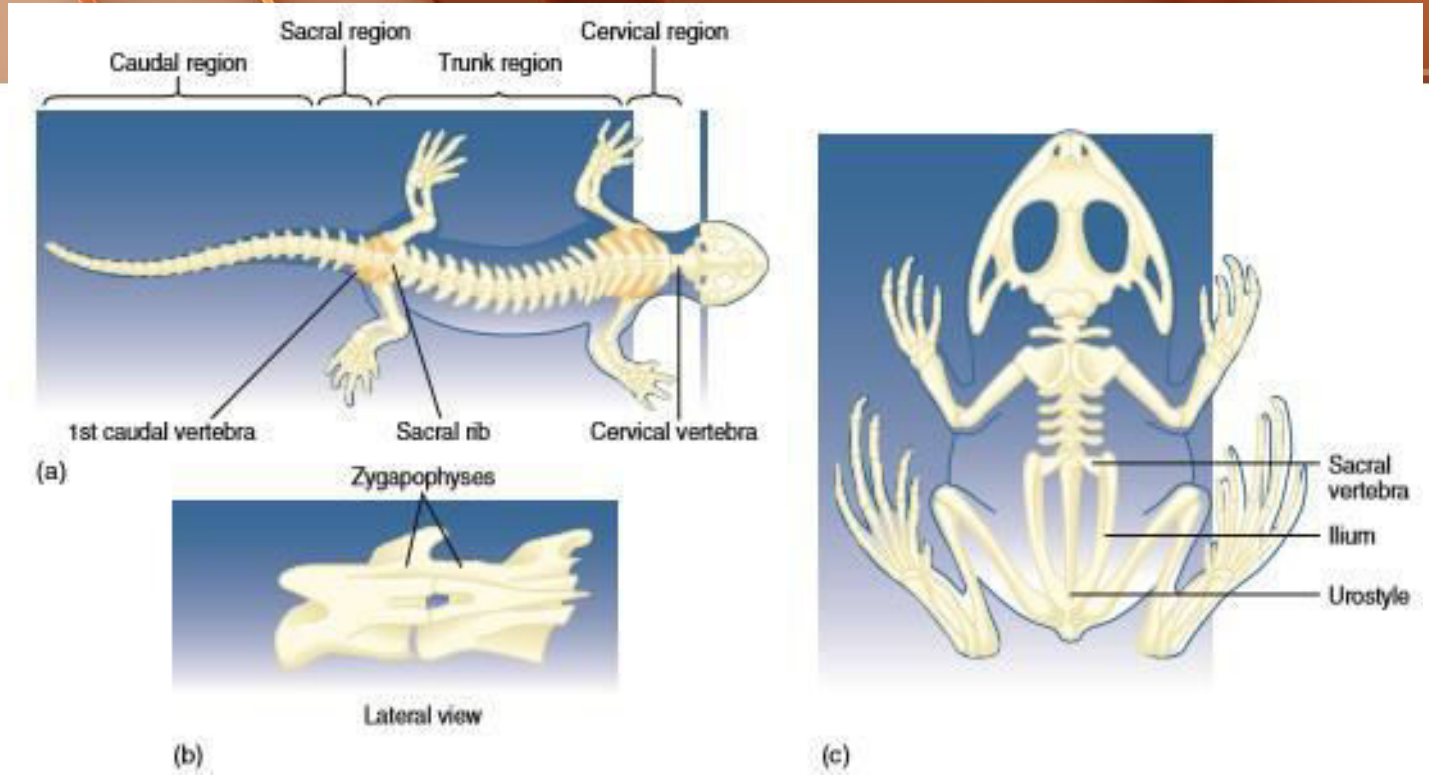
External Structure

- Mostly smooth, although epidermal thickenings may produce warts, claws, or sandpapery textures
- Chromatophores are specialized cells responsible for skin color
- Cryptic coloration, aposematic coloration, and mimicry

Support and Movement

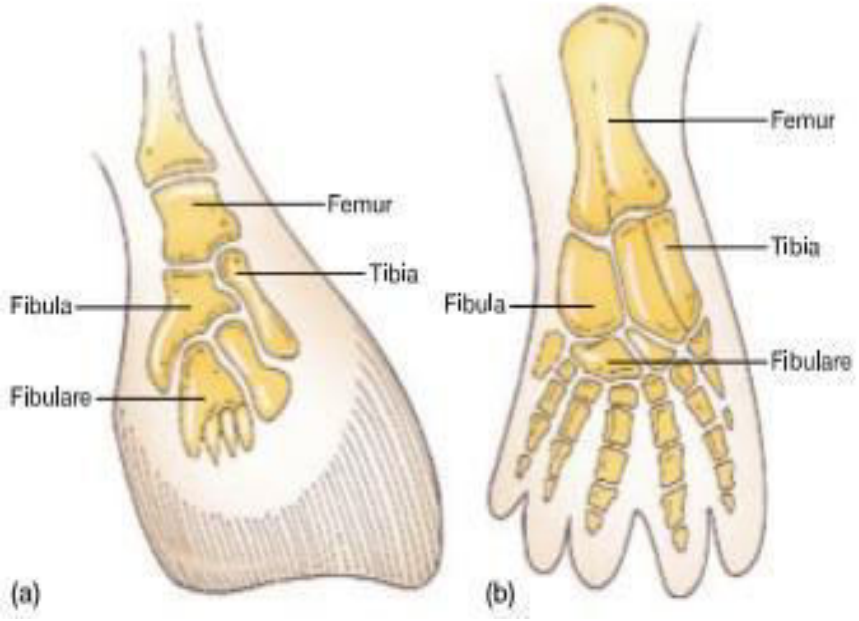
- Water buoys and supports aquatic animals.
- The skeletons of fishes primarily provides protection
- However, terrestrial skeleton is modified to provide support against gravity
- Skull is flattened, smaller, and has fewer bony elements
- These changes lighten the skull
- Changes in jaw structure and musculature allow to crush prey

- Modified vertebral column to provide support and flexibility
- Acts somewhat like the arch of a suspension bridge
- Supportive processes called **zygapophyses** on each vertebra prevent twisting
- Have a neck.
- The first vertebra is a **cervical vertebra**
- The last trunk vertebra is a **sacral vertebra**
- Sternum is present in the **anterior ventral trunk region**
- It is reduced or absent in the Anura.



Skeletons of Amphibians. (a) The salamander skeleton is divided into four regions: cervical, trunk, sacral, and caudal. (b) Interlocking processes, called zygapophyses, prevent twisting between vertebrae. (c) A frog skeleton shows adaptations for jumping. Note the long back legs and the firm attachment of the back legs to the vertebral column through the ilium and urostyle

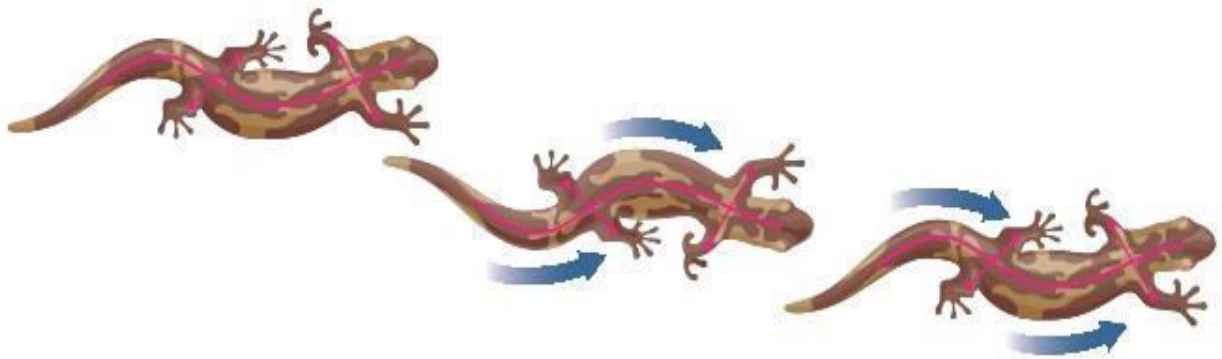
- Bones of vertebrate appendages origin unknown
- Bones of the amphibian appendages and the bones of the fins
- of ancient sarcopterygian fishes suggest possible homologies
- Joints at the shoulder, hip, elbow, knee, wrist, and ankle allow freedom of movement



Origin of Tetrapod Appendages. A comparison of (a) the fin bones of a rhipidistian and (b) the limb bones of a tetrapod

- The pelvic girdle consists of three bones (**the ilium, ischium, and pubis**)
- These bones are important for support on land.
- Tetrapods depend more on appendages than the body wall for locomotion.
- Body-wall musculature reduced, and appendicular
- Unspecialized form of locomotion in Salamenders
- Terrestrial salamanders also move by a pattern of limb and body movements

- Salamanders employ a relatively unspecialized form of locomotion
- Terrestrial salamanders also move by a pattern of limb and body movements



Salamander Locomotion

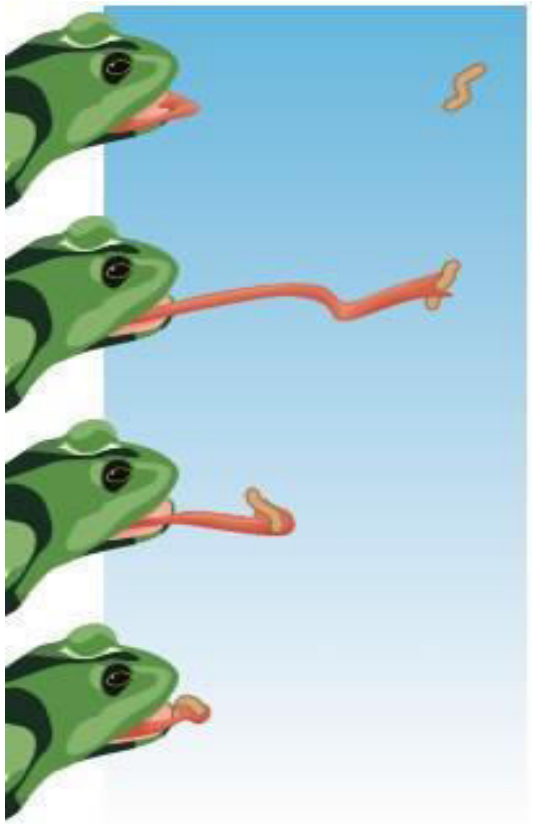
- Caecilians have an **accordion-like movement**
- The long hindlimbs and the pelvic girdle of anurans are modified for jumping.
- The dorsal bone of the pelvis (the ilium) extends anteriorly and securely attaches to the vertebral column
- Urostyle extends posteriorly and attaches to the pelvis

- These skeletal modifications stiffen the posterior half of the anuran.
- Long hindlimbs and powerful muscles form an efficient lever system for jumping.
- Elastic connective tissues and muscles attach the pectoral girdle to the skull and vertebral column
- That function as shock absorbers for landing on the forelimbs.

Lesson 38
NUTRITION AND THE DIGESTIVE SYSTEM

- Adult are carnivores
- The diets of some anurans, however, are more diverse.
- For example, a bullfrog prey on small mammals, birds, and other anurans.
- Prey size and availability
- Larvae are herbivorous
- Locate their prey by sight
- Olfaction plays an important role in prey detection by aquatic salamanders and caecilians.

- Salamanders are relatively unspecialized in their feeding methods
- Anurans and plethodontid salamanders use their tongue and jaws in a flip-and-grab feeding mechanism



Flip-and-Grab Feeding in a Toad

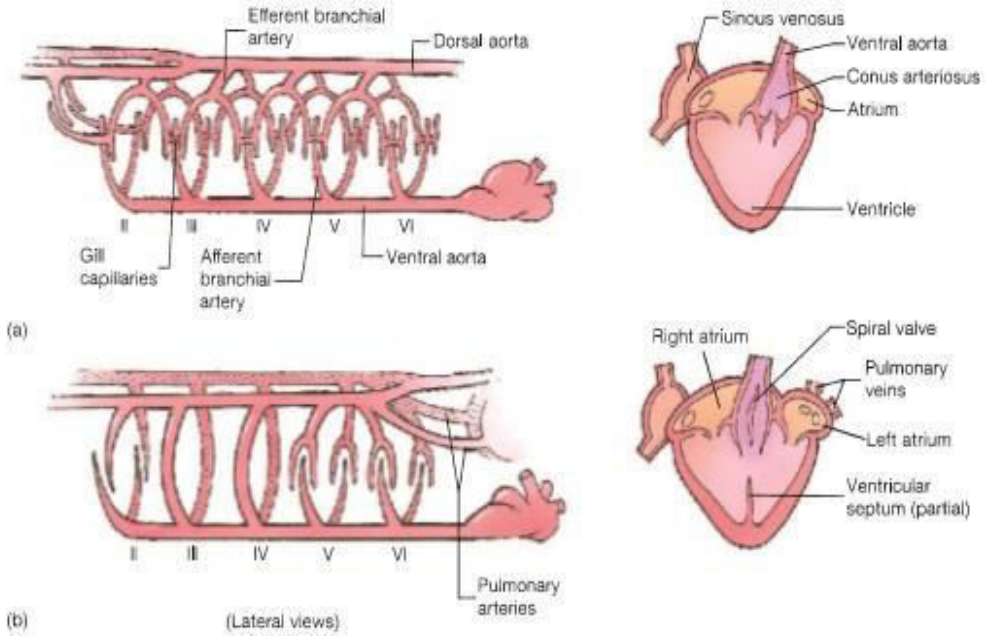
- A true tongue is first seen in amphibians.
- Tongue attaches at the anterior margin of the jaw and folds back over the floor of the mouth.
- Mucous and buccal glands exude sticky secretions.
- Amphibian lunges forward and flicks out its tongue.
- The tongue turns over, and the lower jaw is depressed.
- The head tilts on its single cervical vertebra which helps aim the strike.
- The tip of the tongue entraps the prey

- So the tongue and prey are flicked back inside the mouth.
- All of this may happen in 0.05 to 0.15 second!
- Holds the prey by pressing it against teeth on the roof of the mouth, and the tongue
- Other muscles of the mouth push food toward the esophagus. The eyes sink downward during swallowing and help force food toward the esophagus

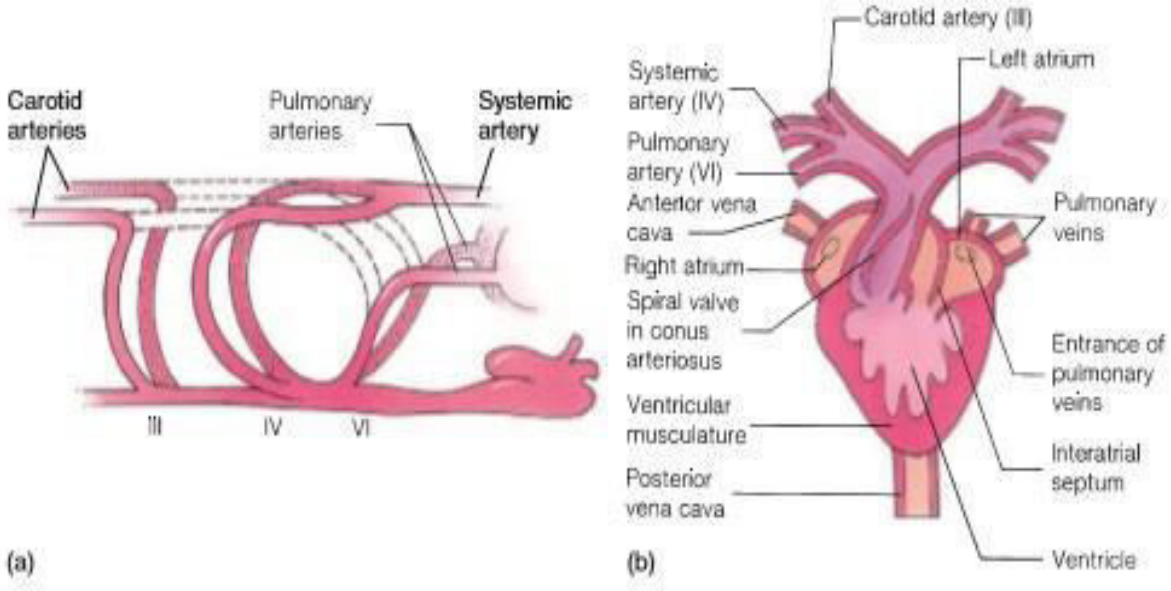
Lesson 39
CIRCULATION, GAS EXCHANGE, AND TEMPERATURE REGULATION

Circulation

- The separation of pulmonary and systemic circuits is less efficient in amphibians than in lungfishes



Circulatory System of Fishes



Anuran Circulatory System

- Atrium is partially divided in urodeles and completely divided in anurans.
- The ventricle has no septum.
- A spiral valve in the conus arteriosus or ventral aorta helps direct blood into pulmonary and systemic circuits.
- Gas exchange occurs across the skin of amphibians, as well as in the lungs.
- When an amphibian is completely submerged, all gas exchange occurs across the skin and other moist surfaces

- Blood vessels leading to the lungs constrict, **reducing blood flow to the lungs and conserving energy.**
- This adaptation is especially valuable for those that overwinter in the mud
- Adult amphibians have **fewer aortic arches than fishes**.
- After leaving the conus arteriosus, blood may enter the carotid artery (aortic arch III), which takes blood to the head; the systemic artery (aortic arch IV), which takes blood to the body; or the pulmonary artery (aortic arch VI).

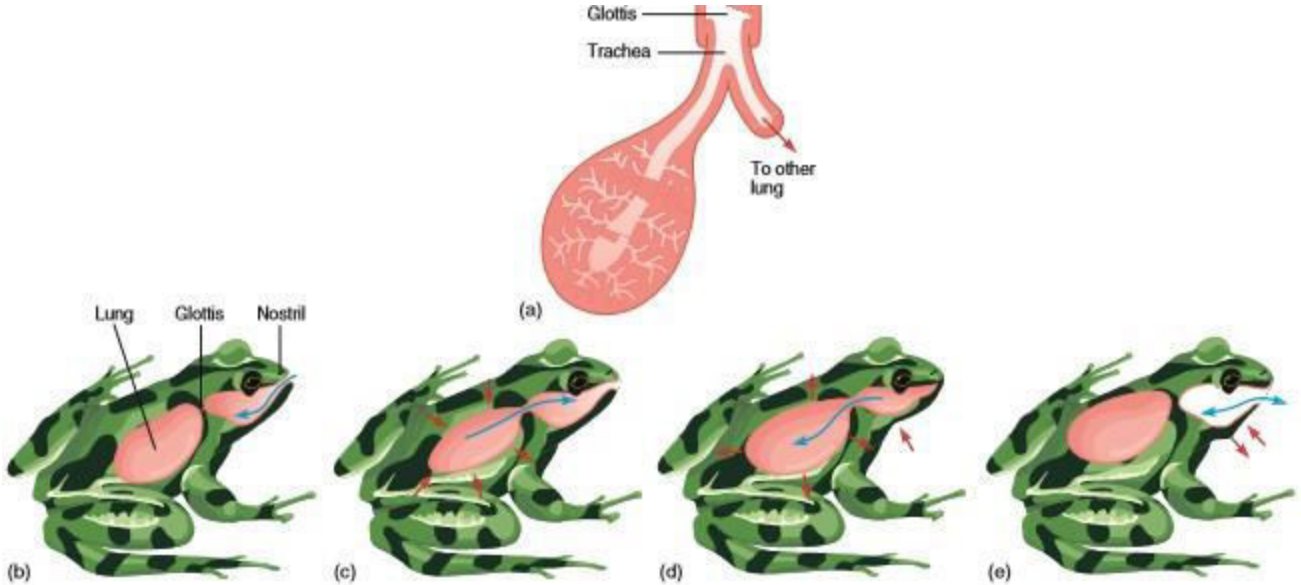
- Amphibians have a well-developed lymphatic system of blindending vessels that returns fluids, proteins, and ions filtered from capillary beds in tissue spaces to the circulatory system.
- The lymphatic system also transports water absorbed across the skin.
- Unlike other vertebrates, amphibians have contractile vessels, called lymphatic hearts.
- Lymphatic spaces between body-wall muscles and the skin transport and store water absorbed across the skin.

Gas Exchange

- Terrestrial animals expend much less energy moving air across gas exchange surfaces
- Exchanges of oxygen and carbon dioxide require moist surfaces
- Amphibian skin is richly supplied with capillary beds.
- These two factors permit the skin to function as a respiratory organ. Gas exchange across the skin is called cutaneous respiration

- In salamanders, 30 to 90% of gas exchange occurs across the skin. Gas exchange also occurs across the moist surfaces of the mouth and pharynx.
- This buccopharyngeal respiration accounts for 1 to 7% of total gas exchange.
- Most amphibians, except for plethodontid salamanders, possess lungs

- The lungs of salamanders are relatively simple sacs.
- The lungs of anurans are subdivided
- Pulmonary (lung) ventilation occurs by a buccal pump mechanism.
- Muscles of the mouth and pharynx create a positive pressure to force air into the lungs



Amphibian Lung Structure, Buccal Pump, and Buccopharyngeal Ventilation.

- Cutaneous and buccopharyngeal respiration contribution to total gas exchange is relatively constant.
- The quantity of gas exchanged across these surfaces cannot be increased when the metabolic rate increases.
- Lungs, however, compensate for this shortcoming.
- As environmental temperature and activity increase, lungs contribute more to total gas exchange.

- At 5° C, approximately 70% of gas exchange occurs across the skin and mouth lining of a frog.
- At 25° C, the absolute quantity of oxygen exchanged across external body surfaces does not change significantly
- Larvae and some adults respire using external gills.
- Cartilaginous rods that form between embryonic pharyngeal slits support three pairs of gills.
- During metamorphosis, the gills are usually reabsorbed, pharyngeal slits close, and lungs become functional.

- Temperature Regulation
- Amphibians are ectothermic.
- Any poorly insulated aquatic animal, regardless of how much metabolic heat it produces, loses heat as quickly as it is produced
- Therefore, when amphibians are in water, they take on the temperature of their environment.
- On land, however, their body temperatures can differ from that of the environment.

- Temperature regulation is mainly behavioral.
- Some cooling results from evaporative heat loss.
- Many are nocturnal and remain in cooler burrows
- Warm themselves by basking in the sun or on warm surfaces.
- Body temperatures may rise 10° C above the air temperature.
- Basking after a meal is common

- Amphibians' daily and seasonal environmental temperatures often fluctuate widely
- They have correspondingly wide temperature tolerances.
- Critical temperature extremes for some salamanders lie between 2 and 27°C, and for some anurans between 3 and 41° C.

Lesson 40
NERVOUS AND SENSORY FUNCTIONS

- The nervous system is similar to that of other vertebrates.
- The brain of adult vertebrates develops from three embryological subdivisions.
- The forebrain contains **olfactory centers** and regions that regulate color change and visceral functions.
- The midbrain contains a region called the **optic tectum**
- The midbrain also processes visual **sensory information**.
- The hindbrain functions in motor coordination and in regulating heart rate and the mechanics of respiration.

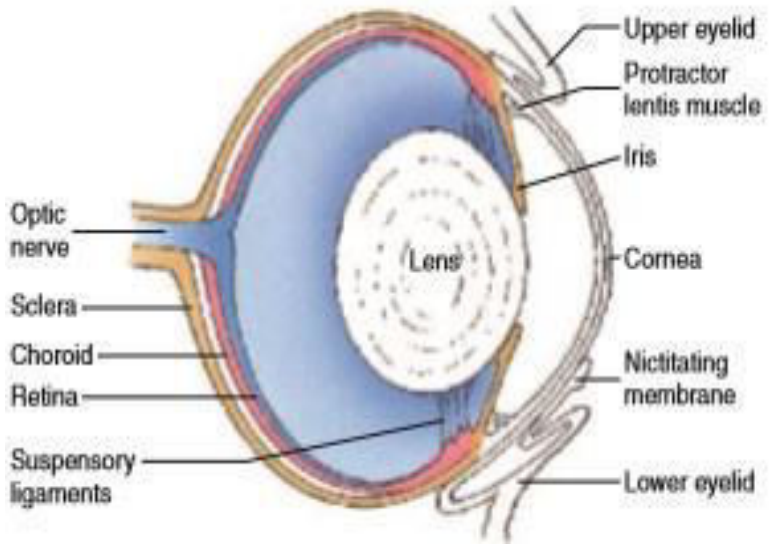
- Many amphibian **sensory receptors** are widely distributed over the skin.
- Some of these are simply **bare nerve endings** that respond to heat, cold, and pain.
- The **lateral-line system** is similar in structure to that found in fishes
- Lateral-line organs are distributed singly or in small groups along the lateral and dorsolateral surfaces of the body
- These receptors respond to **low-frequency vibrations** in the water and movements of the water relative to the animal.
- On land, however, lateral-line receptors are less important.

Chemoreception is an important sense for many amphibians. Chemoreceptors are in the nasal epithelium and the lining of the mouth, on the tongue, and over the skin.

Olfaction is used in mate recognition, as well as in detecting noxious chemicals and in locating food.

Vision is one of the most important senses in amphibians because they are primarily sight feeders. (Caecilians are an obvious exception.)

- A number of adaptations allow the eyes of amphibians to function in terrestrial environments
- The eyes of some amphibians (i.e., anurans and some salamanders) are on the front of the head, providing the binocular vision



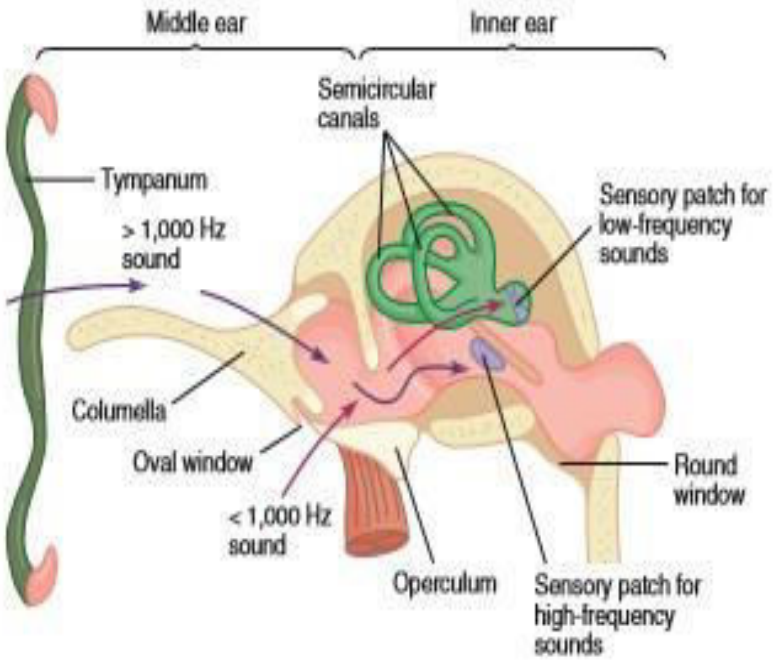
Amphibian Eye

- Well-developed depth perception necessary for capturing prey. Others with smaller lateral eyes lack binocular vision.
- The lower eyelid is movable, and cleans and protects the eye. Much of it is transparent and is called the nictitating membrane.
- When the eyeball retracts into the orbit of the skull, the nictitating membrane is drawn up over the cornea.
- In addition, orbital glands lubricate and wash the eye.

- Together, eyelids and glands keep the eye free of dust and other debris.
- The lens is large and nearly round.
- It is set back from the cornea, and a fold of epithelium called the iris surrounds it.
- The iris can dilate or constrict to control the size of the pupil.
- Focusing, or accommodation, involves bending (refracting) light rays to a focal point on the retina.
- Light waves moving from air across the cornea are refracted because of the change in density between the two media

- The lens provides further refraction.
- The amphibian eye focuses on distant objects when the eye is at rest.
- To focus on near objects, the protractor lentis muscle must move the lens forward
- Rods and cones are in the retina.
- The neuronal interconnections in the retina are complex and allow an amphibian to distinguish between flying insect prey

- The auditory system of amphibians is an evolutionary adaptation
- It transmits both substrate borne vibrations and, in anurans, airborne vibrations.
- The ears of anurans consist of a tympanic membrane, a middle ear, and an inner ear.
- The tympanic membrane is a piece of integument stretched over a cartilaginous ring
- Abutting the tympanic membrane is a middle-ear ossicle (bone) called the stapes (columella)



Ear of an Anuran (Posterior View).

- High-frequency (1,000 to 5,000 Hz) airborne vibrations are transmitted to the inner ear through the tympanic membrane. Low-frequency (100 to 1,000 Hz) substrate borne vibrations are transmitted through the front appendages
- Pectoral girdle to the inner ear through a second ossicle, called the operculum.
- Muscles attached to the operculum and stapes can lock either or both of these ossicles

- Anurans use **low- and high-frequency sounds** in different situations.
- Mating calls are high-frequency sounds
- At other times, low-frequency sounds may warn of approaching predators.
- Salamanders lack a tympanic membrane and middle ear.
- They live in streams, ponds, caves, and beneath leaf litter.
- They have no mating calls

- The inner ear has semicircular canals that help detect rotational movements and other sensory patches that respond to gravity and detect linear acceleration and deceleration.

Lesson 41

EXCRETION AND OSMOREGULATION

Excretion and Osmoregulation

- The kidneys of amphibians lie on either side of the dorsal aorta on the dorsal wall of the body cavity.
- A duct leads to the cloaca
- The urinary bladder is a ventral outgrowth of the cloaca.
- The nitrogenous waste product is either ammonia or urea.

  Amphibians that live in freshwater excrete ammonia.
- It is the immediate end product of protein metabolism.

- The toxic effects of ammonia are avoided
- Amphibians that spend more time on land excrete urea
- Although urea is less toxic than ammonia, it still requires relatively large quantities of water for its excretion.
- Urea can be stored in the urinary bladder.
- Some amphibians excrete ammonia when in water and urea when on land.

- One of the biggest problems is osmoregulation.
- Amphibians face the same osmoregulatory problems as freshwater fishes.
- They must rid the body of excess water and conserve essential ions.
- Amphibian kidneys produce large quantities of hypotonic urine, Skin and walls of the urinary bladder transport Na, Cl and other ions into the blood.
- On land, amphibians must conserve water.
- Adult amphibians do not replace water by intentional drinking
- Instead, amphibians limit water loss by reducing exposure to desiccating conditions.

- Many terrestrial amphibians are **nocturnal**.
- During day light hours, they retreat to areas of high humidity
- Water lost on nighttime trips must be replaced by water uptake across the skin.
- **Diurnal amphibians** usually live in areas of high humidity
- Many amphibians reduce evaporative water loss
- They may **curl their bodies** and tails into tight coils
- Individuals may form closely **packed aggregations** to reduce overall surface area



Daytime sleeping posture of the green tree frog, Hyla cinerea

- Some have **protective coverings** that reduce water loss.
- Hardened regions of skin are resistant to water loss
- Other amphibians prevent water loss by forming **cocoons**
- Cocoons are parchment like.
- These cocoons open only at the nares or the mouth and, in experimental situations, reduce water loss 20 to 50% over non cocooned individuals



The Australian burrowing frog, Cyclorana alboguttatus

- When an amphibian flattens its body, the skin absorbs water.
- The skin's permeability, vascularization, and epidermal sculpturing all promote water reabsorption.
- Minute channels increase surface area
- Amphibians can also **temporarily store water.**
- Water accumulated in the **urinary bladder and lymph sacs** can be selectively reabsorbed to replace evaporative water loss.
- Amphibians of dry environments can store water equivalent to **35% of their total body weight.**

Lesson 42
REPRODUCTION, DEVELOPMENT, AND METAMORPHOSIS

Reproduction

- Amphibians are dioecious
- Fertilization is usually external
- A few anurans have terrestrial nests that are kept moist by being enveloped in foam
- Larval stages are passed in the egg membranes
- The immatures hatch into an adult like body.
- The main exceptions to external fertilization in amphibians are the salamanders.

- All others produce **spermatophores**
- Eggs may be deposited in soil or water or retained in the oviduct during development.
- All caecilians have internal fertilization, and 75% have internal development.
- Amphibian development includes larval stages called tadpoles. Tadpoles often differ from the adults
- These differences reduce competition between adults and larvae.

- Interactions between internal controls and extrinsic factors determine the timing of reproductive activities.
- Temperature seems to be the most important environmental factor
- Breeding periods are seasonal
- In tropical regions, amphibian breeding correlates with rainy seasons.
- Courtship behavior helps individuals locate breeding sites
- Also prepares individuals for reproduction
- Ensures that eggs are fertilized and deposited in locations that promote successful development

- Salamanders rely primarily on **olfactory and visual cues** in courtship and mating
- Male vocalizations and tactile cues are important for anurans.
- Species congregate during times of intense breeding activity.
- Male vocalizations are usually species specific
- After that tactile cues become more important.
- The male grasps the female in a position called **amplexus**
- The male releases sperm as the female releases eggs.
- Little is known of **caecilian breeding** behavior.
- Males have an **intromittent** organ that is a modification of the cloacal wall
- Fertilization is **internal**.



Amplexus

Vocalization

- Sound production is primarily a reproductive function of **male anurans**, **advertisement calls**, and announce to other males
- Advertisement calls are species specific
- The **repertoire of calls** for any one species is limited.
- The calls may also help induce psychological and physiological readiness to breed.
- Females respond by making reciprocation calls
- Release calls inform a partner that a frog is incapable of reproducing.

- **Unresponsive females** give release calls if a male attempts amplexus
- **Distress calls** are not associated with reproduction
- The calls may be loud enough to cause a predator to release the frog.
- The distress call of the South American jungle frog, **Leptodactylus pentadactylus**, is a loud scream similar to the call of a cat in distress.
- Sound-production consists of the larynx and its vocal cords.
- This laryngeal apparatus is well developed in males, who also possess a vocal sac.

- Vocal sacs develop as a diverticulum from the lining of the buccal cavity
- Air from the lungs is forced over the vocal cords and cartilages of the larynx
- Muscles control the tension of the vocal cords
- Vocal sacs act as resonating structures and increase the volume of the sound.
- The use of sound to attract mates is very useful in organisms
- Vocalizations help to **reduce the chaos**.



Anuran Vocalization

Lesson 43
Parental Care and Metamorphosis

- Parental care increases the chances of any one egg developing
- The most common form of parental care in amphibians is attendance of the egg clutch by either parent.
- Maternal care occurs in species with internal fertilization (predominantly salamanders and caecilians)
- Paternal care may occur in species with external fertilization (predominantly anurans).
- It may involve aeration of aquatic eggs, cleaning and/or moistening of terrestrial eggs, protection of eggs from predators, or removal of dead and infected eggs.

- Eggs may be transported when development occurs on land.
- Females of the genus Pipa carry eggs on their backs.
- Two species of Rheobatrachus were discovered in Australia within the last 30 years.
- Rheobatrachus females brooded tadpoles in their stomachs, and the young emerged from the females' mouths
- During brooding, the female's stomach expanded to fill most of her body cavity
- Viviparity and ovoviviparity occur primarily in salamanders and caecilians.



Parental Care of Young

- **Metamorphosis**
- Metamorphosis is a series of abrupt structural, physiological, and behavioral changes.
- Crowding and food availability, influence the time required for metamorphosis.
- Metamorphosis is under the control of **neurosecretions** of the hypothalamus
- Hormones of the anterior lobe of the pituitary gland (the adenohypophysis), and the thyroid gland.

- Metamorphosis of caecilians and salamanders are relatively minor.
- Reproductive structures develop
- Gills are lost
- Caudal fin (when present) is lost.
- In the Anura changes from the tadpole into the small frog are more dramatic
- Limbs and lungs develop, the tail is reabsorbed, the skin thickens, and marked changes in the head and digestive tract occur.
- Metamorphosis explain paedomorphosis in amphibians.

- Some salamanders are paedomorphic
- In some salamander families, paedomorphosis is the rule.
- In other families, the occurrence of paedomorphosis is variable and influenced by environmental conditions.

Reptiles

Lesson 44

RESPTILES: EVOLUTIONARY PERSPECTIVE

- Class reptilia possess amniotic egg.
- Amniotic eggs have extra embryonic membranes that protect the embryo from desiccation, cushion the embryo, promote gas transfer, and store waste materials
- The amniotic eggs of reptiles and birds also have hard or leathery shells that protect the developing embryo, albumen that cushions and provides moisture and nutrients for the embryo, and yolk that supplies food to the embryo. All of these features are adaptations for development on land.
- In amniotic egg, the embryo develops at the surface of a mass of yolk.
- Amniotic egg consists of three layers, The **amnion** encloses the embryo in a fluid-filled sac and protects against shock and desiccation. The **chorion** is nearer the shell and becomes highly vascular and aids in gas exchange. The **allantois** is a ventral outgrowth of the gut and stores nitrogenous wastes (e.g., uric acid).

- Even though the amniotic egg has played an important role in vertebrates' successful invasion of terrestrial habitats, it is one of many reptilian adaptations that have allowed members of this class to flourish on land
- Living representatives of the class Reptilia include the turtles, lizards, snakes, worm lizards, crocodilians, and the tuatara
- There are three classes of reptiles according to opening in temporal region
- Reptiles in the subclass **Anapsida** lack openings or fenestrae region of the skull. The turtles represent this lineage today.
- A second group of reptiles are diapsid. They have upper and lower openings in the temporal region of the skull.
- Third group of reptiles are synapsid. They possess a single dorsal opening in the temporal region of the skull.

Cladistic interpretation of the amniotic lineage

- Cladistic analysis shows that the birds (traditionally the class Aves) and the mammals (traditionally the class Mammalia) share a common ancestor with the reptiles.

Lesson 45

Survey of the reptiles

General characteristics of reptiles

- They have dry skin with epidermal scales
- Skull with one point of articulation with the vertebral column (occipital condyle); respiration via
- Lungs are present in reptiles
- Osmoregulation occurs through metanephric kidneys
- internal fertilization
- Amniotic eggs
- They have four orders named as; 1. Testudines, 2. Rhynchocephalia 3. Squamata, 4. Crocodilia

1. Order testudines (Chelonia)

- Members of the order Testudines or Chelonia are the turtles
- The approximately 225 species of turtles are characterized by a bony shell, limbs articulating internally to the ribs, and a keratinized beak rather than teeth.
- The dorsal portion of the shell is the **carapace**, which forms from a fusion of vertebrae, expanded ribs, and bones in the dermis of the skin. Keratin covers the bone of the carapace.
- The ventral portion of the shell is the **plastron**. It forms from bones of the pectoral girdle and dermal bone, and keratin also covers it

- In some turtles, such as the North American box turtle (Terrapene), the shell has flexible areas, or hinges, that allow the anterior and posterior edges of the plastron to be raised.
- Turtles have eight cervical vertebrae that can be articulated into an S-shaped configuration, which allows the head to be drawn into the shell.
- Turtles have long life spans. Most reach sexual maturity after seven or eight years, and live 14 or more years.
- All turtles are oviparous
- Females use their hindlimbs to excavate nests in the soil. There they lay and cover with soil clutches of 5 to 100 eggs
- Development takes from four weeks to one year, and the parent does not attend to the eggs during this time
- The young are independent of the parent at hatching.

2. Order rhynchocephalia

- The one surviving species of the order is the tuatara (Sphenodon punctatus)
- Tooth attachment and structure distinguish the tuatara from other reptiles.
- It is oviparous and shares under- ground burrows with ground-nesting seabirds
- Tuataras venture out of their burrows at dusk and dawn to feed on insects or, occasionally, small vertebrates.

Lesson 46

ORDER SQUAMATA

Order squamata divided into three suborders.

1.Suborder Sauria—The Lizards

About 3,300 species of lizards are in the suborder Sauria

Lizards usually have two pairs of legs, and their upper and lower jaws unite anteriorly

Lizards vary in length from only a few centimeters to as large as 3 m. Many lizards live on surface substrates and retreat under rocks or logs when necessary

Others are burrowers or tree dwellers.

- Most lizards are oviparous; some are ovoviviparous or viviparous
- They usually deposit eggs under rocks or debris or in burrows. **Geckos**, commonly found on the walls of human dwellings in semitropical areas, are short and stout
- **Iguanas** have robust bodies, short necks, and distinct heads. This group includes the marine iguanas of the Galápagos Islands and the flying dragons (Draco) of Southeast Asia
- **Chameleons** are adapted to arboreal lifestyles and use a long, sticky tongue to capture insects
- The only venomous lizards are the gila monster (Heloderma suspectum) and the Mexican beaded lizard

2. Suborder Serpentes—The Snakes

- About 2,300 species are in the suborder Serpentes
- Three hundred species are venomous
- Snakes are elongate and lack limbs
- The skeleton may contain more than two hundred vertebrae and pairs of ribs
- Joints between vertebrae make the body very flexible
- Snakes possess skull adaptations that facilitate swallowing large prey
- Other differences between lizards and snakes include the mechanism for focusing the eyes and the morphology of the retina.
- Most snakes are oviparous, although a few, such as the New World boas and garter snakes, give birth to live young.

3.Suborder Amphisbaenia—Worm-Lizards

- About 135 species are in the suborder Amphisbaenia
- They are specialized burrowers that live in soils of Africa, South America, the Caribbean, and the Mideast
- Most are legless, and their skulls are wedge or shovel shaped. A single median tooth in the upper jaw distinguishes amphisbaenians from all other vertebrates.
- The skin of amphisbaenians has ringlike folds called **annuli**.
- Muscles of the skin cause it to telescope and bulge outward, forming an anchor against a burrow wall
- Amphisbaenians move easily forward or backward—thus, the suborder name. They feed on worms and small insects and are oviparous

Lesson 47

ORDER CROCODILIAN

The order Crocodilia has 21 species

Along with dinosaurs, crocodilians are derived from the archosaurs and distinguished from other reptiles by certain skull characteristics: openings in the skull in front of the eye, triangular rather than circular eye orbits, and laterally compressed teeth

Living crocodilians include the alligators, crocodiles, gavials, and caimans.

Crocodilians have not changed much over their 170 million-year history

The snout is elongate and often used to capture food

The nostrils are at the tips of the snout, so the animal can breathe while mostly submerged

Air passageways of the head lead to the rear of the mouth and throat, and a flap of tissue near the back of the tongue forms a watertight seal that allows breathing without inhaling water in the mouth

A plate of bone, called the **secondary palate**, evolved in the archosaurs and separates the nasal and mouth passage ways.

The muscular, elongate, and laterally compressed tail is used for swimming, offensive and defensive maneuvers, and attacking prey.

Teeth are used only for seizing prey

Food is swallowed whole, but if a prey item is too large, crocodilians tear it apart by holding onto a limb and rotating their bodies wildly until the prey is dismembered

The stomach is **gizzardlike**, and crocodilians swallow rocks and other objects as abrasives for breaking apart ingested food

Crocodilians are oviparous and display parental care of hatchlings that parallels that of birds. Nesting behavior and parental care may be traced back to the common ancestor of both groups

Comparison Table Crocodile vs Alligator

Characters	Crocodile	Alligator
Distribution	Asia, Africa, Central America, Malaya, Indonesia and North Australia.	Mostly found in North America and China.
Habitats	Crocodile live in freshwater, They can also live in salt water.	Alligator live in fresh water. Lakes, swamps and streams have fresh water.
Habits	More aggressive, dangerous to man.	Less aggressive, attack when provoked.
Four mandibular tooth	First into a maxillary groove and visible externally.	Fits into a pit and not visible externally.
Teeth	Unequal.	Very unequal.
Body length	8 metres long.	3 metres long.
Color	Olive green with black spot or bands.	Steel grey.
Snout	Moderately long and pointed.	Short and broad.
	Long pointed jaw - V shaped.	Wide round jaw - U shaped.
	 V shaped snout designed for snatching prey.	 U shaped snout designed for more leverage for crushing.

www.majordifferences.com

Lesson 48

EVOLUTIONARY PRESSURES: EXTERNAL STRUCTURE AND LOCOMOTION

Evolutionary pressures

The lifestyles of most reptiles reveal striking adaptations for terres-trialism

EXTERNAL STRUCTURE AND LOCOMOTION

➢ **External structure of reptiles**

- Reptilian skin is thick, dry, and keratinized
- Scales may be modified for various functions. For example, the large belly scales of snakes provide contact with the substrate during locomotion
- Skin secretions include pheromones that function in sex recognition and defense
- All reptiles periodically shed the outer, epidermal layers of the skin in a process called **ecdysis**
- The frequency of ecdysis varies from one species to another, and it is greater in juveniles than adults
- Cryptic coloration, mimicry, and aposematic coloration occur in reptiles. Color and color change also function in sex recognition and thermoregulation

Support and movement

- The skull of reptiles is longer than that of amphibians, and a plate of bone, the **secondary palate**, partially separates the nasal passages from the mouth cavity
- Reptiles have more cervical vertebrae. The first two cervical vertebrae (atlas and axis) provide greater freedom of movement for the head
- The ribs of reptiles may be highly modified
- The caudal vertebrae of many lizards possess a vertical fracture plane When a lizard is grasped by the tail, caudal vertebrae can be broken, and a portion of the tail is lost. Tail loss, or **autotomy**, is an adaptation that allows a lizard to escape from a predator's grasp, or the disconnected, wiggling piece of tail may distract a predator from the lizard
- The lizard later regenerates the lost portion of the tail

- Primitive reptiles move by crawling on earth surface
- Many prehistoric reptiles were bipedal, meaning that they walked on hindlimbs
- They had a narrow pelvis and a heavy, outstretched tail for balance
- Bipedal locomotion freed the front appendages, which became adapted for prey cap- ture or flight in some animals.

Lesson 49

NUTRITION AND THE DIGESTIVE SYSTEM

Nutrition and the digestive system

- The tongues of turtles and crocodilians are non-protrusible and aid in swallowing
- Like some anurans, some lizards and the tuatara have sticky tongues for capturing prey
- Skull and jaws of reptiles help to capture large prey
- Posteriorly pointing teeth prevent prey escape and help force the food into the esophagus
- The glottis, the respiratory opening, is far forward so that the snake can breathe while slowly swallowing its prey

- Vipers possess hollow fangs on the maxillary bone at the anterior margin of the upper jaw.
- These fangs connect to venom glands that inject venom when the viper bites
- The maxillary bone (upper jaw bone) of vipers is hinged so that when the snake's mouth is closed, the fangs fold back and lie along the upper jaw. When the mouth opens, the maxillary bone rotates and causes the fangs to swing down
- Because the fangs project outward from the mouth, vipers may strike at objects of any size
- Rear-fanged snakes possess grooved rear teeth. In those that are venomous, venom is channeled along these grooves and worked into prey to quiet them during swallowing

- Coral snakes, sea snakes, and cobras have fangs that rigidly attach to the upper jaw in an erect position. Fangs are grooved or hollow, and contraction of muscles associated with venom glands injects venom into the fangs. Some cobras can “spit” venom at their prey; if not washed from the eyes, the venom may cause blindness.
- Venom glands are modified **salivary glands**. Most snake venoms are mixtures of neurotoxins and hemotoxins. The venoms of coral snakes, cobras, and sea snakes are primarily neurotoxins that attack nerve centers and cause respiratory paralysis
- The venoms of vipers are primarily hemotoxins. They break up blood cells and attack blood vessel linings.

- Like amphibians, reptiles possess two atria that are completely separated in the adult and have veins from the body and lungs emptying into them
- Except for turtles, the sinus venosus is no longer a chamber but has become a patch of cells that acts as a pacemaker
- The ventricle of most reptiles is incompletely divided (figure 20.14). (Only in crocodilians is the ventricular septum complete

Lesson 50

Circulation

Circulation

- Because reptiles are, on average, larger than amphibians, their blood must travel under higher pressures to reach distant body parts
- Like amphibians, reptiles possess two atria that are completely separated in the adult and have veins from the body and lungs emptying into them
- Except for turtles, the sinus venosus is no longer a chamber but has become a patch of cells that acts as a pacemaker. The ventricle of most reptiles is incompletely divided (Only in crocodilians is the ventricular septum complete.)
- The ventral aorta and the conus arteriosus divide during development and become three major arteries that leave the heart. A pulmonary artery leaves the ventral side of the ventricle and takes blood to the lungs

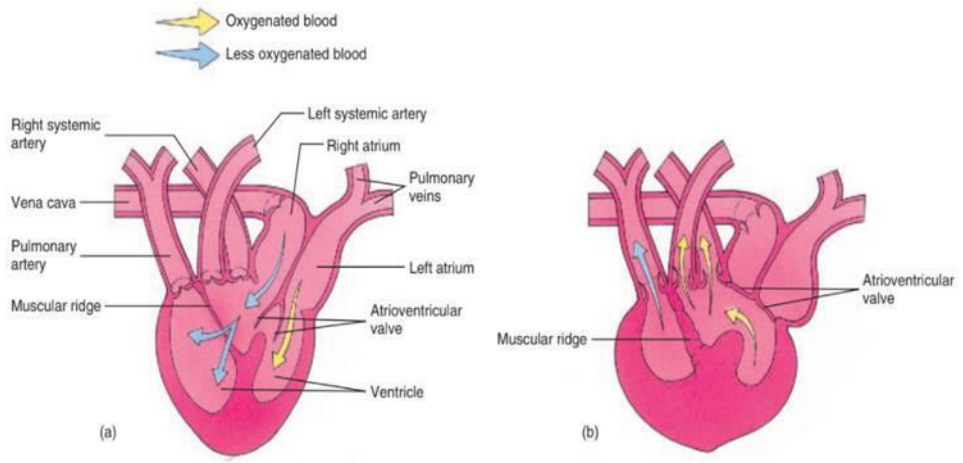


Figure : Heart and Major Arteries of a Lizard. (a) When the atria contract, blood enters the ventricle. An atrioventricular valve prevents the mixing of oxy-genated and less oxygenated blood across the incompletely separated ventricle. (b) When the ventricle contracts, a muscular ridge closes to direct oxygenated blood to the systemic arteries and less oxygenated blood to the pulmonary artery

- Two systemic arteries, one from the ventral side of the heart and the other from the dorsal side of the heart, take blood to the lower body and the head
- Blood low in oxygen enters the ventricle from the right atrium and leaves the heart through the pulmonary artery and moves to the lungs
- Blood high in oxygen enters the ventricle from the lungs via pulmonary veins and the left atrium, and leaves the heart through left and right systemic arteries
- The incomplete separation of the ventricle permits shunting of some blood away from the pulmonary circuit to the systemic circuit by constriction of muscles associated with the pulmonary artery
- This is advantageous because virtually all reptiles breathe intermittently.

Lesson 51

Gas exchange and temperature regulation

Gas exchange

- Reptiles exchange respiratory gases across internal respiratory surfaces to avoid losing large quantities of water
- In most reptiles, **a negative-pressure mechanism** is responsible for lung ventilation. A posterior movement of the ribs and the body wall expands the body cavity, decreasing pressure in the lungs and drawing air into the lungs. Air is expelled by elastic recoil of the lungs and forward movements of the ribs and body wall, which compress the lungs
- The ribs of turtles are a part of their shell; thus, movements of the body wall to which the ribs attach are impossible. Turtles exhale by contracting muscles that force the viscera upward, compressing the lungs. They inhale by contracting muscles that increase the volume of the visceral cavity, creating negative pressure to draw air into the lungs

Temperature regulation

- Most reptiles use external heat sources for thermoregulation and are, therefore, ectotherms
- To sustain activity, however, body temperatures are regulated within a narrow range, between 25 and 37° C. If that is not possible, the reptile usually seeks a retreat where body temperatures are likely to remain within the range compatible with life.
- Most thermoregulatory activities of reptiles are behavioral, and they are best known in the lizards. To warm itself, a lizard orients itself at right angles to the sun's rays, often on a surface inclined toward the sun, and presses its body tightly to a warm surface to absorb heat by conduction. To cool itself, a lizard orients its body parallel to the sun's rays, seeks shade or burrows, or assumes an erect posture (legs extended and tail arched) to reduce conduction from warm surfaces
- In hot climates, many reptiles are nocturnal.

- Various physiological mechanisms also regulate body temperature. As temperatures rise, some reptiles begin panting, which releases heat through evaporative cooling.
- Marine iguanas divert blood to the skin while basking in the sun and warm up quickly. On diving into the cool ocean, however, marine iguanas reduce heart rate and blood flow to the skin, which slows heat loss
- Chromatophores also aid in temperature regulation. Dispersed chromatophores (thus, a darker body) increase the rate of heat absorption
- In temperate regions, many reptiles withstand cold winter temperatures by entering torpor when body temperatures and metabolic rates decrease
- Individuals that are usually solitary may migrate to a common site to spend the winter. Heat loss from these groups, called **hibernacula**, is reduced because the total surface area of many individuals clumped together is reduced compared to widely separated animals

Lesson 52

NERVOUS AND SENSORY FUNCTIONS

- The brain of reptiles is similar to the brains of other vertebrates. The cerebral hemispheres are somewhat larger than those of amphibians. This increased size is associated with an improved sense of smell
- The optic lobes and the cerebellum are also enlarged, which reflects increased reliance on vision and more refined coordination of muscle functions
- Binocular vision then provides the depth perception for determining whether or not the insect is within range of the chameleon’s tongue
- Vision is the dominant sense in most reptiles, and their eyes are similar to those of amphibians
- Snakes focus on nearby objects by moving the lens forward.

- All other reptiles focus on nearby objects when the normally elliptical lens is made more spherical
- Reptiles have a greater number of cones than do amphibians and probably have well-developed color vision
- Upper and lower eyelids, a nictitating membrane, and a blood sinus protect and cleanse the surface of the eye
- In snakes and some lizards, the upper and lower eyelids fuse in the embryo to form a protective window of clear skin, called the **spectacle**. (During ecdysis, the outer layers of the spectacle become clouded and impair the vision of snakes.)
- The blood sinus, which is at the base of the nictitating membrane, swells with blood to help force debris to the corner of the eye, where it may be rubbed out. Horned lizards squirt blood from their eyes by rupturing this sinus in a defensive maneuver to startle predators. Some reptiles possess a median (parietal) eye that develops from outgrowths of the roof of the forebrain.

- Some reptiles possess a median (parietal) eye that develops from outgrowths of the roof of the forebrain
- In the tuatara, it is an eye with a lens, a nerve, and a retina
- In other reptiles, the parietal eye is less developed
- Parietal eyes are covered by skin and probably cannot form images. They can, however, differentiate light and dark periods and are used in orientation to the sun
- The structure of reptilian ears varies. The ears of snakes detect substrate vibrations.
- They lack a middle-ear cavity, an auditory tube, and a tympanic membrane. A bone of the jaw articulates with the stapes and receives substrate vibrations
- Snakes can also detect airborne vibrations. In other reptiles, a tympanic membrane may be on the surface or in a small depression in the head. The inner ear of reptiles is similar to that of amphibians

- Olfactory senses are better developed in reptiles than in amphibians. In addition to the partial secondary palate providing more surface for olfactory epithelium, many reptiles possess blind-ending pouches that open through the secondary palate into the mouth cavity. These pouches, called Jacobson's (vomeronasal) organs.
- Jacobson's organs develop in embryonic crocodilians but are not present in adults of this group. Anapsids (turtles) lack these olfactory organs
- The protrusible, forked tongues of snakes and lizards are accessory olfactory organs for sampling air borne chemicals. A snake's tongue flicks out and then moves to the Jacobson's organs, which perceive odor molecules
- Tuataras use Jacobson's organs to taste objects held in the mouth
- Rattlesnakes and other pit vipers have heat-sensitive pit organs on each side of the face between the eye and nostril. These depressions are lined with sensory epithelium and are used to detect objects with temperatures different from the snake's surroundings. Pit vipers are usually nocturnal, and their pits help them to locate small, warm-blooded prey.

Lesson 53

Excretion and osmoregulation

Excretion and osmoregulation

- The kidneys of embryonic reptiles are similar to those of fishes and amphibians
- Life on land, increased body size, and higher metabolic rates, however, require kidneys capable of processing wastes with little water loss
- A kidney with many more blood-filtering units, called **nephrons**, replaces the reptilian embryonic kidney during development
- The functional kidneys of adult reptiles are called **metanephric kidneys**. Their function depends on a circulatory system that delivers more blood at greater pressures to filter large quantities of blood
- Most reptiles excrete uric acid. It is nontoxic, and being relatively insoluble in water, it precipitates in the excretory system.
- The urinary bladder or the cloacal walls reabsorb water, and the uric acid can be stored in a paste like form.

- Nocturnal habits and avoiding hot surface temperatures during the day by burrowing reduce water loss
- When water is available, many reptiles (e.g., chuckwallas) store large quantities of water in lymphatic spaces under the skin or in the urinary bladder
- Many lizards possess salt glands below the eyes for ridding the body of excess salt.

Lesson 54

REPRODUCTION AND DEVELOPMENT, FURTHER PHYLOGENETIC CONSIDERATIONS

Reproduction and development

- Parental care occurs in some reptiles and may involve maintaining relatively high humidity around the eggs
- The eggs are often supplied with large quantities of yolk for long developmental periods, and parental energy and time are sometimes invested in the post hatching care of dependent young.
- Development of amniotic eggs is the necessity for internal fertilization. Fertilization must occur in the reproductive tract of the female before protective egg membranes are laid down around an egg
- All male reptiles, except tuataras, possess an intromittent organ for introducing sperm into the female reproductive tract
- Lizards and snakes possess paired **hemipenes** at the base of the tail that are erected by being turned inside out, like the finger of a glove.

- Gonads lie in the abdominal cavity
- In males, a pair of ducts delivers sperm to the cloaca
- After copulation, sperm may be stored in a seminal receptacle in the female reproductive tract. Secretions of the seminal receptacle nourish the sperm and arrest their activity
- Sperm may be stored for up to **four years** in some turtles, and up to **six years** in some snakes! In temperate latitudes, sperm can be stored over winter.
- Copulation may take place in the fall, when individuals congregate in hibernacula, and fertilization and development may occur in the spring, when temperatures favor successful development.
- Fertilization occurs in the upper regions of the oviduct, which leads from the ovary to the cloaca. Glandular regions of the oviduct secrete albumen and the eggshell
- The shell is usually tough yet flexible. In some crocodilians, the eggshell is calcareous and rigid, like the eggshells of birds.

- Parthenogenesis has been described in six families of lizards and one species of snakes. In these species, no males have been found
- Populations of parthenogenetic females have higher reproductive potential than bisexual populations
- Reptiles often have complex reproductive behaviors that may involve males actively seeking out females
- As in other animals, courtship functions in sexual recognition and behavioral and physiological preparation for reproduction
- Lizards and snakes also use sex pheromones
- Vocalizations are important only in crocodilians
- During the breeding season, males are hostile and may bark or cough as territorial warnings to other males. Roaring vocalizations also attract females, and mating occurs in the water.

- After they are laid, reptilian eggs are usually abandoned (figure a). Virtually all turtles bury their eggs in the ground or in plant debris
- Other reptiles lay their eggs under rocks, in debris, or in burrows. About one hundred species of reptiles have some degree of parental care of eggs
- One example is the American alligator, Alligator mississippiensis (figure b). The female builds a mound of mud and vegetation about 1 m high and 2 m in diameter. She hollows out the center of the mound, partially fills it with mud and debris, deposits her eggs in the cavity, and then covers the eggs
- Temperature within the nest influences the sex of the hatchlings. Temperatures at or below **31.5° C** result in female off-spring.





a) This young giant Madagascar day gecko (Phelsuma madagascariensis) is hatching from its egg b) A female American alligator (Alligator mis-sissippiensis) tending to her nest.

- Temperatures between 32.5 and 33° C result in male off- spring
- Temperatures around 32° C result in both male and female offspring.
- The female remains in the vicinity of the nest throughout development to protect the eggs from predation. She frees hatchlings from the nest in response to their high-pitched calls and picks them up in her mouth to transport them to water.
- She may scoop shallow pools for the young and remain with them for up to two years. Young feed on scraps of food the female drops when she feeds and on small vertebrates and invertebrates that they catch on their own.

Further phylogenetic considerations

- The archosaur and synapsid lineages of ancient reptiles diverged from ancient amniotes
- The archosaur lineage not only included the dinosaurs and gave rise to crocodilians, but it also gave rise to two groups of fliers. The pterosaurs ranged from sparrow size to animals with wingspans of 13 m. An elongation of the fourth finger supported their membranous wings, their sternum was adapted for the attachment of flight muscles, and their bones were hollow to lighten the skeleton for flight.
- These adaptations are paralleled by, though not identical to, adaptations in the birds—the descendants of the second lineage of flying archosaurs

- The synapsid lineage eventually gave rise to the mammals. The legs of synapsids were relatively long and held their bodies off the ground. Teeth and jaws were adapted for effective chewing and tearing. Additional bones were incorporated into the middle ear. These and other mammal-like characteristics developed between the Carboniferous and Triassic periods.

Lesson 55

Survey of Birds

Characteristics of birds

1. Fossils of the earliest birds clearly show reptilian features. Fossils of Archaeopteryx, Sinornis, and Eoalulavis give clues to the origin of flight in birds
2. Integumentary, skeletal, muscular, and gas exchange systems of birds are adapted for flight and endothermic temperature regulation
3. Large regions of the brain of birds are devoted to integrating sensory information
4. Birds' complex mating systems and behavioral patterns increase the chances of offspring survival
5. Migration and navigation allow birds to live, feed, and reproduce in environments
6. The body of the birds are covered with feathers
7. Birds have horny beaks and scaly feet
8. Birds are endothermic vertebrates
9. They are oviparous which means they lay eggs to develop their young ones
10. Birds are the species in the animal kingdom that walk on two legs. (they are Bipedal)
11. Like humans birds have four chambered heart
12. The shape of the body is streamlined which helps them to fly high in the sky
13. Birds have a very strong vision
14. Depending on the species the birds can be said to be carnivores or herbivores creature.

PHYLOGENETIC RELATIONSHIPS

- The major characteristics of this class are adaptations for flight, including appendages modified as wings, feathers, endothermy, a high metabolic rate, a vertebral column modified for flight, and bones lightened by numerous air spaces. In addition, modern birds possess a horny bill and lack teeth
- Birds are traditionally classified as members of the class Aves
- The similarities between birds and reptiles are so striking that birds are often referred to as “glorified reptiles.”
- Like the crocodilians, birds have descended from ancient archosaurs
- Modern birds are derived from the Saurischian lineage of dinosaurs that also included bipedal carnivorous dinosaurs like Tyrannosaurus. According to cladistic interpretations the birds can be thought of as modern dinosaurs.

ANCIENT BIRDS AND THE EVOLUTION OF FLIGHT

- In 1861, one of the most important vertebrate fossils was found in a slate quarry in Bavaria, Germany (figure 1). It was a fossil of a pigeon-sized animal that lived during the Jurassic period, about 150 million years ago. It had a long, reptilian tail and clawed fingers. The complete head of this specimen was not preserved, but imprints of feathers on the tail and on short, rounded wings were the main evidence that this was the fossil of an ancient bird. It was named **Archaeopteryx**
- Most zoologists consider Archaeopteryx to be the oldest bird yet discovered and very close to the main line of evolution between the reptiles and birds
- A second ancient bird (Sinornis), discovered in China, fits well with the view that Archaeopteryx was closely related to ancestral bird stocks. Sinornis fossils are 135 million years old—only 15 million years younger than Archaeopteryx



Figure 1: Archaeopteryx an ancient bird a) Archaeopteryx fossil b) Artists representation Some zoologists think that Archaeopteryx was a ground dweller rather than the tree dweller depicted here.

- A very recent discovery of a fossil bird from early Cretaceous deposits in Spain provides additional important information on the origin of flight. This bird, **Eoalulavis**, was found in 115- million-year-old deposits and had a wingspan of 17 cm (roughly the same as a goldfinch). This fossil showed that Eoalulavis had a wing structure called an alula. Alula is present in many modern birds that engage in slow, hovering flight. Its presence in this fossil indicates that complex flight mechanisms associated with slow, hovering, and highly maneuverable flight evolved at least 115 million years ago.

Lesson 56

DIVERSITY OF MODERN BIRDS

DIVERSITY OF MODERN BIRDS

- Archaeopteryx, Sinornis, and Eoalulavis provide the only evidence of the transition between reptiles and birds. Zoologists do not know, however, whether or not one of these birds is the direct ancestor of modern birds. A variety of fossil birds for the period between 100 million and 70 million years ago has been found. Some of these birds were large, flightless birds; others were adapted for swimming and diving; and some were fliers. Most, like Archaeopteryx, had reptilelike teeth. Most of the lineages that these fossils represent became extinct, along with the dinosaurs, at the end of the Mesozoic era.

- Some of the few birds that survived into the Tertiary period were the ancestors of modern, toothless birds. The phylogeny of modern birds is controversial. It is sufficient to say that adaptive radiation has resulted in about 9,100 species of living birds, which are divided into about 27 orders (table 21.1). (The number of orders varies, depending on the classification system used.) Characteristic behaviors, songs, anatomical differences, and ecological niches distinguish the orders.

TABLE 21.1
CLASSIFICATION OF THE BIRDS

Class Aves (a′ves) (L. *avis*, bird)*
Adaptations for flight include: foreappendages modified as feathered wings, endothermic, high metabolic rate, flexible neck, fused posterior vertebrae, and bones lightened by numerous air spaces. The skull is lightened by a reduction in bone and the presence of a horny bill that lacks teeth. The birds.

Order Sphenisciformes (sfe-nis″i-for′mez)
Heavy bodied; flightless, flipperlike wings for swimming; well insulated with fat. Penguins.

Order Struthioniformes (stroo″the-on-i-for′mez)
Large, flightless birds; wings with numerous fluffy plumes. Ostriches.

Order Rheiformes (re″i-for′mez)
Large, flightless birds; degenerate wings with soft, loose plumes. Rheas.

Order Casuariiformes (kaz″u-ar″ e-i-for′mez)
Reduced wings; coarse and hairlike plumage. Cassowaries, emus.

Order Gaviiformes (ga″ve-i-for′mez)
Strong, straight bill; diving adaptations include legs far back on body, bladelike tarsus, webbed feet; and heavy bones. Loons.

Order Podicipediformes (pod″i-si-ped″i-for′mez)
Short wings; soft and dense plumage; feet webbed with flattened nails. Grebes.

Order Procellariiformes (pro-sel-lar-e-i-for′mez)
Tubular nostrils, large nasal glands; long and narrow wings. Albatrosses, shearwaters, petrels.

Order Pelecaniformes (pel″e-can-i-for′mez)
Four toes joined in common web; nostrils rudimentary or absent; large gular sac. Pelicans, boobies, cormorants, anhingas, frigate-birds.

Order Ciconiiformes (si-ko″ne-i-for′mez)
Long neck, often folded in flight; long-legged waders. Herons, egrets, storks, wood ibises, flamingos.

Order Anseriformes (an″ser-i-for′mez)
South American screamers, ducks, geese, and swans; the latter three groups possess a wide, flat bill and an undercoat of dense down; webbed feet.

Order Falconiformes (fal″ko-ni-for′mez)
Strong, hooked beak; large wings; raptorial feet. Vultures, secretarybirds, hawks, eagles, ospreys, falcons.

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Order Galliformes (gal″li-for′mez)
Short beak; short, concave wings; strong feet and claws. Curassows, grouse, quail, pheasants, turkeys.
Order Gruiformes (gru″i-for′mez)
Order characteristics variable and not diagnostic. Marsh birds, including cranes, limpkins, rails, coots.
Order Charadriiformes (ka-rad″re-i-for′mez)
Order characteristics variable. Shorebirds, gulls, terns, auks.
Order Columbiformes (co-lum″bi-for′mez)
Dense feathers loosely set in skin; well-developed crop. Pigeons, doves, sandgrouse.
Order Psittaciformes (sit″ta-si-for′mez)
Maxilla hinged to skull; thick tongue; reversible fourth toe; usually brightly colored. Parrots, lories, macaws.
Order Cuculiformes (ku-koo″li-for′mez)
Reversible fourth toe; soft, tender skin; Plantaineaters, roadrunners, cuckoos.
Order Strigiformes (strij″i-for′mez)
Large head with fixed eyes directed forward; raptorial foot. Owls.
Order Caprimulgiformes (kap″ri-mul″ji-for′mez)
Owl-like head and plumage, but weak bill and feet; beak with wide gape; insectivorous. Whippoorwills, other goatsuckers.
Order Apodiformes (a-pod″i-for′mez)
Long wings; weak feet. Swifts, hummingbirds.
Order Coraciiformes (kor″ah-si″ah-for′mez)
Large head; large beak; metallic plumage. Kingfishers, todies, bee eaters, rollers.
Order Piciformes (pis″i-for′mez)
Usually long strong beak; strong legs and feet with fourth toe permanently reversed in woodpeckers. Woodpeckers, toucans, honeyguides, barbets.
Order Passeriformes (pas″er-i-for′mez)
Largest avian order; 69 families of perching birds; perching foot; variable external features. Swallows, larks, crows, titmice, nuthatches, and many others.

*Selected bird orders are described.

Lesson 57

EVOLUTIONARY PRESSURES: EXTERNAL STRUCTURE AND LOCOMOTION

Evolutionary pressure

- Virtually every body system of a bird shows some adaptation for flight. Endothermy, feathers, acute senses, long, flexible necks, and lightweight bones are a few of the many adaptations described in this section
- **External structure and locomotion**
- The covering of feathers on a bird is called the **plumage**. Feathers have two primary functions essential for flight. They form the flight surfaces that provide lift and aid steering, and they prevent excessive heat loss, permitting the endothermic maintenance of high metabolic rates. Feathers also have roles in courtship, incubation, and waterproofing.
- Feathers develop in a fashion similar to the epidermal scales of reptiles, and this similarity is one source of evidence that demonstrates the evolutionary ties between reptiles and birds (figure 21.3). Only the inner pulp of feathers contains dermal elements, such as blood vessels, which supply nutrients and pigments for the growing feather. As feathers mature, their blood supply is cut off, and the feathers become dead, keratinized, epidermal structures seated in epidermal invaginations of the skin called feather follicles

- The most obvious feathers are contour feathers, which cover the body, wings, and tail (Figure 21.4a, c). Contour feathers consist of a vane with its inner and outer webs, and a supportive shaft. Feather barbs branch off the shaft, and barbules branch off the barbs. Barbules of adjacent barbs overlap one another. The ends of barbules lock with hooklike hamuli (sing., hamulus). Interlocking barbs keep contour feathers firm and smooth. Other types of feathers include down feathers, which function as insulating feathers, and filoplume feathers (pinfeathers), which have sensory functions (figure 21.4b, d)
- Birds maintain a clean plumage to rid the feathers and skin of parasites. **Preening**, which is done by rubbing the bill over the feathers, keeps the feathers smooth, clean, and in place. Hamuli that become dislodged can be rehooked by running a feather through the bill
- Secretions from an oil gland at the base of the tail of many birds are spread over the feathers during preening to keep the plumage water repellent and supple. The secretions also lubricate the bill and legs to prevent chafing

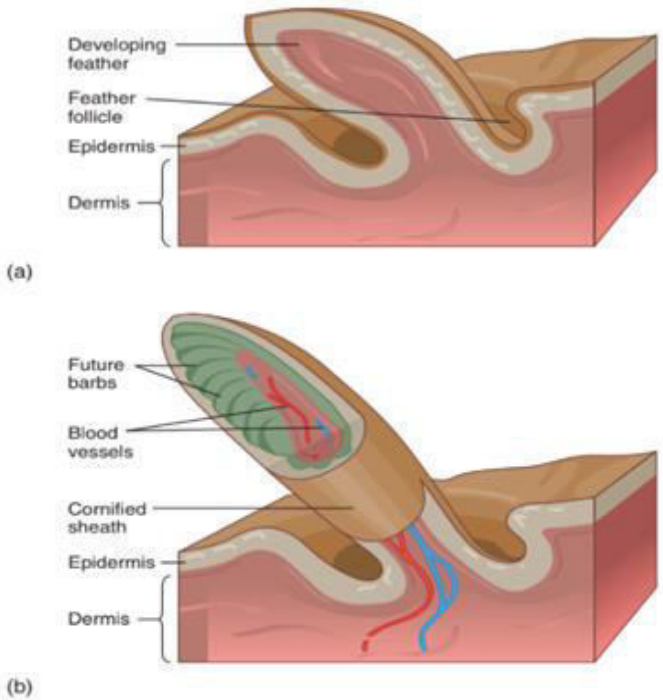


FIGURE 21.3

Formation of Bird Feathers during Embryonic Development. *(a)* Feathers form from epidermal evaginations *(b)* Blood flow to the feather supports initial feather development. Later in development, the blood supply to the feather is cut off and the feather becomes a dead, keratinized, epidermal structure seated in a feather follicle.

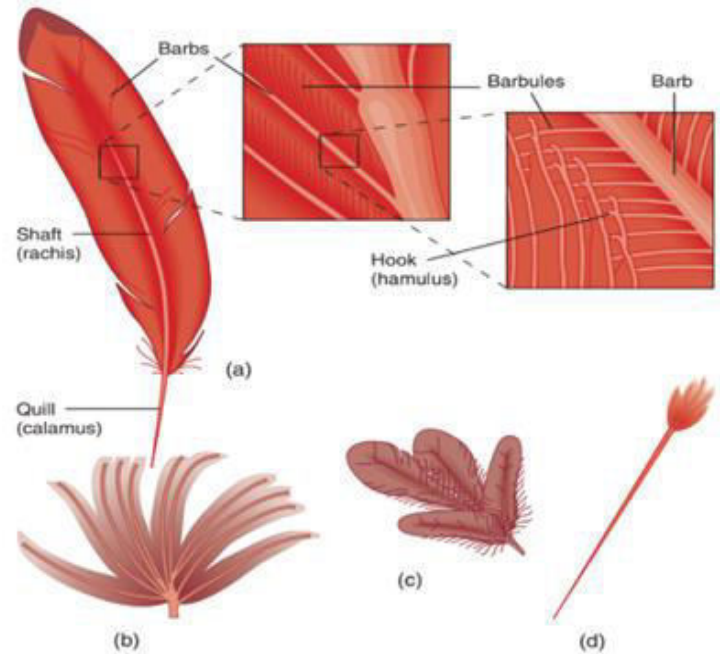


FIGURE 21.4

Anatomy of Selected Feather Types. *(a)* Anatomy of a contour feather, showing enlargements of barbs and barbules. *(b)* Down feather. Various types of down feathers provide insulation for adult and immature birds. *(c)* Contour feather. *(d)* Filoplume. Filoplume feathers are usually covered by contour feathers and are associated with nerve endings in the skin, thus serving as sensory structures.

- Anting is a maintenance behavior common to many songbirds and involves picking up ants in the bill and rubbing them over the feathers. The formic acid that ants secrete is apparently toxic to feather mites.
- Feather pigments deposited during feather formation produce most colors in a bird's plumage
- Other colors, termed structural colors, arise from irregularities on the surface of the feather that diffract white light. For example, blue feathers are never blue because of the presence of blue pigment
- A porous, non-pigmented outer layer on a barb reflects blue wavelengths of light

- The other wavelengths pass into the barb and are absorbed by the dark pigment melanin. Iridescence results from the interference of light waves caused by a flattening and twisting of barbules
- Color patterns are involved in cryptic coloration, species and sex recognition, and sexual attraction.
- Mature feathers receive constant wear; thus, all birds periodically shed and replace their feathers in a process called **molting**
- The timing of molt periods varies in different taxa.

Lesson 58

The Skeleton

The Skeleton

- The bones of most birds are lightweight yet strong
- Some bones, such as the humerus (forearm bone), have large air spaces and internal strutting (reinforcing bony bars), which increase strength (figure 21.5c)
- Birds also have a reduced number of skull bones, and a lighter, keratinized sheath called a **bill** replaces the teeth
- The appendages involved in flight cannot manipulate nesting materials or feed young. The bill and very flexible neck and feet make these activities possible
- The cervical vertebrae have saddle-shaped articular surfaces that permit great freedom of movement. In addition, the first cervical vertebra (the atlas) has a single point of articulation with the skull (the occipital condyle), which permits a high degree of rotational movement between the skull and the neck. (The single occipital condyle is another characteristic that birds share with reptiles.) This flexibility allows the bill and neck to function as a fifth appendage.

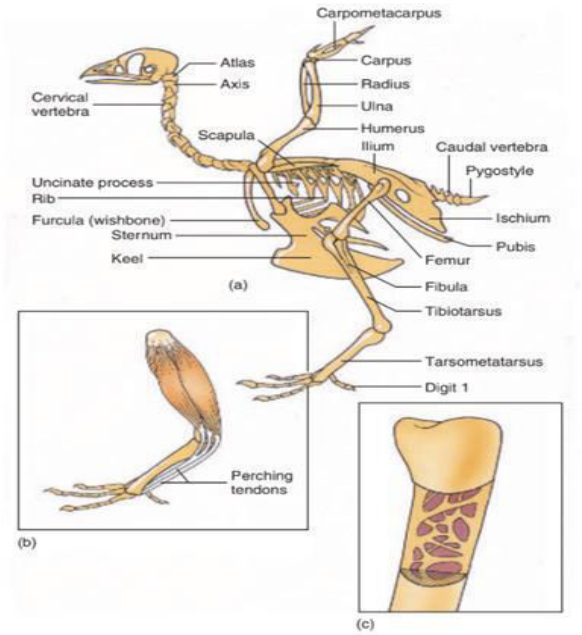


FIGURE 21.5 Bird Skeleton. (a) Skeleton of a pigeon. (b) Perching tendons run from the toes across the back of the ankle joint, which cause the foot to grip a perch. (c) Internal structure of the humerus. Note the air spaces in this pneumatic bone.

- The pelvic girdle, vertebral column, and ribs are strengthened for flight.
- The thoracic region of the vertebral column contains ribs, which attach to thoracic vertebrae
- Most ribs have posteriorly directed uncinate processes that overlap the next rib to strengthen the rib cage (figure 21.5a).
- Posterior to the thoracic region is the lumbar region. The synsacrum forms by the fusion of the posterior thoracic vertebrae, all the lumbar and sacral vertebrae, and the anterior caudal vertebrae. Fusion of these bones helps to maintain the proper flight posture and supports the hind appendages during landing, hopping, and walking
- The posterior caudal vertebrae are fused into a **pygostyle**, which helps support the tail feathers that are important in steering.

- The sternum of most birds bears a large, median keel for the attachment of flight muscles. (Exceptions to this include some flightless birds, such as ostriches.)
- The keel attaches firmly to the rest of the axial skeleton by the ribs
- Paired clavicles fuse medially and ventrally into a furcula (wishbone)
- The appendages of birds have also been modified. Some bones of the front appendages have been lost or fused, and they are points of attachment for flight feathers. The rear appendages are used for hopping, walking, running, and perching
- Perching tendons run from the toes across the back of the ankle joint to muscles of the lower leg. When the ankle joint is flexed, as in landing on a perch, tension on the perching tendons increases, and the foot grips the perch (figure 21.5b). This automatic grasp helps a bird perch even while sleeping.
- The muscles of the lower leg can increase the tension on these tendons, for example, when an eagle grasps a fish in its talons.

- The largest, strongest muscles of most birds are the flight muscles
- They attach to the sternum and clavicles and run to the humerus
- The muscles of most birds are adapted physiologically for flight
- Flight muscles must contract quickly and fatigue very slowly. These muscles have many mitochondria and produce large quantities of ATP to provide the energy required for flight, especially long-distance migrations
- Domestic fowl have been selectively bred for massive amounts of muscle (“white meat”) that humans like as food but that is poorly adapted for flight because it contains fibers that can rapidly contract but feature few mitochondria and poor vascularization.

Lesson 59

Flight

Flight

- The wings of birds are adapted for different kinds of flight. However, regardless of whether a bird soars, glides, or has a rapid flapping flight, the mechanics of staying aloft are similar
- Bird wings form an airfoil. The anterior margin of the wing is thicker than the posterior margin
- The upper surface of the wing is slightly convex, and the lower surface is flat or slightly concave
- Air passing over the wing travels farther and faster than air passing under the wing, decreasing air pressure on the upper surface of the wing and creating lift (figure 21.6a)
- The lift the wings create must overcome the bird's weight, and the forces that propel the bird forward must overcome the drag that the bird moving through the air creates
- Increasing the angle that the leading edge of the wing makes with the oncoming air (the angle of attack) increases lift. As the angle of attack increases, however, the flow of air over the upper surface becomes turbulent, reducing lift (figure 21.6b).

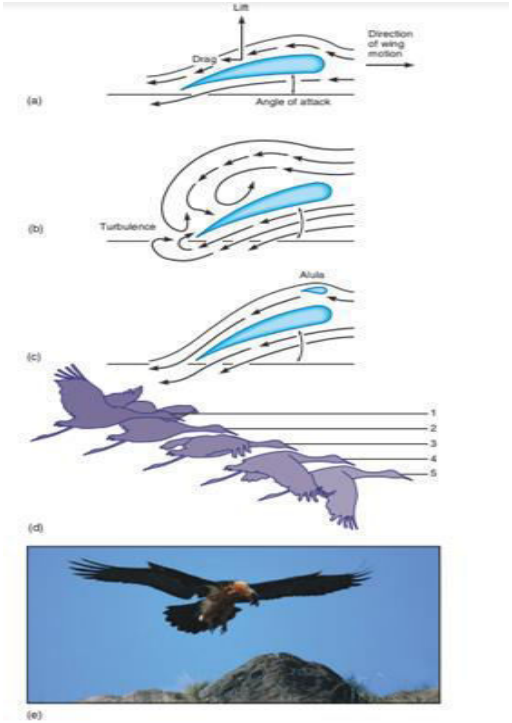


Figure 21.6: Mechanism of bird flight (a) A bird's wing acts as an airfoil. Air pass over the top of the wing travels farther and faster than air passing under the wing, creating lift. (b) Increasing the angle of attack increases lift but also increases turbulence. (c) The alula reduces turbulence. (d) Wing orientation during a downstroke. (e) Note the alula on the wings of this bearded vulture (Gypaetus barbatus). This species was widely distributed in the Alps. Excessive hunting nearly eliminated this species around 1900. It is now making a comeback, inhabiting large territories above the Alps forest line.

- Turbulence is reduced if air can flow rapidly through slots at the leading edge of the wing
- Slotting the feathers at the wing tips and the presence of an alula on the anterior margin of the wing reduce turbulence
- The alula is a group of small feathers that bones of the medial digit support
- During takeoff, landing, and hovering flight, the angle of attack increases, and the alula is elevated (figure 21.6c,e)
- During soaring and fast flight, the angle of attack decreases, and slotting is reduced. The distal part of the wing generates most of the propulsive force of flight. Because it is farther from the shoulder joint, the distal part of the wing moves farther and faster than the proximal part of the wing. During the downstroke (the powerstroke), the leading edge of the distal part of the wing is oriented slightly downward and creates a thrust somewhat analogous to the thrust that an airplane propeller creates (figure 21.6d).
- During the upstroke (the recovery stroke), the distal part of the wing is oriented upward to decrease resistance.

- Feathers on a wing overlap so that, on the downstroke, air presses the feathers at the wing margins together, allowing little air to pass between them, enhancing both lift and propulsive forces
- Feathers part slightly on the upstroke, allowing air to pass between them, which reduces resistance during this recovery stroke
- The tail of a bird serves a variety of balancing, steering, and braking functions during flight. It also enhances lift that the wings produce during low-speed flight
- During horizontal flight, spreading the tail feathers increases lift at the rear of the bird and causes the head to dip for descent
- Closing the tail feathers has the opposite effect. Tilting the tail sideways turns the bird. When a bird lands, its tail deflects downward, serving as an air brake.

- Different birds, or the same bird at different times, use different kinds of flight
- During **gliding flight**, the wing is stationary, and a bird loses altitude. Waterfowl coming in for a landing use gliding flight
- **Flapping flight** generates the power for flight and is the most common type of flying. Many variations in wing shape and flapping patterns result in species-specific speed and maneuverability
- **Soaring flight** allows some birds to remain airborne with little energy expenditure. During soaring, wings are essentially stationary, and the bird utilizes updrafts and air currents to gain altitude. Hawks, vultures, and other soaring birds are frequently observed circling along mountain valleys, soaring down-wind to pick up speed and then turning upwind to gain altitude.
- As the bird slows and begins to lose altitude, it turns downwind again. The wings of many soarers are wide and slotted to provide maximum maneuverability at relatively low speeds. Oceanic soarers, such as albatrosses and frigate birds, have long, narrow wings that provide maximum lift at high speeds, but they compromise maneuverability and ease of takeoff and landing.
- Hummingbirds perform **hovering flight**. They hover in still air by fanning their wings back and forth (50 to 80 beats per second) to remain suspended in front of a flower or feeding station

Lesson 60

Nutrition and digestive system

Nutrition and digestive system

- Most birds have ravenous appetites! This appetite supports a high metabolic rate that makes endothermy and flight possible. For example, hummingbirds feed almost constantly during the day. In spite of high rates of food consumption, they often cannot sustain their rapid metabolism overnight, and they may become torpid, with reduced body temperature and respiratory rate, until they can feed again in the morning
- Bird bills and tongues are modified for a variety of feeding habits and food sources (figures 21.7 and 21.8)
- For example, a woodpecker's tongue is barbed for extracting grubs from the bark of trees (see figure 27.6d)
- Sapsuckers excavate holes in trees and use a brush like tongue for licking the sap that accumulates in these holes
- The tongues of hummingbirds and other nectar feeders roll into a tube for extracting nectar from flowers



Figure 21.8: (a) The bill of a bald eagle (Haliacetus leucocephalus) is specialized for tearing prey. (b) The thick, powerful bill of this cardinal (Cardinalis cardinalis) cracks tough seeds. (c) The bill of a flamingo (Phoenicopterus ruber) strains food from the water in a head-down feeding posture. Large bristles fringe the upper and lower mandibles. As water is sucked into the bill, larger particles are filtered and left outside. Inside the bill, tiny inner bristles filter smaller algae and animals. The tongue removes food from the bristles.

- In many birds, a diverticulum of the esophagus, called the **crop**, is a storage structure that allows birds to quickly ingest large quantities of locally abundant food. They can then seek safety while digesting their meal
- The crop of pigeons produces "**pigeon's milk**," a cheesy secretion formed by the proliferation and sloughing of cells lining the crop. Young pigeons (squabs) feed on pigeon's milk until they are able to eat grain
- Cedar waxwings, vultures, and birds of prey use their esophagus for similar storage functions
- Crops are less well developed in insect-eating birds because insectivorous birds feed throughout the day on sparsely distributed food

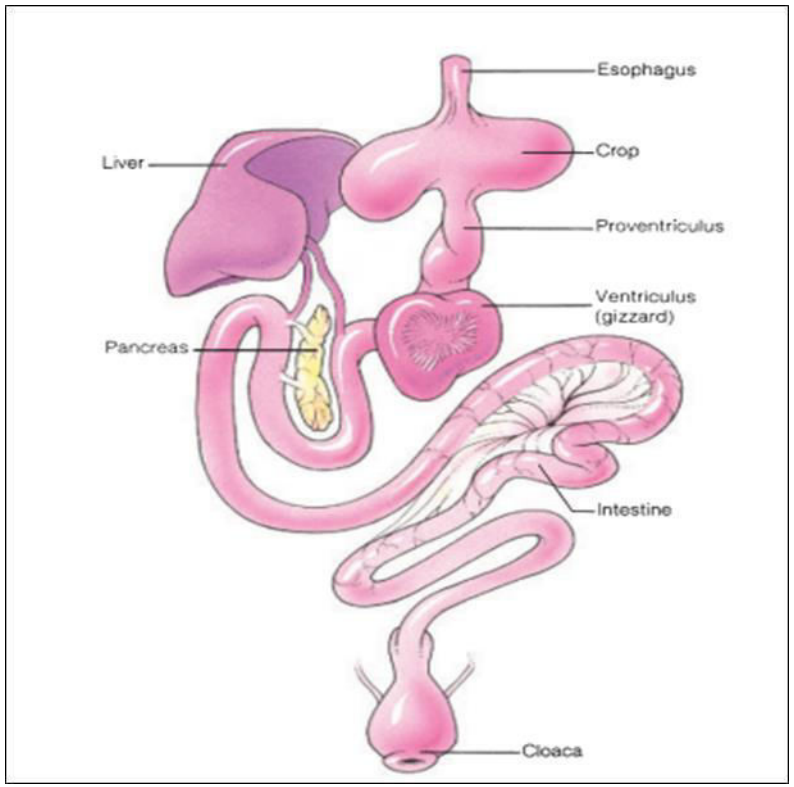


Figure 21.9: Digestive System of a Pigeon. Birds have high metabolic rates that require a nearly constant nutrient supply

- The stomach of birds is modified into two regions. **The proventriculus** secretes gastric juices that initiate digestion (figure 21.9). The **ventriculus** (gizzard) has muscular walls to abrade and crush seeds or other hard materials
- Birds may swallow sand and other abrasives to aid digestion
- The bulk of enzymatic digestion and absorption occurs in the small intestine, aided by secretions from the pancreas and liver
- Paired ceca may be located at the union of the large and small intestine. These blind-ending sacs contain bacteria that aid in cellulose digestion.
- Birds usually eliminate undigested food through the cloaca; however, owls form pellets of bone, fur, and feathers that are ejected from the ventriculus through the mouth. Owl pellets accumulate in and around owl nests and are useful in studying their food habits.

Feeding habits of birds

- Birds are often grouped by their feeding habits
- Robins, for example, feed largely on worms and other invertebrates when these foods are available. In the winter, however, robins may feed on berries
- In some of their feeding habits, birds directly conflict with human interests. Bird damage to orchard and grain crops is tallied in the millions of dollars each year.
- Flocking and roosting habits of some birds, such as European starlings and redwing blackbirds, concentrate millions of birds in local habitats, where they devastate fields of grain
- Birds of prey have minimal impact on populations of poultry and game birds, and on commercial fisheries. Unfortunately, the mistaken impression that they are responsible for significant losses has led humans to poison and shoot them.

Lesson 61

CIRCULATION AND GAS EXCHANGE

Circulation

- The circulatory system of birds is similar to that of reptiles, except that the heart has completely separated atria and ventricles, resulting in separate pulmonary and systemic circuits
- This separation prevents any mixing of highly oxygenated blood with less oxygenated blood
- In vertebrate evolution, the sinus venosus has gradually decreased in size. It is a separate chamber in fishes, amphibians, and turtles and receives blood from the venous system. In other reptiles, it is a group of cells in the right atrium that serves as the pacemaker for the heart. In birds, the sinus venosus also persists only as a patch of pacemaker tissue in the right atrium
- The bird heart is relatively large (up to 2.4% of total body weight), and it beats rapidly. Rates in excess of 1,000 beats per minute have been recorded for hummingbirds under stress
- Larger birds have relatively smaller hearts and slower heart rates. The heart rate of an ostrich, for example, varies between 38 and 176 beats per minute
- A large heart, rapid heart rate, and complete separation of highly oxygenated from less oxygenated blood are important adaptations for delivering the large quantities of blood required for endothermy and flight.

Gas exchange

- The respiratory system of birds is extremely complex and efficient. It consists of external nares, which lead to nasal passageways and the pharynx
- Bone and cartilage support the trachea
- A special voice box, called the **syrinx**, is located where the trachea divides into bronchi. The muscles of the syrinx and bronchi, as well as characteristics of the trachea, produce bird vocalizations
- The bronchi lead to a complex system of air sacs that occupy much of the body and extend into some of the bones of the skeletal system (figure 21.10a)
- The air sacs and bronchi connect to the lungs. The lungs of birds are made of small air tubes called **parabronchi**

- Air capillaries about 10 mm in diameter branch from the parabronchi and are associated with capillary beds for gas exchange
- Inspiration and expiration result from increasing and decreasing the volume of the thorax and from alternate expansion and compression of air sacs during flight and other activities
- During breathing, the movement of the sternum and the posterior ribs compresses the thoracic air sacs.
- During inspiration, air moves into the abdominal air sacs. At the same time, air already in the lungs moves through parabronchi into the thoracic air sacs (figure 21.10b, **cycle 1**).
- During expiration, the air in the thoracic air sacs moves out of the respiratory system, and the air in the abdominal air sacs moves into parabronchi (figure 21.10c)
- At the next inspiration (figure 21.10b, cycle 2), the air moves into the thoracic air sacs, and it is expelled during the next expiration. It takes two ventilatory cycles to move a particular volume of air through the respiratory system of a bird.

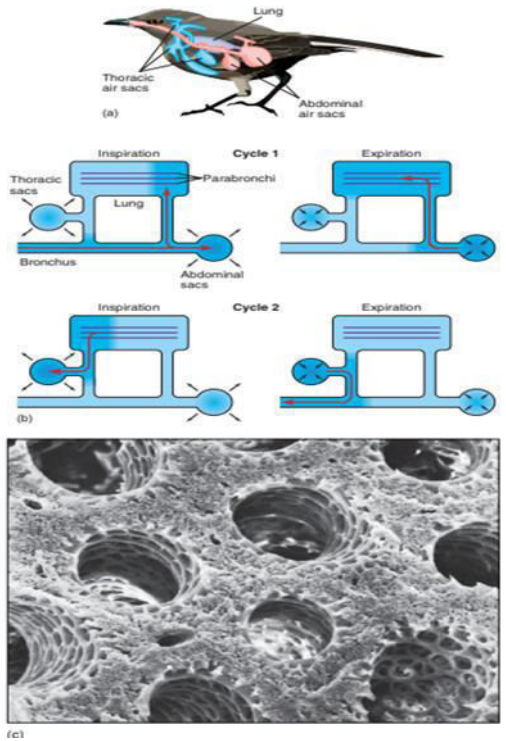


Figure 21.10 Respiratory System of a Bird. (a) Air sacs branch from the respiratory tree. (b) Air flow during inspiration and expiration. Air flows through the parabronchi during both inspiration and expiration. The shading represents the movement through the lungs of one inspiration. (c) Scanning electron micrograph of parabronchi.

- Because of high metabolic rates associated with flight, birds have a greater rate of oxygen consumption than any other vertebrate
- When other tetrapods inspire and expire, air passes into and out of respiratory passageways in a simple back-and-forth cycle. Ventilation is interrupted during expiration, and much “dead air” (air not forced out during expiration) remains in the lungs. Because of the unique system of air sacs and parabronchi, bird lungs have a nearly continuous movement of oxygen-rich air over respiratory surfaces during both inspiration and expiration. The quantity of “dead air” in the lungs is sharply reduced, compared with other vertebrates
- This avian system of gas exchange is more efficient than that of any other tetrapod. In addition to supporting high metabolic rates, this efficient gas exchange system probably also explains how birds can live and fly at high altitudes, where oxygen tensions are low. During their migrations, bar-headed geese fly over the peaks of the Himalayas at altitudes of 9,200 m. A human mountain climber begins to feel the symptoms of altitude sickness and must carry auxiliary oxygen supplies above 2,500 m.

Lesson 62

Temperature regulation

Thermoregulation

- Birds maintain body temperatures between 38 and 45° C
- Lethal extremes are lower than 32 and higher than 47° C
- On a cold day, a resting bird fluffs its feathers to increase their insulating properties, as well as the dead air space within them. It also tucks its bill into its feathers to reduce heat loss from the respiratory tract
- The most exposed parts of a bird are the feet and tarsi, which have neither fleshy muscles nor a rich blood supply. Temperatures in these extremities are allowed to drop near freezing to prevent heat loss
- Countercurrent heat exchange between the warm blood flowing to the legs and feet, and the cooler blood flowing to the body core from the legs and feet, prevents excessive heat loss at the feet
- Heat is returned to the body core before it goes to the extremities and is lost to the environment. Shivering also generates heat in extreme cold. Increases in metabolism during winter months require additional food.

- Some birds become torpid and allow their body temperatures to drop on cool nights. For example, whippoorwills allow their body temperatures to drop from about 40° C to near 16° C, and respiratory rates become very slow
- Muscular activity during flight produces large quantities of heat, which birds can dissipate by panting
- Evaporative heat loss from the floor of the mouth is enhanced by fluttering vascular membranes of this region.

Lesson 63

Nervous and sensory function

Nervous and sensory function

- The forebrain of birds is much larger than that of reptiles due to the enlargement of the cerebral hemispheres, including a region of gray matter, the corpus striatum
- The **corpus striatum** functions in visual learning, feeding, courtship, and nesting
- A **pineal body** on the roof of the forebrain appears to stimulate ovarian development and to regulate other functions influenced by light and dark periods
- The **optic tectum** (the roof of the mid brain), along with the corpus striatum, plays an important role in integrating sensory functions
- The midbrain also receives sensory input from the eyes
- As in reptiles, the hindbrain includes the cerebellum and the medulla oblongata, which coordinate motor activities and regulate heart and respiratory rates, respectively.

- Vision is an important sense for most birds
- The structures of bird eyes are similar to those of other vertebrates, but bird eyes are much larger relative to body size than those of other vertebrates
- The eyes are usually somewhat flattened in an antero-posterior direction; however, the eyes of birds of prey protrude anteriorly because of a bulging cornea
- Birds have a unique double-focusing mechanism
- Padlike structures (similar to those of reptiles) control the curvature of the lens, and ciliary muscles change the curvature of the cornea
- Double, nearly instantaneous focusing allows an osprey or other bird of prey to remain focused on a fish throughout a brief, but breathtakingly fast, descent
- The retina of a bird's eye is thick and contains both rods and cones. Rods are active under low light intensities, and cones are active under high light intensities
- Cones are especially concentrated (1,000,000/mm2) at a focal point called the **fovea**.

- Unlike other vertebrates, some birds have two foveae per eye
- The one at the center of the retina is sometimes called the “**search fovea**” because it gives the bird a wide angle of monocular vision
- The other fovea is at the posterior margin of the retina. It functions with the posterior fovea of the other eye to allow binocular vision. The posterior fovea is called the **“pursuit fovea”** because binocular vision produces depth perception, which is necessary to capture prey.
- Other birds use the “search fovea” to observe the landscape below them during flight and the “pursuit” fovea when depth perception is needed, as in landing on a branch of a tree

- The position of the eyes on the head also influences the degree of binocular vision (figure 21.12)
- Pigeons have eyes located well back on the sides of their head, giving them a nearly 360° monocular field, but a narrow binocular field. They do not have to pursue their food (grain), and a wide monocular field of view helps them stay alert to predators while feeding on the ground
- Hawks and owls have eyes farther forward on the head. This increases their binocular field of view and correspondingly decreases their monocular field of view.

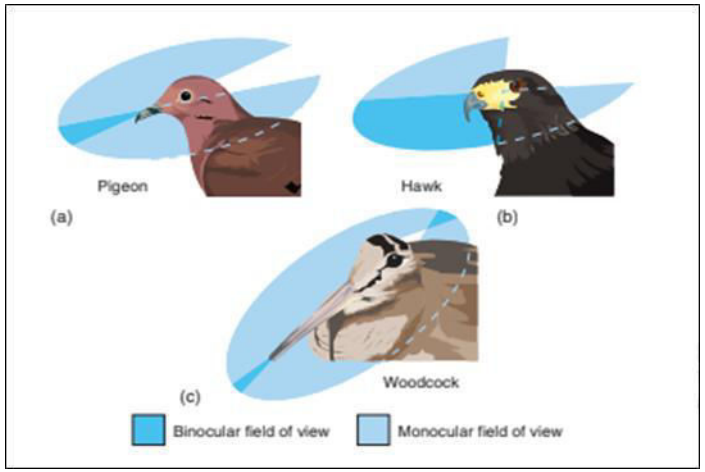


Figure 21.12: Avian Vision. The fields of view of (a) a pigeon, (b) a hawk, and (c) a woodcock. Woodcocks have eyes located far posteriorly and have a narrow field of binocular vision in front and behind. They can focus on predators circling above them while probing mud with their long beaks.

- Like reptiles, birds have a nictitating membrane that is drawn over the surface of the eye to cleanse and protect the eye
- Olfaction apparently plays a minor role in the lives of most birds.
- External nares open near the base of the beak, but the olfactory epithelium is poorly developed. Exceptions include turkey vultures, which locate their dead and dying prey largely by smell.
- In contrast, most birds have well-developed hearing. Loose, delicate feathers called **auriculars** cover the external ear opening
- Middle and inner ear structures are similar to those of reptiles
- The sensitivity of the avian ear (**100 to 15,000 Hz**) is similar to that of the human ear (16 to 20,000 Hz)

Lesson 64

EXCRETION AND OSMOREGULATION, REPRODUCTION AND DEVELOPMENT

Excretion and osmoregulation

- Birds and reptiles face essentially identical excretory and osmoregulatory demands
- Like reptiles, birds excrete **uric acid**, which is temporarily stored in the cloaca
- Water is also reabsorbed in the cloaca
- As with reptiles, the excretion of uric acid conserves water and promotes embryo development in terrestrial environments
- In addition, some birds have **supraorbital salt glands** that drain excess sodium chloride through the nasal openings to the outside of the body. These are especially important in marine birds that drink seawater and feed on invertebrates containing large quantities of salt in their tissues
- Salt glands can secrete salt in a solution that is two to three times more concentrated than other body fluids. Salt glands, therefore, compensate for the kidney's in ability to concentrate salts in the urine.

Reproduction and development

- Sexual activities of birds have been observed more closely than those of any other group of animals. These activities include establishing territories, finding mates, constructing nests, incubating eggs, and feeding young
- All birds are oviparous
- Gonads are in the dorsal abdominal region, next to the kidneys
- Testes are paired, and coiled tubules (vasa deferentia) conduct sperm to the cloaca
- An enlargement of the vasa deferentia, the **seminal vesicle**, is a site for temporary storage and maturation of sperm prior to mating. Testes enlarge during the breeding season
- Except for certain waterfowl and ostriches, birds have no intromittent organ, and sperm are transferred by cloacal contact when the male briefly mounts the female

- In females, two ovaries form during development, but usually only the left ovary fully develops (figure 21.13)
- A large, funnel-shaped opening (the ostium) of the oviduct envelops the ovary and receives eggs after ovulation
- The egg is fertilized in the upper portions of the oviduct, and albumen that glandular regions of the oviduct wall secrete gradually surrounds the zygote as it completes its passage
- A shell gland in the lower region of the oviduct adds a shell
- The oviduct opens into the cloaca.

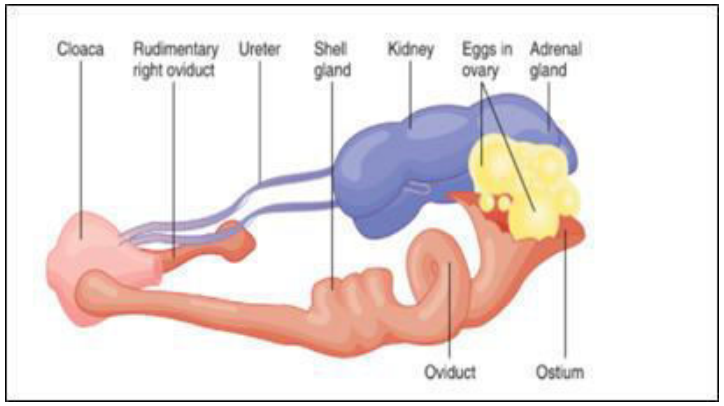


Figure 21.13: Urogenital System of a Female Pigeon. The right ovary and oviduct are rudimentary in most female birds.

- Many birds establish territories prior to mating
- Although size and function vary greatly among species, territories generally allow birds to mate without interference. They provide nest locations and sometimes food resources for adults and offspring
- Breeding birds defend their territories and expel intruders of the same sex and species
- Threats are common, but actual fighting is minimal. Mating may follow the attraction of a mate to a territory. For example, male woodpeckers drum on trees to attract females.
- Mating occurs when a mate’s call or posture signals readiness. It happens quickly but repeatedly to assure fertilization of all the eggs that will be laid.

- Most birds are monogamous. A single male pairs with a single female during the breeding season. Some birds (swans, geese, eagles) pair for life
- Frequent mating apparently strengthens the pair bonds that develop.
- Monogamy is common when resources are widely and evenly distributed, and one bird cannot control access to resources.
- Monogamy is also advantageous because both parents usually participate in nest building and care of the young
- One parent incubates and protects the eggs or chicks while the other searches for food.

- Some birds are polygynous. Males mate with more than one female, and the females care for the eggs and chicks
- Polygyny tends to occur in species whose young are less dependent at hatching and in situations where patchy resource distribution may attract many females to a relatively small breeding area
- Prairie chickens are polygynous, and males display in groups called **leks**. In prairie chicken leks, the males in the center positions are preferred and attract the majority of females (figure 21.14)
- A few bird species are polyandrous, and the females mate with more than one male. For example, female spotted sandpipers are larger than males, and they establish and defend their territories from other females. They lay eggs for each male that is attracted to and builds a nest in their territory. If a male loses his eggs to a predator, the female replaces them



Figure 21.14: Courtship Displays. A male greater prairie chicken (Tympanuchus cupido) displaying in a lek.

- Polyandry results in the production of more eggs than in monogamous matings
- It is thought to be advantageous when food is plentiful but, because of predation or other threats, the chances of successfully rearing young are low
- Nest construction usually begins after pair formation. The female usually initiates this instinctive behavior.

Lesson no 65

Nesting activities

Nesting Activities

- The nesting behavior of birds is often species specific
- Some birds choose nest sites away from other members of their species, and other birds nest in large flocks
- The group of eggs laid and chicks produced by a female is called a **clutch**. Clutch size usually varies
- Most birds incubate their eggs, and some birds have a featherless, vascularized incubation or brood patch that helps keep the eggs at temperatures between 33 and 37°C
- Birds turn the eggs to prevent egg membranes from adhering in the egg and deforming the embryo
- Adults of some species sprinkle the eggs with water to cool and humidify them.
- The incubation period lasts between 10 and 80 days and correlates with egg size and degree of development at hatching
- One or two days before hatching, the young bird penetrates an air sac at the blunt end of its egg, inflates its lungs, and begins breathing
- Hatching occurs as the young bird pecks the shell with a keratinized egg tooth on the tip of the upper jaw and struggles to free itself.

- Some birds are helpless at hatching; others are more independent. Those that are entirely dependent on their parents are said to be **altricial** and they are often naked at hatching (figure 21.15a)
- Altricial young must be brooded constantly at first because endothermy is not developed. They grow rapidly, and when they leave the nest, they are nearly as large as their parents. For example, American robins weigh 4 to 6 g at hatching and leave the nest 13 days later weighing 57 g
- Precocial young are alert and lively at hatching (figure 21.15b). They are usually covered with down and can walk, run, swim, and feed themselves—although one parent is usually present to lead the young to food and shelter



FIGURE 21.15 Altricial and Precocial Chicks. (a) An American robin (Turdus migratorius) feeding nestlings. Robins have altricial chicks that are helpless at hatching. (b) Killdeer (Charadrius vociferus) have precocial chicks that are down covered and can move about.

Lesson 66

MIGRATION AND NAVIGATION

Migration

- Migration (as used here) refers to periodic round trips between breeding and nonbreeding areas
- Most migrations are annual, with nesting areas in northern regions and wintering grounds in the south
- Migrations occasionally involve east/west movements or altitude changes
- Migration allows birds to avoid climatic extremes and to secure adequate food, shelter, and space throughout the year
- Birds migrate in response to species-specific physiological conditions
- Innate (genetic) clocks and environmental factors influence preparation for migration.

- The photoperiod is an important migratory cue for many birds, particularly for birds in temperate zones
- The changing photoperiod initiates seasonal changes in gonadal development that often serve as migratory stimuli
- Increasing day length in the spring promotes gonadal development, and decreasing day length in the fall initiates gonadal regression
- In many birds, the changing photoperiod also appears to promote fat deposition, which acts as an energy reserve for migration
- The anterior lobe of the pituitary gland and the pineal body have been implicated in mediating photoperiod responses

- The mechanics of migration are species specific
- Some long distance migrants may store fat equal to 50% of their body weight and make nonstop journeys
- Other species that take a more leisurely approach to migration begin their journeys early and stop frequently to feed and rest
- In clear weather, many birds fly at altitudes greater than 1,000 m, which reduces the likelihood of hitting tall obstacles

Navigation

- Birds use two forms of navigation
- **Route-based navigation** involves keeping track of landmarks (visual or auditory) on an outward journey so that those landmarks can be used in a reverse sequence on the return trip
- **Location-based navigation** is based on establishing the direction of the destination from information available at the journey's site of origin. It involves the use of sun compasses, other celestial cues, and/or the earth's magnetic field
- Birds' lenses are transparent to ultraviolet light, and their photoreceptors respond to it, allowing them to orient using the sun, even on cloudy days. This orientation cue is called a **sun compass**
- Celestial cues other than the sun can be used to navigate. Humans recognize that in the Northern Hemisphere, the North Star lines up with the axis of rotation of the earth. The angle between the North Star and the horizon decreases as you move toward the equator. Birds may use similar information to determine latitude

Lesson 67

Characteristics of mammals

Characteristics of Mammals

- Any member of the group of vertebrate animals in which the young are nourished with milk from special mammary glands of the mother
- In addition to these characteristic milk glands, mammals are distinguished by several other unique features. Hair is a typical mammalian feature, although in many whales it has disappeared except in the fetal stage
- A chain of three tiny bones transmits sound waves across the middle ear
- A muscular diaphragm separates the heart and the lungs from the abdominal cavity. Only the left aortic arch persists. (In birds the right aortic arch persists; in reptiles, amphibians, and fishes both arches are retained.)
- Mature red blood cells (erythrocytes) in all mammals lack a nucleus; all other vertebrates have nucleated red blood cells

- Mammalian characteristics evolved gradually over a 200-million-year period in the synapsid lineage.
- Two subclasses of mammals evolved during the Mesozoic era—the Prototheria and the Theria. Modern mammals include monotremes, marsupial mammals, and placental mammals.
- The skin of mammals is thick and protective and has an insulating covering of hair.
- Adaptations of teeth and the digestive tract allow mammals to exploit a wide variety of food resources.
- Efficient systems for circulation and gas exchange support the high metabolic rate associated with mammalian endothermy

- The brain of mammals has an expanded cerebral cortex that processes information from various sensory structures.
- Metanephric kidneys permit mammals to excrete urea without excessive water loss.
- Complex behavior patterns enhance mammalian survival
- Most mammals are viviparous and have reproductive cycles that help ensure internal fertilization and successful development.

Lesson 68

Mammals: Evolutionary Perspective

- The beginning of the Tertiary period, about 70 million years ago, was the start of the “age of mammals.
- It coincided with the extinction of many reptilian lineages, which led to the adaptive radiation of the mammals. Tracing the roots of the mammals, however, requires returning to the Carboniferous period, when the synapsid lineage diverged from other amniote lineages
- Mammalian characteristics evolved gradually over a period of 200 million years
- The early synapsids were the pelycosaurs. Some were herbivores; others showed skeletal adaptations that reflect increased effectiveness as predators

- The anterior teeth of the upper jaw were large and were separated from the posterior teeth by a gap that accommodated the enlarged anterior teeth of the lower jaw when the jaw closed
- The palate was arched, which strengthened the upper jaw and allowed air to pass over prey held in the mouth
- The legs were longer and slimmer than those of earlier amniotes
- By the middle of the Permian period, other successful mammal-like reptiles had arisen from the pelycosaurs. They were a diverse group known as the **therapsids**
- Some were predators, and others were herbivores.

- The teeth of the herbivorous therapsids were also mammal-like.
- Some had a large space, called the diastema, separating the anterior and posterior teeth
- The posterior teeth had ridges (cusps) and cutting edges that were probably used to shred plant material
- Unlike other reptiles, therapsids held hindlimbs directly beneath the body and moved them parallel to the long axis of the body.

- Changes in the size and shape of the ribs suggest the separation of the trunk into thoracic and abdominal regions and a breathing mechanism similar to that of mammals
- The last therapsids were a group called the cynodonts (figure 22.2b)
- Some of these were as large as a big dog, but most were small and little different from the earliest mammals
- The first mammals were small (less than 10 cm long) with delicate skeletons.

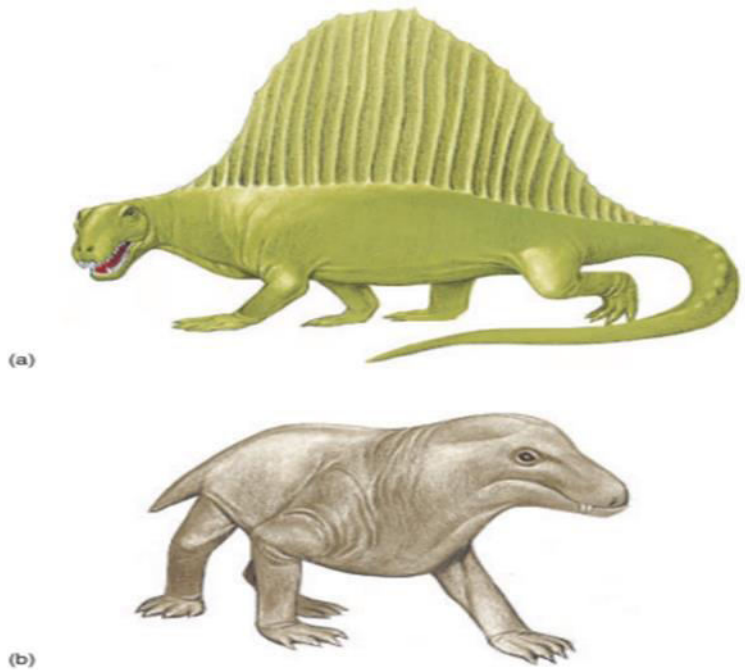


Figure 22.2: Members of the Subclass Synapsidia. (a) Dimetrodon was a 3 m long pelycosaur. It probably fed on other reptiles and amphibians. The large sail may have been a recognition signal and a thermoregulatory device. (b) Cynognathus was a mammal-like reptile that probably foraged for small animals, much like a badger does today. The badger-sized animal was a cynodont within the order Therapsida, the stock from which mammals arose during the mid-Triassic period.

Lesson 69

Diversity of Mammals

- Hair, mammary glands, specialized teeth, three middle-ear ossicles, and other characteristics listed in table 22.1
- The subclass Theria diverged into three infraclasses by the late Cretaceous period. The infraclass **Ornithodelphia** contains the monotremes. This name refers to the fact that monotremes, unlike other mammals, possess a **cloaca**.
- Unlike other mammals, monotremes are oviparous (figure 22.4a,b).
- The six species of monotremes are found in Australia and New Guinea.

- The infraclass **Metatheria** contains the marsupial mammals. They are viviparous but have very short gestation periods. A protective pouch, called the marsupium, covers the female mammary glands. The young crawl into the marsupium after birth, where they feed and complete development. About 250 species of marsupials live in the Australian region and the Americas (figure 22.4c).
- The other therian infraclass, Eutheria contains the placental mammals. They are usually born at an advanced stage of development, having been nourished within the uterus. Exchanges between maternal and fetal circulatory systems occur by diffusion across an organ called the placenta, which is composed of both maternal and fetal tissue. The approximately 3,800 species of eutherians are classified into 17 orders (figures 22.5).



Figure 22.5:

Order Edentata. (a) A giant anteater (Myrmecophaga tridactyla). Anteaters lack teeth. They use powerful forelimbs to tear into an insect nest and a long tongue covered with sticky saliva to capture prey. (b) An armadillo (Dasypus novemcinctus).

TABLE 22.1
CLASSIFICATION OF MAMMALS

Class Mammalia (mah-ma'le-ah)
Mammary glands; hair; diaphragm; three middle-ear ossicles; heterodont dentition; sweat, sebaceous, and scent glands; four-chambered heart; large cerebral cortex.

Subclass Prototheria (pro"to-ther'e-ah)
This subclass formerly contained the monotremes. Monotremes have recently been reclassified, and this subclass now contains only extinct species.

Subclass Theria (ther'e-ah)
Technical characteristics of the skull distinguish members of this subclass.

Infraclass Ornithodelphia (or"ne-tho-del'fe-ah)
Technical characteristics of the skull distinguish members of this infraclass, Monotremes.

Infraclass Metatheria (met"ah-ther'e-ah)
Viviparous; primitive placenta; young are born early and often are carried in a marsupial pouch on the female's belly Marsupials.

Infraclass Eutheria (u-ther'-e-ah)*
Complex placenta; young develop to advanced stage prior to birth. Placentals.

Order Insectivora (in-sec-tiv'or-ah)
Diverse group of small, primitive mammals; third largest mammalian order. Hedgehogs, tenrecs, moles, shrews.

Order Chiroptera (ki-rop'ter-ah)
Cosmopolitan, but especially abundant in the tropics; bones of the arm and hand are elongate and slender; flight membranes extend from the body, between digits of forelimbs, to the hindlimbs; most are insectivorous, but some are fruit eaters, fish eaters, and blood feeders; second largest mammalian order. Bats.

Order Primates (pri-ma'tez)
Adaptations of primates reflect adaptations for increased agility in arboreal (tree-dwelling) habitats; omnivorous diets; unspecialized teeth; grasping digits; freely movable limbs; nails on digits; reduced nasal cavity; enlarged eyes and cerebral hemispheres. Lemurs (Madagascar and the Comoro Islands), tarsiers (jungles of Sumatra and the East Indies), monkeys, gibbons, great apes (apes and humans).

Order Edentata (e"den-ta'tah) or **Xenarthra** (ze'nar-thra)
Incisors and canines absent; cheek teeth, when present, lack enamel; braincase is long and cylindrical; hindfoot is four toed; forefoot has two or three prominent toes with large claws; limbs are specialized for climbing or digging; xenarthrous lumbar vertebrae. Anteaters, tree sloths, armadillos.

Order Lagomorpha (lag"o-mor'fah)
Two pairs of upper incisors; one pair of lower incisors; incisors are ever-growing and slowly worn down by feeding on vegetation. Rabbits, pikas.

Order Rodentia (ro-den'che-ah)
Largest mammalian order; upper and lower jaws bear a single pair of ever-growing incisors. Squirrels, chipmunks, rats, mice, beavers, porcupines, woodchucks, lemmings.

Order Cetacea (se-ta'she-ah)
Streamlined, nearly hairless, and insulated by thick layers of fat (blubber); no sebaceous glands; forelimbs modified into paddlelike flippers for swimming; hindlimbs reduced and not visible externally; tail fins (flukes) flattened horizontally; external naris (blowhole) on top of skull. Toothed whales (beaked whales, narwhals, sperm whales, dolphins, porpoises, killer whales); toothless, filter-feeding whales (right whales, gray whales, blue whales, and humpback whales).

Order Carnivora (kar-niv'o-rah)
Predatory mammals; usually have a highly developed sense of smell and a large braincase; premolars and molars modified into carnassial apparatus; three pairs of upper and lower incisors usually present, and canines are well developed. Dogs, cats, bears, raccoons, minks, sea lions, seals, walruses, otters.

Order Proboscidea (pro"bah-sid'e-ah)
Long, muscular proboscis (trunk) with one or two finger-like processes at the tip; short skull with the second incisor on each side of the upper jaw modified into tusks; six cheek teeth are present in each half of each jaw; teeth erupt (grow into place) in sequence from front to rear, so that one tooth in each jaw is functional. African and Indian elephants.

Order Sirenia (si-re'ne-ah)
Large, aquatic herbivores that weigh in excess of 600 kg; nearly hairless, with thick, wrinkled skin; heavy skeleton; forelimb is flipperlike, and hindlimb is vestigial; horizontal tail fluke is present; horizontally oriented diaphragm; teeth lack enamel. Manatees (coastal rivers of the Americas and Africa), dugongs (western Pacific and Indian Oceans).

Order Perissodactyla (pe-ris"so-dak'ti-lah).
Skull usually elongate, large molars and premolars; primarily grazers. (The Artiodactyla also have hoofs. Artiodactyls and perissodactyls are, therefore, called ungulates. [L. *ungula*, hoof] Horses, rhinoceroses, zebras, tapirs.

Order Artiodactyla (ar"re-o-dak'ti-lah)
Hoofed; axis of support passes between third and fourth digits; digits one, two, and five reduced or lost; primarily grazing and browsing animals (pigs are an obvious exception). Pigs, hippopotamuses, camels, antelope, deer, sheep, giraffes, cattle.

Lesson 70

Evolutionary Pressure: External Structure and Locomotion

External Structure and Locomotion

Evolutionary perspective

- Mammals are naturally distributed on all continents except Antarctica, and they live in all oceans
- **EXTERNAL STRUCTURE AND LOCOMOTION**
- The skin of a mammal, like that of other vertebrates, consists of epidermal and dermal layers. It protects from mechanical injury, invasion by microorganisms, and the sun's ultraviolet light
- Skin is also important in temperature regulation, sensory perception, excretion, and water regulation (see figure)
- **Hair** is a keratinized derivative of the epidermis of the skin and is uniquely mammalian. It is seated in an invagination of the epidermis, called a **hair follicle**.
- A coat of hair, called **pelage**, usually consists of two kinds of hair
- Long guard hairs protect a dense coat of shorter, insulating under hairs. Because hair is composed largely of dead cells, it must be periodically molted.
- In some mammals (e.g., humans), molting occurs gradually and may not be noticed. In others, hair loss occurs rapidly and may result in altered pelage characteristics.

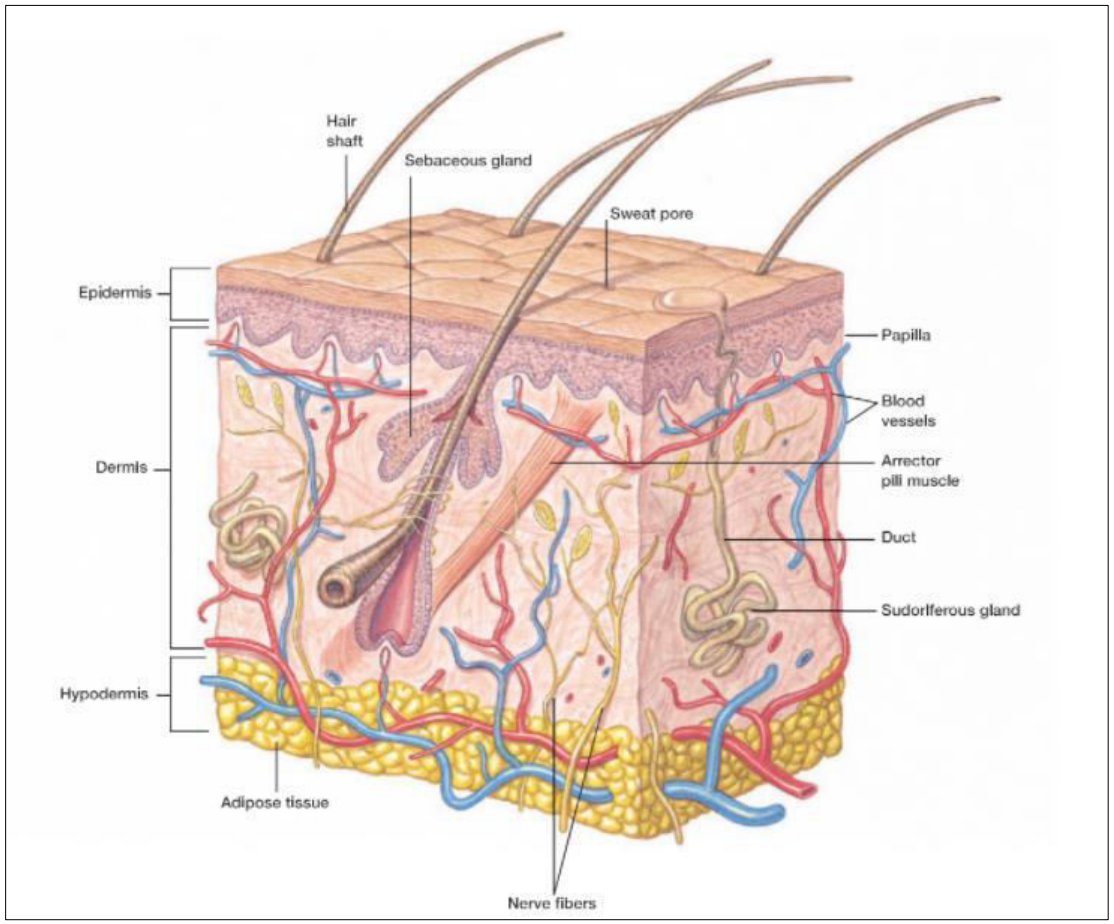


Figure: Skin of Mammals. Notice the various structures in the dermis of human skin.

- In the fall, many mammals acquire a thick coat of insulating under hair, and he pelage color may change. For example, the Arctic fox takes on a white or cream color with its autumn molt, which helps conceal the fox in a snowy environment. With its spring molt, the Arctic fox acquires a gray and yellow pelage (see figure 22.6)
- Hair is also important for the sense of touch
- Mechanical displacement of a hair stimulates nerve cells associated with the hair root
- Guard hairs may sometimes be modified into thick-shafted hairs called **vibrissae**. Vibrissae occur around the legs, nose, mouth, and eyes of many mammals. Their roots are richly innervated and very sensitive to displacement.



Figure 22.6: Order Carnivora. An Arctic fox (Alopex lagopus) with its winter pelage. With its spring molt, the Arctic fox acquires a gray and yellow colored coat.

- Air spaces in the hair shaft and air trapped between hair and the skin provide an effective insulating layer
- A band of smooth muscle, called the **arrector pili muscle**, runs between the hair follicle and the lower epidermis
- When the muscle contracts, the hairs stand upright, increasing the amount of air trapped in the pelage and improving its insulating properties. Arrector pili muscles are under the control of the autonomic nervous system, which also controls a mammal's "fight-or-flight" response
- In threatening situations, the hair (especially on the neck and tail) stands on end and may give the perception of increased size and strength.

Lesson 71

Mammal: External structure and locomotion 1

- Hair color depends on the amount of pigment (melanin) deposited in it and the quantity of air in the hair shaft
- The pelage of most mammals is dark above and lighter underneath. This pattern makes them less conspicuous under most conditions
- Some mammals advertise their defenses using aposematic (warning) coloration. The contrasting markings of a skunk are a familiar example.
- **Pelage** is reduced in large mammals from hot climates (e.g., elephants and hippopotamuses) and in some aquatic mammals (e.g., whales) that often have fatty insulation.
- A few mammals (e.g., naked mole rats) have almost no pelage

- Claws are present in all amniote classes. They are used for locomotion and offensive and defensive behavior. Claws form from accumulations of keratin that cover the terminal phalanx (bone) of the digits. In some mammals, they are specialized to form nails or hooves (figure 22.7).
- **Glands** develop from the epidermis of the skin
- **Sebaceous (oil) glands** are associated with hair follicles, and their oily secretion lubricates and waterproofs the skin and hair

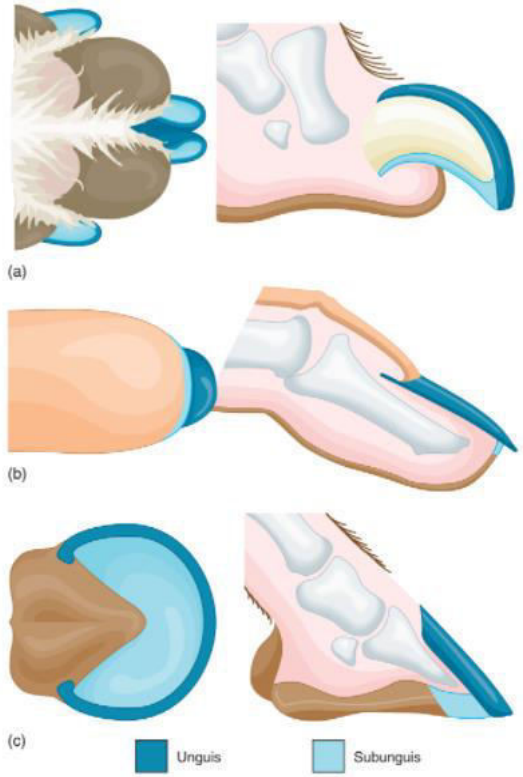


Figure 22.7: Structure of Claws, Nails, and Hooves. (a) Claws. (b) Nails are flat, broad claws found on the hands and feet of primates and are an adaptation for arboreal habits, where grasping is essential. (c) Hooves are characteristic of ungulate mammals. The number of toes is reduced, and the animals walk or run on the tips of the remaining digits. The unguis is a hard, keratinized dorsal plate, and the subunguis is a softer ventral plate

- Most mammals also possess **sudoriferous (sweat) glands**
- Small sudoriferous glands (**eccrine glands**) release watery secretions used in evaporative cooling
- Larger sudoriferous glands (**apocrine glands**) secrete a mixture of salt, urea, and water, which microorganisms on the skin convert to odorous products.
- **Scent or musk glands** are around the face, feet, or anus of many mammals
- These glands secrete pheromones, which may be involved with defense, species and sex recognition, and territorial behavior.

- **Mammary glands** are functional in female mammals and are present, but nonfunctional, in males
- The milk that mammary glands secrete contains water, carbohydrates (especially the sugar lactose), fat, protein, minerals, and antibodies.
- Mammary glands are probably derived evolutionarily from apocrine glands and usually contain substantial fatty deposits
- Monotremes have mammary glands that lack nipples. The glands discharge milk into depressions on the belly, where the young lap it up
- In other mammals, mammary glands open via nipples or teats, and the young suckle for their nourishment (figure 22.8).

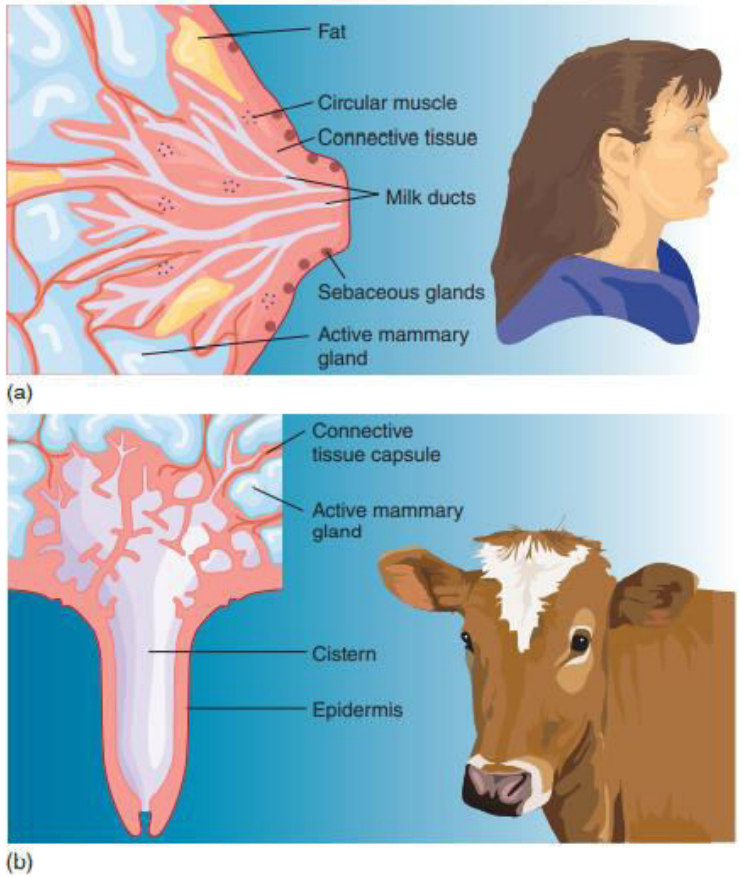


Figure 22.8: Mammary Glands. Mammary glands are specialized to secrete milk following the birth of young. (a) In humans, many ducts lead from the glands to a nipple. Parts of the duct system are enlarged to store milk. Suckling by an infant initiates a hormonal response that causes the mammary glands to release milk. (b) Some mammals (e.g., cattle) have teats that form by the extension of a collar of skin around the opening of mammary ducts. Milk collects in a large cistern prior to its release. The number of nipples or teats varies with the number of young produced

Lesson no. 72

The Skull and Teeth

- The skulls of mammals show important modifications of the reptilian pattern
- One feature that zoologists use to distinguish reptilian from mammalian skulls is the method of **jaw articulation**
- In reptiles, the jaw articulates at two small bones at the rear of the jaw. In mammals, these bones have moved into the middle ear, and along with the stapes, form the middle-ear ossicles
- A single bone of the lower jaw articulates the mammalian jaw.

- A **secondary palate** evolved twice in vertebrates—in the archosaur lineage and in the synapsid lineage
- In some therapsids, small, shelflike extensions of bone (the hard palate) partially separated the nasal and oral passageways
- In mammals, the secondary palate extends posteriorly by a fold of skin, called the **soft palate**, which almost completely separates the nasal passages from the mouth cavity. Unlike other vertebrates that swallow food whole or in small pieces, some mammals chew their food
- The more extensive secondary palate allows mammals to breathe while chewing. Breathing needs to stop only briefly during swallowing (figure 22.9).

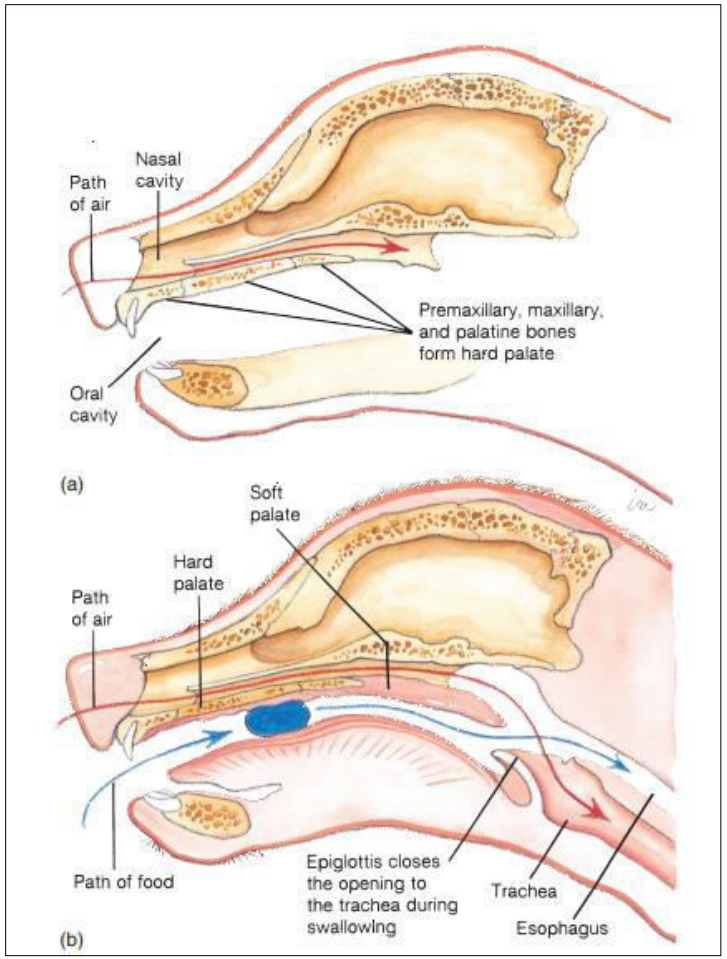


Figure 22.9: Secondary Palate. (a) The secondary palate (consisting of the hard and soft palates) of a mammal almost completely separates the nasal and oral cavities. (b) Breathing stops only momentarily during swallowing.

- The structure and arrangement of teeth are important indicators of mammalian lifestyles
- In reptiles, the teeth are uniformly conical, a condition referred to as **homodont**
- In mammals, the teeth are often specialized for different functions, a condition called **heterodont**
- Reptilian teeth attach along the top or inside of the jaw, whereas in mammals, the teeth are set into sockets of the jaw
- Most mammals have two sets of teeth during their lives.

- The first teeth emerge before or shortly after birth and are called **deciduous or milk teeth**
- These teeth are lost, and permanent teeth replace them
- Adult mammals have up to four kinds of teeth
- 1. **Incisors** are the most anterior teeth in the jaw. They are usually chisel-like and used for gnawing or nipping
- 2. **Canines** are often long, stout, and conical, and are usually used for catching, killing, and tearing prey. Canines and incisors have single roots
- Premolars are positioned next to canines, and have one or two roots and truncated surfaces for chewing

- Molars have broad chewing surfaces and two (upper molars) or three (lower molars) roots
- Mammalian species have characteristic numbers of each kind of adult tooth
- Zoologists use a dental formula to characterize taxa. It is an expression of the number of teeth of each kind in one-half of the upper and lower jaws.
- The teeth of the upper jaw are listed above those of the lower jaw and in the following order: incisors, canine, premolars, and molars

- For example:

Human	Beaver
2•1•2•3	1•0•1•3
2•1•2•3	1•0•1•3

- Mammalian teeth (dentition) may be specialized for particular diets
- In some mammals, the dentition is reduced, sometimes to the point of having no teeth
- For example, armadillos and the giant anteater (order Edentata) feed on termites and ants, and their teeth are reduced

- Some mammals (e.g., humans, order Primates; and pigs, order Artiodactyla) are omnivorous; they feed on a variety of plant and animal materials
- They have anterior teeth with sharp ripping and piercing surfaces, and posterior teeth with flattened grinding surfaces for rupturing plant cell walls
- Mammals that eat plant material often have flat, grinding posterior teeth and incisors, and sometimes have canines modified for nipping plant matter (e.g., horses, order Perissodactyla; deer, order Artiodactyla) or gnawing (e.g., rabbits, order Lagomorpha; beavers, order Rodentia)

- In rodents, the incisors grow throughout life
- Although most mammals have enamel covering the entire tooth, rodents have enamel only on the front surfaces of their incisors
- The teeth are kept sharp by slower wear in front than in back. A gap called the diastema separates the anterior food-procuring teeth from the posterior grinding teeth.
- The diastema results from an elongation of the snout that allows the anterior teeth to reach close to the ground or into narrow openings to procure food

- The posterior teeth have a high, exposed surface (crown) and continuous growth, which allows these teeth to withstand years of grinding tough vegetation
- Predatory mammals use canines and incisors for catching, killing, and tearing prey
- In members of the order Carnivora (e.g., coyotes, dogs, and cats), the fourth upper premolars and first lower molars form a scissor like shearing surface, called the carnassial apparatus, that is used for cutting flesh from prey.

Lesson 73

The Vertebral Column and Appendicular Skeleton

- The vertebral column of mammals is divided into five regions
- As with reptiles and birds, the first two cervical vertebrae are the atlas and axis
- Five other cervical vertebrae usually follow
- Even the giraffe and the whale have seven neck vertebrae, which are greatly elongate or compressed, respectively
- In contrast, tree sloths have either six or nine cervical vertebrae, and the manatee has six cervical vertebrae.

- The trunk is divided into thoracic and lumbar regions, as is the case for birds
- In mammals, the division is correlated with their method of breathing. The thoracic region contains the ribs
- Most ribs connect to the thoracic vertebrae and to the sternum via costal cartilages. Other ribs attach only to the thoracic vertebrae or to other ribs through cartilages.
- All ribs protect the heart and lungs. The articulation between the thoracic vertebrae provides the flexibility needed in turning, climbing, and lying on the side to suckle young.

- Lumbar vertebrae have interlocking processes that give support, but little freedom of movement
- The appendicular skeleton of mammals rotates under the body so that the appendages are usually directly beneath the body
- Joints usually limit the movement of appendages to a single anteroposterior plane, causing the tips of the appendages to move in long arcs

- The bones of the pelvic girdle are fused in the adult, a condition that is advantageous for locomotion but presents problems during the birth of offspring
- In a pregnant female, the ventral joint between the halves of the pelvis—the pubic symphysis— loosens before birth, allowing the pelvis to spread during birth.

Muscles

- Because the appendages are directly beneath the body of most mammals, the skeleton bears the weight of the body
- Muscle mass is concentrated in the upper appendages and girdles
- Many running mammals (e.g., deer, order Artiodactyla) have little muscle in their lower leg that would slow leg movement
- Instead, tendons run from muscles high in the leg to cause movement at the lower joints.

Lesson 74

Mammals: Nutrition and Digestive System

Nutrition

- The digestive tract of mammals is similar to that of other vertebrates but has many specializations for different feeding habits
- Some specializations of teeth have already been described
- The feeding habits of mammals are difficult to generalize. Feeding habits reflect the ecological specializations that have evolved
- For example, most members of the order Carnivora feed on animal flesh and are, therefore, carnivores
- Other members of the order, such as bears, feed on a variety of plant and animal products and are omnivores

- Some carnivorous mammals are specialized for feeding on arthropods or soft-bodied invertebrates, and are often referred to (rather loosely) as insectivores. These include animals in the orders Insectivora (e.g., shrews), Chiroptera (bats), and Edentata (anteaters) (see figure 22.5a)
- Herbivores such as deer (order Artiodactyla) and zebras (order Perissodactyla) feed mostly on vegetation, but their diet also includes invertebrates inadvertently ingested while feeding.

Digestive System

- Specializations in the digestive tract of most herbivores reflect the difficulty of digesting food rich in cellulose
- Horses, rabbits, and many rodents have an enlarged cecum at the junction of the large and small intestines
- A **cecum** is a fermentation pouch where microorganisms aid in cellulose digestion. Sheep, cattle, and deer are called **ruminants** (L. ruminare, to chew the cud). Their stomachs are modified into four chambers

- The first three chambers are storage and fermentation chambers and contain microorganisms that synthesize a cellulose-digesting enzyme (cellulase)
- Gases that fermentation produces are periodically belched, and some plant matter (cud) is regurgitated and rechewed
- Other microorganisms convert nitrogenous compounds in the food into new proteins.

Lesson 75

Circulation

- The hearts of birds and mammals are superficially similar
- Both are four-chambered pumps that keep blood in the systemic and pulmonary circuits separate, and both evolved from the hearts of ancient reptiles
- Their similarities, however, are a result of adaptations to active lifestyles
- The evolution of similar structures in different lineages is called **convergent evolution**

- The mammalian heart evolved in the synapsid reptilian lineage, whereas the avian heart evolved in the archosaur lineage (figure 22.12).
- One of the most important adaptations in the circulatory system of eutherian mammals concerns the distribution of respiratory gases and nutrients in the fetus (figure 22.13a)
- Exchanges between maternal and fetal blood occur across the placenta. Although maternal and fetal blood vessels are intimately associated, no blood actually mixes

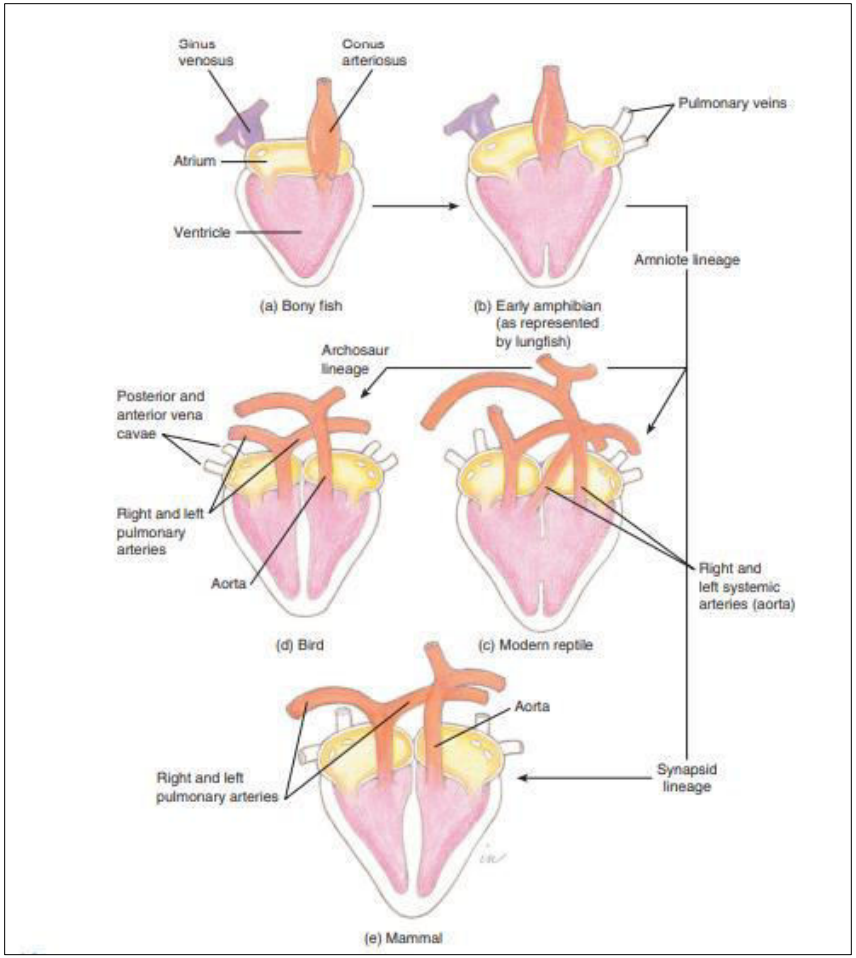


Figure 22.12: Possible Sequence in the Evolution of the Vertebrate Heart. (a) Diagrammatic representation of a bony fish heart. (b) In lungfish, partially divided atria and ventricles separate pulmonary and systemic circuits. This heart was probably similar to that in primitive amphibians and early amniotes. (c) The hearts of modern reptiles were derived from the pattern in (b). (d) The archosaur and (e) synapsid lineages resulted in completely separated, four-chambered hearts

- Nutrients, gases, and wastes simply diffuse between fetal and maternal blood supplies
- Blood entering the right atrium of the fetus is returning from the placenta and is highly oxygenated.
- Because fetal lungs are not inflated, resistance to blood flow through the pulmonary arteries is high
- Therefore, most of the blood entering the right atrium bypasses the right ventricle and passes instead into the left atrium through a valved opening between the atria (the foramen ovale).

- Some blood from the right atrium, however, does enter the right ventricle and the pulmonary artery
- Because of the resistance at the un inflated lungs, most of this blood is shunted to the aorta through a vessel connecting the aorta and the left pulmonary artery (the ductus arteriosus)
- At birth, the placenta is lost, and the lungs are inflated
- Resistance to blood flow through the lungs is reduced, and blood flow to them increases

- Flow through the ductus arteriosus decreases, and the vessel is gradually reduced to a ligament
- Blood flow back to the left atrium from the lungs correspondingly increases, and the valve of the foramen ovale closes and gradually fuses with the tissue separating the right and left atria (figure 22.13b).

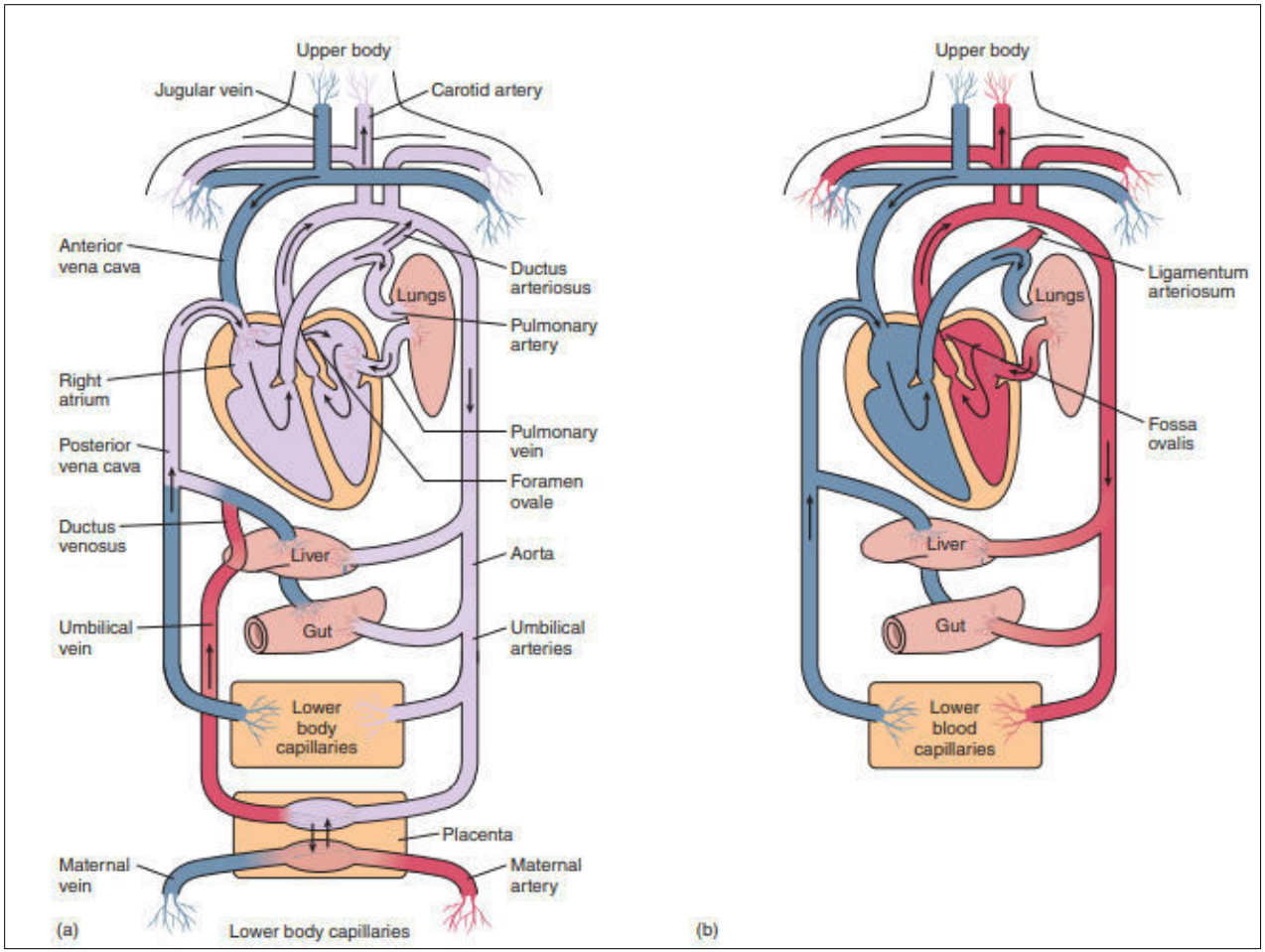


Figure 22.13: Mammalian Circulatory Systems. The circulatory patterns of (a) fetal and (b) adult mammals. Highly oxygenated blood is shown in red, and less oxygenated blood is shown in blue. In fetal circulation, highly oxygenated blood from the placenta mixes with less oxygenated blood prior to entering the right atrium. Thus, most arterial blood of the fetus is moderately oxygenated. The pale lavender color in (a) symbolizes this state of oxygenation.

Lesson 76

Gas Exchange

- High metabolic rates require adaptations for efficient gas exchange
- Most mammals have separate nasal and oral cavities and longer snouts, which provide an increased surface area for warming and moistening inspired air
- Respiratory passageways are highly branched, and large surface areas exist for gas exchange
- Mammalian lungs resemble a highly vascular sponge, rather than the saclike structures of amphibians and a few reptiles

- Mammalian lungs, like those of reptiles, inflate using a negative-pressure mechanism
- Unlike reptiles and birds, however, mammals possess a muscular diaphragm that separates the thoracic and abdominal cavities
- **Inspiration** results from contraction of the diaphragm and expansion of the rib cage, both of which decrease the intrathoracic pressure and allow air to enter the lungs

- **Expiration** is normally by elastic recoil of the lungs and relaxation of inspiratory muscles, which decreases the volume of the thoracic cavity
- The contraction of other thoracic and abdominal muscles can produce forceful exhalation.

Lesson 77

Temperature regulation

Temperature regulation

- Mammals are widely distributed over the earth, and some face harsh environmental temperatures
- Nearly all face temperatures that require them to dissipate excess heat at some times and to conserve and generate heat at other times
- Heat-producing mechanisms of mammals are divided into two categories
- **Shivering thermogenesis** is muscular activity that generates large amounts of heat but little movement

- **Nonshivering thermogenesis** involves heat production by general cellular metabolism and the metabolism of special fat deposits called brown fat
- Heat production is effective in thermoregulation because mammals are insulated by their pelage and/or fat deposits
- Fat deposits are also sources of energy to sustain high metabolic rates

- Mammals without a pelage can conserve heat by allowing the temperature of surface tissues to drop
- A walrus in cold, arctic waters has a surface temperature near 0° C; however, a few centimeters below the skin surface, body temperatures are about 35° C
- Upon emerging from the icy water, the walrus quickly warms its skin by increasing peripheral blood flow

- Most tissues cannot tolerate such rapid and extreme temperature fluctuations
- Further investigations are likely to reveal some unique biochemical characteristics of these skin tissues
- Even though most of the body of an arctic mammal is unusually well insulated, appendages often have thin coverings of fur as an adaptation to changing thermoregulatory needs
- Even in winter, an active mammal sometimes produces more heat than is required to maintain body temperature. Patches of poorly insulated skin allow excess heat to be dissipated.

- During periods of inactivity or extreme cold, however, arctic mammals must reduce heat loss from these exposed areas, often by assuming heat-conserving postures
- Mammals sleeping in cold environments conserve heat by tucking poorly insulated appendages and their faces under well-insulated body parts

- **Countercurrent heat-exchange systems** may help regulate heat loss from exposed areas (figure 22.14)
- Arteries passing peripherally through the core of an appendage are surrounded by veins that carry blood back toward the body
- When blood returns to the body through these veins, heat transfers from arterial blood to venous blood and returns to the body rather than being lost to the environment
- When excess heat is produced, blood is shunted away from the countercurrent veins toward peripheral vessels, and excess heat is radiated to the environment

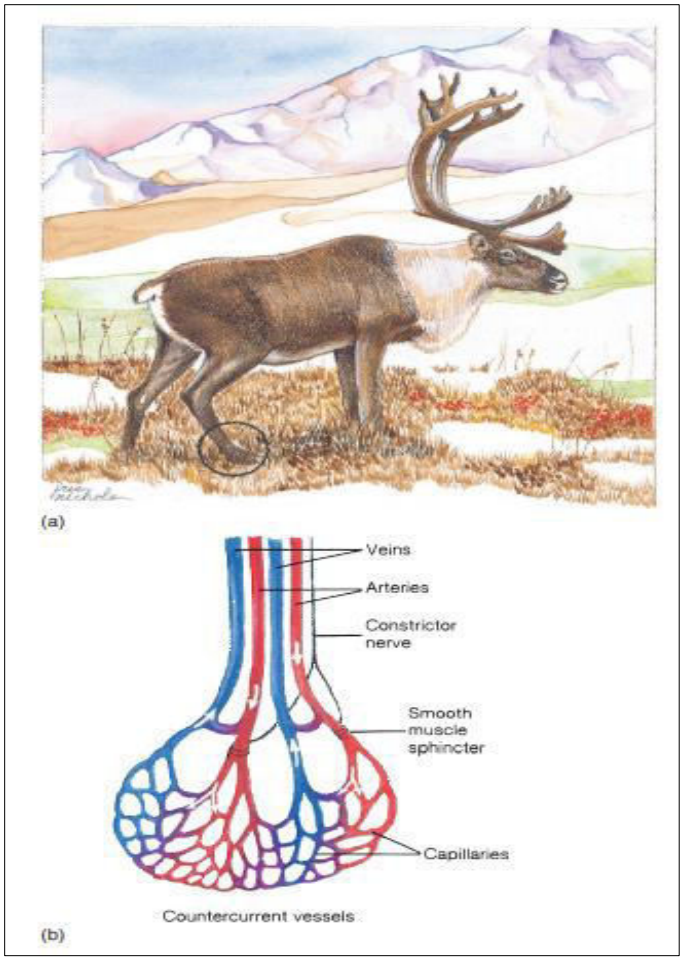


Figure 22.14: Countercurrent Heat Exchange. Countercurrent heat exchangers conserve body heat in mammals adapted to cold environments. (a) Systems similar to the one depicted here are found in the legs of reindeer (Rangifer tarandus) and in the flippers of dolphins. (b) Heat transferred from blood moving peripherally in arteries warms venous blood returning from an extremity. During winter, the lower part of a reindeer's leg may be at 10°C, while body temperature is about 40°C. White arrows indicate direction of blood flow

- Mammals have few problems getting rid of excess heat in cool, moist environments
- Heat can be radiated into the air from vessels near the surface of the skin or lost by evaporative cooling from either sweat glands or respiratory surfaces during panting
- Hot, dry environments present far greater problems, because evaporative cooling may upset water balances.
- Jackrabbits and elephants use their long ears to radiate heat. Small mammals often avoid the heat by remaining in burrows during the day and foraging for food at night
- Other mammals seek shade or watering holes for cooling.

Lesson 78

Winter sleep and Hibernation

Winter Sleep

- Mammals react in various ways to environmental extremes
- Caribou migrate to avoid extremes of temperature, and wildebeest migrate to avoid seasonal droughts
- Other mammals retreat to burrows under the snow, where they become less active but are still relatively alert and easily aroused—a condition called winter sleep.

- For example, bears and raccoons retreat to dens in winter
- Their body temperatures and metabolic rates decrease somewhat, but they do not necessarily remain inactive all winter

Hibernation

- **Hibernation** is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates
- True hibernators include the monotremes (echidna and duck-billed platypus) and many members of the Insectivora (e.g., moles and shrews), Rodentia (e.g., chipmunks and woodchucks), and Chiroptera (bats)
- In preparation for hibernation, mammals usually accumulate large quantities of body fat.

- After a hibernating mammal retreats to a burrow or a nest, the hypothalamus sets the body's thermostat to about 2° C
- The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about four breaths per minute
- The heart rate falls from 200 to 300 beats per minute to about 20 beats per minute
- During hibernation, a mammal may lose a third to half of its body weight

- Arousal from hibernation occurs by metabolic heating, frequently using brown fat deposits, and it takes several hours to raise body temperature to near 37° C.

Lesson 79

NERVOUS AND SENSORY FUNCTIONS

- The basic structure of the vertebrate nervous system is retained in mammals
- The development of complex nervous and sensory functions goes hand-in-hand with active lifestyles and is most evident in the enlargement of the cerebral hemispheres and the cerebellum of mammals
- Most integrative functions shift to the enlarged cerebral cortex (neocortex)

- In mammals, the sense of touch is well developed. Receptors are associated with the bases of hair follicles and are stimulated when a hair is displaced
- Olfaction was apparently an important sense in early mammals, because fossil skull fragments show elongate snouts, which would have contained olfactory epithelium
- Cranial casts of fossil skulls show enlarged olfactory regions.

- Olfaction is still an important sense for many mammals. Mammals can perceive olfactory stimuli over long distances during either the day or night to locate food, recognize members of the same species, and avoid predators.

- Auditory senses were similarly important to early mammals
- More recent adaptations include an ear flap (the pinna) and the external ear canal leading to the **tympanum** that directs sound to the middle ear
- The middle ear contains three ear ossicles that conduct vibrations to the inner ear
- The sensory patch of the inner ear that contains the sound receptors is long and coiled and is called the **cochlea**
- This structure provides more surface area for receptor cells and gives mammals greater sensitivity to pitch and volume than is present in reptiles
- Cranial casts of early mammals show well-developed auditory regions.

- Vision is an important sense in many mammals, and eye structure is similar to that described for other vertebrates
- Accommodation occurs by changing the shape of the lens
- Color vision is less well developed in mammals than in reptiles and birds
- Rods dominate the retinas of most mammals, which supports the hypothesis that early mammals were nocturnal
- Primates, squirrels, and a few other mammals have well-developed color vision.

Lesson 80

Excretion and Osmoregulation

- Mammals, like all amniotes, have a metanephric kidney
- Unlike reptiles and birds, which excrete mainly uric acid, mammals excrete urea
- Urea is less toxic than ammonia and does not require large quantities of water in its excretion
- Unlike uric acid, however, urea is highly water soluble and cannot be excreted in a semisolid form; thus, some water is lost
- Excretion in mammals is always a major route for water loss.

- In the nephron of the kidney, fluids and small solutes are filtered from the blood through the walls of a group of capillary-like vessels, called the **glomerulus**
- The remainder of the nephron consists of tubules that reabsorb water and essential solutes and secrete particular ions into the filtrate
- The primary adaptation of the mammalian nephron is a portion of the tubule system called the **loop of the nephron**.

- The transport processes in this loop and the remainder of the tubule system allow mammals to produce urine that is more concentrated than blood
- For example, **beavers** produce urine that is **twice** as concentrated as blood, while **Australian hopping mice** produce urine that is **22 times** more concentrated than blood
- This accomplishes the same function that nasal and orbital salt glands do in reptiles and birds.

- Water loss varies greatly, depending on activity, physiological state, and environmental temperature
- Water is lost in urine, feces, and evaporation from sweat glands and respiratory surfaces, and during nursing
- Mammals in very dry environments have many behavioral and physiological mechanisms to reduce water loss

- The kangaroo rat, named for its habit of hopping on large hind legs, is capable of extreme water conservation (figure 22.15)
- It is native to the southwestern deserts of the United States and Mexico, and it survives without drinking water
- Its feces are almost dry, and its nocturnal habits reduce evaporative water loss.



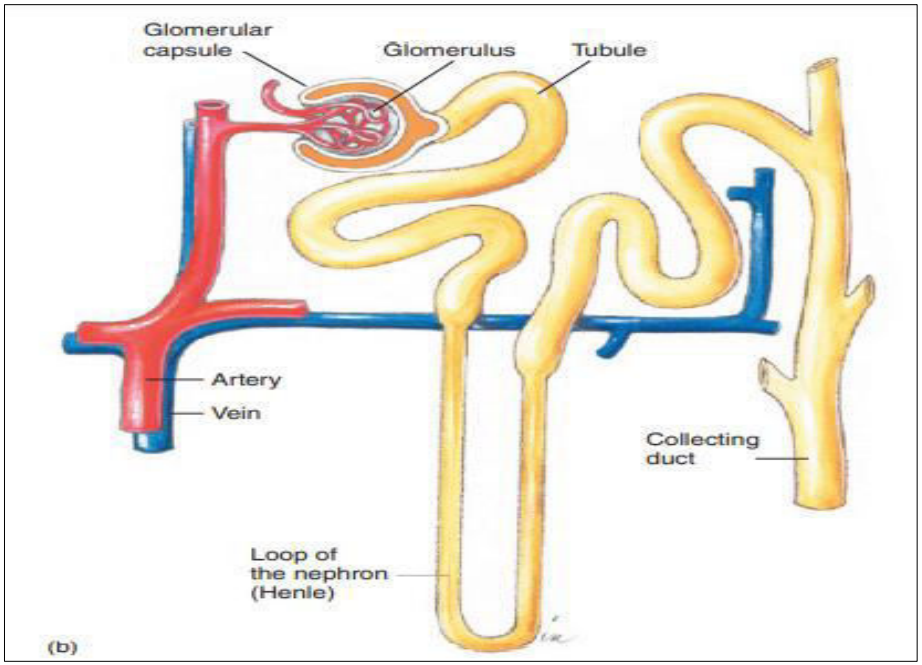


Figure 22.15
Order Rodentia. (a) The kangaroo rat (Dipodomys ordii). (b) The long loop of the nephron of this desert animal conserves water, preventing dehydration.

- Condensation as warm air in the respiratory passages encounters the cooler nasal passages minimizes respiratory water loss
- A low-protein diet, which reduces urea production, minimizes excretory water loss
- The nearly dry seeds that the kangaroo rat eats are rich sources of carbohydrates and fats
- Metabolic oxidation of carbohydrates produces water as a by-product.

Lesson 81

Behaviour of Mammals

- Mammals have complex behaviors that enhance survival
- Visual cues are often used in communication
- The bristled fur, arched back, and open mouth of a cat communicate a clear message to curious dogs or other potential threats
- A tail-wagging display of a dog has a similarly clear message
- A wolf defeated in a fight with other wolves lies on its back and exposes its vulnerable throat and belly.

- Similar displays may allow a male already recognized as being subordinate to another male to avoid conflict within a social group
- Pheromones are used to recognize members of the same species, members of the opposite sex, and the reproductive state of a member of the opposite sex
- Pheromones may also induce sexual behavior, help establish and recognize territories, and ward off predators

- The young of many mammalian species recognize their parents, and parents recognize their young, by smell
- Bull elk smell the rumps of females during the breeding season to recognize those in their brief receptive period
- They also urinate on their own bellies and under hair to advertise their reproductive status to females and other males.

- Male mammals urinate on objects in the environment to establish territories and to allow females to become accustomed to their odors
- Rabbits and rodents spray urine on a member of the opposite sex to inform the second individual of the first's readiness to mate
- Skunks use chemicals to ward off predators.

- Auditory and tactile communication are also important in the lives of mammals
- Herd animals stay together and remain calm as long as familiar sounds (e.g., bellowing, hooves walking over dry grasses and twigs, and rumblings from ruminating stomachs) are uninterrupted
- Unfamiliar sounds may trigger alarm and flight

- Vocalizations and tactile communication are important in primate social interactions
- Tactile communication ranges from precopulatory “nosing” that occurs in many mammals to grooming
- Grooming helps maintain a healthy skin and pelage, but also reinforces important social relationships within primate groups.

Lesson 82

Territoriality

- Many mammals mark and defend certain areas from intrusion by other members of the same species
- When cats rub their face and neck on humans or on furniture, the behavior is often interpreted as affection
- Cats, however, are really staking claim to their territory, using odors from facial scent glands.

- Some territorial behavior attracts females to, and excludes other males from, favorable sites for mating and rearing young
- Male California sea lions (Zalophus californianus) establish territories on shorelines where females come to give birth to young
- For about two weeks, males engage in vocalizations, displays, and sometimes serious fighting to stake claim to favorable territories

- Older, dominant bulls are usually most successful in establishing territories, and young bulls generally swim and feed just offshore
- When they arrive at the beaches, females select a site for giving birth
- Selection of the birth site also selects the bull that will father next year's offspring
- Mating occurs approximately two weeks after the birth of the previous year's off-spring.

- Development is arrested for the three months during which the recently born young do most of their nursing. This mechanism is called **embryonic diapause**
- Thus, even though actual development takes about nine months, the female carries the embryo and fetus for a period of one year.

Lesson 83

Reproduction and development

- In no other group of animals has viviparity developed to the extent it has in mammals
- Mammalian viviparity requires a large expenditure of energy on the part of the female during development and on the part of one or both parents caring for young after they are born
- Viviparity is advantageous because females are not necessarily tied to a single nest site, but can roam or migrate to find food or a proper climate

- Viviparity is accompanied by the evolution of a portion of the reproductive tract where the young are nourished and develop
- In viviparous mammals, the oviducts are modified into one or two uteri

Reproductive Cycles

- Most mammals have a definite time or times during the year in which ova (eggs) mature and are capable of being fertilized
- Reproduction usually occurs when climatic conditions and resource characteristics favor successful development
- Mammals living in environments with few seasonal changes and those that exert considerable control over immediate environmental conditions (e.g., humans) may reproduce at any time of the year
- However, they are still tied to physiological cycles of the female that determine when ova can be fertilized.

- Most female mammals undergo an estrus (Gr. oistros, a vehement desire) cycle, which includes a time during which the female is behaviorally and physiologically receptive to the male
- During the estrus cycle, hormonal changes stimulate the maturation of ova in the ovary and induce ovulation (release of one or more mature ova from an ovarian follicle)
- A few mammals (e.g., rabbits, ferrets, and mink) are induced ovulators; coitus (copulation) induces ovulation.

Lesson 84

Reproductive cycles 1

- Hormones also mediate changes in the uterus and vagina
- As the ova are maturing, the inner lining of the uterus proliferates and becomes more vascular in preparation for receiving developing embryos
- External swelling in the vaginal area and increased glandular discharge accompany the proliferation of vaginal mucosa

- During this time, males show heightened interest in females, and females are receptive to males
- If fertilization does not occur, the changes in the uterus and vagina are reversed until the next cycle begins
- No bleeding or sloughing of uterine lining usually occurs.

- Many mammals are **monestrus** and have only a single yearly estrus cycle that is sharply seasonal
- Wild dogs, bears, and sea lions are monestrus; domestic dogs are diestrus
- Other mammals are polyestrus. Rats and mice have estrus cycles that repeat every four to six days
- The menstrual cycle of female humans, apes, and monkeys is similar to the estrus cycle in that it results in a periodic proliferation of the inner lining of the uterus and correlates with the maturation of an ovum

- If fertilization does not occur before the end of the cycle, menses—the sloughing of the uterine lining—occurs

Lesson 85

Fertilization in Mammals

- Fertilization-usually occurs in the upper third of the oviduct within hours of copulation
- In a few mammals, fertilization may be delayed
- In some bats, for example, coitus occurs in autumn, but fertilization is delayed until spring
- Females store sperm in the uterus for periods in excess of two months

- This delayed fertilization is apparently an adaptation to winter dormancy
- Fertilization can occur immediately after females emerge from dormancy rather than waiting until males attain their breeding state
- In many other mammals, fertilization occurs right after coitus, but development is arrested after the first week or two. This embryonic diapause, which was described previously for sea lions, also occurs in some bats, bears, martens, and marsupials.

- The adaptive significance of embryonic diapause varies with species. In the sea lion, embryonic diapause allows the mother to give birth and mate within a short interval, but not have her resources drained by both nursing and pregnancy
- It also allows young to be born at a time when resources favor their survival
- In some bats, fertilization occurs in the fall before hibernation, but birth is delayed until resources become abundant in the spring.

Lesson 86

Modes of Development

- Monotremes are oviparous
- The ovaries release ova with large quantities of yolk
- After fertilization, shell glands in the oviduct deposit a shell around the ovum, forming an egg
- Female echidnas incubate eggs in a ventral pouch
- Platypus eggs are laid in their burrows.

- All other mammals have a placenta through which young are nourished for at least a portion of their development
- The maternal bloodstream, not yolk, supplies nutrients
- In marsupials, most nourishment for the fetus comes from “uterine milk” that uterine cells secrete
- Some nutrients diffuse from maternal blood into a highly vascular yolk sac that makes contact with the uterus. This connection in marsupials is a primitive placenta.

- The marsupial gestation period (the length of time young develop within the female reproductive tract) varies between 8 and 40 days in different species
- The gestation period is short because of marsupials' inability to sustain the production of hormones that maintain the uterine lining
- After birth, tiny young crawl into the marsupium and attach to a nipple, where they suckle for an additional 60 to 270 days (figure 22.17).



(a) (b)

Figure 22.17
Order Marsupialia. (a) The Virginia opossum (Didelphis virginiana) is found throughout Central America and is the only opossum found in North America. (b) Opossum young nursing in a marsupial pouch.

- In eutherian mammals, the embryo implants deeply into the uterine wall
- Embryonic and uterine tissues grow rapidly and be-come highly folded and vascular, forming the placenta
- Although maternal and fetal blood do not mix, nutrients, gases, and wastes diffuse between the two bloodstreams
- Gestation periods of eutherian mammals vary from 20 days (some rodents) to 19 months (the African elephant)

- Following birth, the placenta and other tissues that surrounded the fetus in the uterus are expelled as “afterbirth.” The newborns of many species (e.g., humans) are helpless at birth; others (e.g., deer and horses) can walk and run shortly after birth.